

Infinitesimal study of group schemes. Formal groups

SGA 3, Exposé VIIB

English draft

P. Gabriel

About this draft

This is an unofficial English translation of Exposé VIIIB, “Infinitesimal study of group schemes. Formal groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp7B-13oct24.pdf (99 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

The translator’s contribution is licensed under the MIT License. The French original remains copyright of its authors and publishers.

Contents

About this draft	iii
Exposé VIIB. Infinitesimal study of group schemes	1
0. Recollections on pseudocompact rings and modules	1
1. Formal varieties over a pseudocompact ring	13
2. Generalities on formal groups	33
3. Phenomena particular to characteristic 0	51
4. Phenomena particular to characteristic $p > 0$	57
5. Homogeneous spaces of infinitesimal formal groups over a field	62
Bibliography	76

Exposé VIIB. Infinitesimal study of group schemes

by P. Gabriel

B) Formal groups

⁽⁰⁾The study of formal groups is usually extremely simple. If this does not appear clearly in the pages which follow, the responsibility lies with an arithmetician who claims to know formal groups over “something other than fields”.⁽¹⁾ We have therefore developed, for formal groups “locally free over projective limits of artinian rings”, the generalities usually stated for formal groups defined over a field. For a more detailed study of the latter, we refer to the 1964/65 Heidelberg–Strasbourg seminar in algebraic geometry.⁽²⁾

0. Recollections on pseudocompact rings and modules

This section contains some technical preliminaries; in it we recall and supplement some results of [CA] (*Des catégories abéliennes*, Bull. Soc. Math. France **90**, 1962).

0.1. — A *left pseudocompact ring* is a separated and complete topological ring with identity which has a base of neighborhoods of 0 consisting of left ideals $\mathfrak{l}$ of finite colength (i.e. $\text{length}_A(A/\mathfrak{l}) < +\infty$). Here we shall suppose that A is commutative, so that there is no need to distinguish “between left and right”. In particular, the quotients $A/\mathfrak{l}$ are artinian rings and A is identified with the topological projective limit of these rings equipped with the discrete topology. A complete noetherian local ring $(A, \mathfrak{m})$ is evidently pseudocompact.⁽³⁾

0.1.1. — Every closed ideal I of A is the intersection of the open ideals containing it.⁽⁴⁾ Every maximal closed ideal is therefore open. Moreover, if $\mathfrak{l}$ is an open ideal of A , the maximal ideals of $A/\mathfrak{l}$ correspond bijectively to the maximal ideals $\mathfrak{m}$ containing $\mathfrak{l}$; the latter are therefore open and closed. Consequently, every maximal closed ideal is an open maximal ideal (and hence closed); the converse is evident. The set of these ideals will be denoted by $\Upsilon(A)$. If $\mathfrak{l}$ is an open ideal of A and $\mathfrak{m} \in \Upsilon(A)$, the localization $(A/\mathfrak{l})_{\mathfrak{m}}$ is thus a local ring if $\mathfrak{m}$ contains $\mathfrak{l}$, and is zero otherwise. Since $A/\mathfrak{l}$ is artinian, it is the direct product of finitely many local rings, which one can write

$$A/\mathfrak{l} \simeq \prod_{\mathfrak{m} \in \Upsilon(A)} (A/\mathfrak{l})_{\mathfrak{m}}.$$

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** The interest of formal groups over a complete noetherian local ring appears, for example, in the work of Lubin and Tate (cf. [LT65]). The study of formal groups over an arbitrary base, and of questions of lifting and deformation, in particular for Barsotti–Tate groups (“ p -divisible groups”), has given rise to an abundant literature; cf. for example [LT66, Ta67, Gr74, Me72, La75, Fo77, Il85, Br00]; in particular, the results of the present exposé are largely taken up in Chapter I of [Fo77].

⁽²⁾**N.D.E.** The editors found only such a seminar dated 1965/66 and entitled “Groupes algébriques linéaires”, in which the notion of a formal group does not appear; see, however, [De72].

⁽³⁾**N.D.E.** (when it is equipped with the $\mathfrak{m}$ -adic topology)

⁽⁴⁾**N.D.E.** Indeed, if $x \notin I$, there exists an open ideal $\mathfrak{l}$ such that $(x + \mathfrak{l}) \cap I = \emptyset$; then $I + \mathfrak{l}$ is an open ideal not containing x . Moreover, in what follows we have made explicit the fact that every “maximal closed” ideal is maximal and open.

From this one obtains “canonical” isomorphisms

$$\begin{aligned} A &\simeq \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} \prod_{\mathfrak{m}} (A/\mathfrak{l})_{\mathfrak{m}} \\ &\simeq \prod_{\mathfrak{m}} \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} A_{\mathfrak{m}}, \end{aligned}$$

where one has put

$$A_{\mathfrak{m}} = \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}}.$$

This *local component* $A_{\mathfrak{m}}$ is a filtered projective limit of artinian local rings equipped with the discrete topology; it is therefore a local ring which is pseudocompact for the projective limit topology.⁽⁵⁾

0.1.2. — Let $\mathfrak{r}(A)$ be the intersection of the open maximal ideals of A , that is, the cartesian product of the ideals $\mathfrak{m}A_{\mathfrak{m}}$ when A is identified with $\prod_{\mathfrak{m}} A_{\mathfrak{m}}$. For every open ideal $\mathfrak{l}$ of A , the image of $\mathfrak{r}(A)$ in $A/\mathfrak{l}$ is contained in the radical of $A/\mathfrak{l}$. Some power of this image is therefore zero, so that $\mathfrak{r}(A)^n$ is contained in $\mathfrak{l}$ when n is sufficiently large. The sequence of the $\mathfrak{r}(A)^n$ therefore tends to 0. The same holds for the sequence of the x^n when x belongs to $\mathfrak{r}(A)$. In other words, every element of $\mathfrak{r}(A)$ is topologically nilpotent, and the converse is clear. It follows that the sequence with general term $1 + x + \cdots + x^n$ converges, and converges to $1/(1 - x)$ when $x \in \mathfrak{r}(A)$. This shows that $\mathfrak{r}(A)$ is the Jacobson radical of A , i.e. the intersection of *all* the maximal ideals of A (cf. Bourbaki, *Algèbre*, Chap. 8, § 6, Th. 1).⁽⁶⁾

Remarks. ⁽⁷⁾

- (a) If $\mathfrak{p}$ is an *open* prime ideal of A , then, since $A/\mathfrak{p}$ is artinian, $\mathfrak{p}$ is a maximal ideal. Consequently, $\Upsilon(A)$ is the set of open prime ideals of A .
- (b) Each $\mathfrak{m}A_{\mathfrak{m}}$ is an *ideal of definition* of $A_{\mathfrak{m}}$, i.e. an *open* ideal I such that the sequence of the I^n tends to 0 (cf. EGA 0_I, 7.1.2). Consequently, $\text{Spec}(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})$, equipped with the topological ring $A_{\mathfrak{m}}$, is an *affine formal scheme* in the sense of EGA I, 10.1.2.
- (c) The topological ring A is *admissible* in the sense of EGA 0_I, 7.1.2 if and only if $\mathfrak{r}(A)$ is open (hence an ideal of definition), and this holds if and only if $\Upsilon(A)$ is *finite*. In this case the affine formal scheme $\text{Spf}(A) = \text{Spec}(A/\mathfrak{r}(A))$ (cf. EGA I, 10.1.2) has $\Upsilon(A)$, equipped with the discrete topology, as its underlying space, and the ring of sections of the structure sheaf over a subset E of $\Upsilon(A)$ is the product $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$.
- (d) Let A be an arbitrary pseudocompact ring. In 1.1 below, the space $\Upsilon(A)$ is equipped with the discrete topology and with the sheaf of rings whose ring of sections over any subset E is $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$. By b), every point then has an open neighborhood which is an affine formal scheme, so this defines a *formal scheme* (EGA I, 10.4.2), which will be denoted by $\text{Spf}(A)$. (For this formal scheme to be affine, it must be quasi-compact, hence $\Upsilon(A)$ must be finite, and in this case $\text{Spf}(A)$ coincides with the definition of EGA I, 10.1.2.)

0.1.3. — If A and B are two pseudocompact rings, a homomorphism from A to B is, by definition, a *continuous* map compatible with addition, multiplication, and the identity

⁽⁵⁾**N.D.E.** Observe that $A_{\mathfrak{m}}$ is the localization of A at the maximal ideal $\mathfrak{m}$. Indeed, the identity element $e_{\mathfrak{m}}$ of $A_{\mathfrak{m}}$ is an idempotent of A not belonging to $\mathfrak{m}$, and since $A = A_{\mathfrak{m}} \times B$, where $B = A(1 - e_{\mathfrak{m}})$, $A_{\mathfrak{m}}$ is identified with the localization $A_{e_{\mathfrak{m}}}$, and hence also with the localization $S^{-1}A$, where $S = A - \mathfrak{m}$. Moreover, since $e_{\mathfrak{m}}(1 - e_{\mathfrak{m}}) = 0$, $\mathfrak{m}$ contains $1 - e_{\mathfrak{m}}$, hence also B , and thus $\mathfrak{m}$ is identified with $\mathfrak{m}A_{\mathfrak{m}} \times B$.

⁽⁶⁾**N.D.E.** Indeed, let $x \in \mathfrak{r}(A)$; if $\mathfrak{m}$ is a maximal ideal not containing x , there exists $y \in A$ such that $1 - xy \in \mathfrak{m}$, whereas $xy \in \mathfrak{r}(A)$ and hence $1 - xy$ is invertible, a contradiction. Note the following consequence: if $\Upsilon(A)$ is a finite set $\{\mathfrak{m}_1, \dots, \mathfrak{m}_r\}$, the $\mathfrak{m}_i$ are all the maximal ideals of A .

⁽⁷⁾**N.D.E.** We have added these remarks in order to compare the definition of the formal spectrum $\text{Spf}(A)$ given in 1.1 with those of EGA I, 10.1.2 and 10.4.2.

elements. Such a homomorphism sends a topologically nilpotent element of A to a topologically nilpotent element of B ; it therefore maps the radical $\mathfrak{r}(A)$ of A into the radical $\mathfrak{r}(B)$ of B .

0.2. — Let A be a (commutative) pseudocompact ring. A *pseudocompact A -module* M is a separated and complete topological A -module which has a base of neighborhoods of 0 consisting of submodules M' such that M/M' has finite length over A . If M and N are two pseudocompact A -modules, a morphism from M to N is by definition a continuous A -linear map. The group of these morphisms will be denoted by $\mathrm{Hom}_c(M, N)$.

Proposition 0.2.B. ⁽⁸⁾

- (i) *Pseudocompact A -modules form an abelian category, which will be denoted by $\mathbf{PC}(A)$. (In particular, for every morphism $f : M \rightarrow N$, $\mathrm{Im}(f)$ is a complete submodule, hence closed in N .)*
- (ii) *Pseudocompact A -modules of finite length (whose topology is therefore discrete) form a system of cogenerators of $\mathbf{PC}(A)$.*
- (iii) *Infinite products and filtered projective limits are exact, i.e. $\mathbf{PC}(A)$ satisfies axiom $(\mathrm{AB5}^*)$.* ⁽⁹⁾

For the reader's convenience, let us briefly indicate the steps of the proof. First, one has the following lemma ([CA] § IV.3, Lemma 1; for the proof, see [BEs], III, § 7.4, Th. 1 and Example 2):

Lemma 0.2.C. *Let B be a ring, I a filtered ordered set, and (M_i) and (N_i) two projective systems of left B -modules indexed by I . Let (s_i) be a morphism of projective systems $(M_i) \rightarrow (N_i)$ such that s_i is surjective and has artinian kernel for every i . Then the map*

$$\varprojlim s_i : \varprojlim M_i \rightarrow \varprojlim N_i$$

is surjective.

Corollary 0.2.D. ([CA] § IV.3, Prop. 10 & 11). *Let M be a pseudocompact A -module.*

- (i) *Let K be a closed submodule of M . Then M/K , equipped with the quotient topology, is a pseudocompact A -module.*
- (ii) *Let (M_i) be a decreasing filtered family of closed submodules of M .*
 - (a) *The canonical map $M \rightarrow \varprojlim M/M_i$ is surjective and has kernel $\bigcap_i M_i$.*
 - (b) *For every closed submodule N of M , one has $N + \bigcap_i M_i = \bigcap_i (N + M_i)$.*

PROOF. Let (L_j) be the decreasing filtered family of open submodules of M . Equip M/K with the quotient topology, i.e. a base of neighborhoods of 0 consists of the open submodules $(K + L_j)/K$. Since K is closed, it equals the intersection of the $K + L_j$, so the map

$$\tau : M/K \rightarrow \varprojlim_j M/(K + L_j)$$

is injective. It is also open, the right-hand term being the projective limit of the discrete modules $M/(K + L_j)$. Moreover, for each j , the map $t_j : M/L_j \rightarrow M/(K + L_j)$ is surjective, with artinian kernel, so by the preceding lemma the map t in the commutative diagram below is surjective:

$$\begin{array}{ccc} M & \xrightarrow[\sim]{p} & \varprojlim_j M/L_j \\ \downarrow & & \downarrow t \\ M/K & \xrightarrow{\tau} & \varprojlim_j M/(K + L_j). \end{array}$$

⁽⁸⁾**N.D.E.** We have brought out the results of this paragraph in the following proposition, and have then indicated the steps of the proof; cf. [CA], § IV.3 or [DG70], § V.2.

⁽⁹⁾**N.D.E.** Cf. [Gr57], I § 1.5 and Prop. 1.8; one may also consult, for example, [Po73], § 2.8 or [We95], Append. A.4.

Since p is an isomorphism because M is complete, it follows that τ is surjective, hence an isomorphism. This proves (i). Let us prove (ii)(a). By the foregoing, for every i one has an isomorphism $M/M_i \xrightarrow{\sim} \varprojlim_j M/(M_i + L_j)$, and thus both horizontal arrows in the commutative diagram below are isomorphisms:

$$\begin{array}{ccc} M & \xrightarrow[p \sim]{} & \varprojlim_j M/L_j \\ \downarrow g & & \downarrow s \\ \varprojlim_i M/M_i & \xrightarrow[\sim]{} & \varprojlim_{i,j} M/(M_i + L_j). \end{array}$$

Moreover, for each j , the family of submodules $(M_i + L_j)/L_j$ has a least element, since M/L_j is artinian; hence the morphism $s_j : M/L_j \rightarrow \varprojlim_i M/(L_j + M_i)$ is surjective; therefore, by the preceding lemma, s is surjective. It follows that g is surjective. Finally, the kernel of g is the projective limit of the M_i , i.e. their intersection. This proves (a). Let us deduce (b). Since N is a closed submodule (hence separated and complete), it is a pseudocompact module for the topology induced by that of M . Hence, by (a), the morphisms f and g in the commutative exact diagram below are surjective:

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & M/N \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & \varprojlim_i N/(N \cap M_i) & \longrightarrow & \varprojlim_i M/M_i & \longrightarrow & \varprojlim_i M/(N + M_i) \end{array} \quad .$$

Then, by the “snake lemma”, the sequence $0 \rightarrow \text{Ker } f \rightarrow \text{Ker } g \rightarrow \text{Ker } h \rightarrow 0$ is exact, and the equality $N + \bigcap_i M_i = \bigcap_i (N + M_i)$ follows. $\square$

One can now prove Proposition 0.2.B. Let $f : M \rightarrow N$ be a morphism of pseudocompact A -modules. Then $K = \text{Ker}(f)$ is a closed submodule of M , hence separated and complete, so K is a pseudocompact module for the topology induced by that of M . By 0.2.D(i), M/K equipped with the quotient topology is pseudocompact. Let us show that the continuous bijective morphism $M/K \rightarrow \text{Im}(f)$ is bicontinuous. Identifying M/K with $\text{Im}(f)$, one has to show that the quotient topology $\mathcal{Q}$ is finer than the topology $\mathcal{T}$ induced by that of N . Denote by (L_j) (resp. (N_i)) the decreasing filtered set of open submodules of M (resp. N), and put $N'_i = N_i \cap \text{Im}(f)$. Let $P = (K + L_j)/K$ be a submodule of M/K open for $\mathcal{Q}$. Since $M/(K + L_j)$ is artinian, the family $N'_i + P$ has a least element $N'_{i_0} + P$. Since the N'_i are open, hence closed, for $\mathcal{T}$ and thus also for $\mathcal{Q}$, it follows from 0.2.D(ii)(b) that

$$N'_{i_0} + P = \bigcap_i (N'_i + P) = P + \bigcap_i N'_i = P,$$

hence $N'_{i_0} \subset P$. This shows that P is open for $\mathcal{T}$, and $M/K \rightarrow \text{Im}(f)$ is therefore an isomorphism of pseudocompact modules. In particular, $\text{Im}(f)$ is complete for $\mathcal{T}$, hence closed in N . Then, by 0.2.D(i) again, $\text{Coker}(f)$ equipped with the quotient topology is pseudocompact. This proves that $\mathbf{PC}(A)$ is an abelian category. Moreover, arbitrary projective limits exist in $\mathbf{PC}(A)$: if (M_i) is a projective system of pseudocompact modules, the projective limit of the M_i has as its underlying module the projective limit of the underlying modules, and as its topology the projective limit topology. Furthermore, if one has a family of exact sequences in $\mathbf{PC}(A)$:

$$0 \longrightarrow K_i \longrightarrow M_i \longrightarrow Q_i \longrightarrow 0,$$

then the sequence $0 \rightarrow \prod_i K_i \rightarrow \prod_i M_i \rightarrow \prod_i Q_i \rightarrow 0$ is exact. Part (iii) of 0.2.B follows, since in any abelian category in which arbitrary products exist, conditions (a) and (b) of 0.2.D are equivalent and equivalent to exactness of filtered projective limits (cf. [Mi65], III 1.2–1.9 or [Po73], Chap. 2, Th. 8.6). Finally, every pseudocompact module M is a submodule of the product $\prod_L M/L$, where L runs through the open submodules of M , so the objects of

finite length form a system of cogenerators of $\mathbf{PC}(A)$. (Moreover, every object of length n is isomorphic to a quotient A^n/L , where L is an open submodule of A^n of colength n ; these quotients therefore form a *set* of cogenerators.) This completes the proof of 0.2.B. ⁽¹⁰⁾Let $(\mathbf{Ab})$ be the category of abelian groups and $\mathbf{LF}(A)$ the full subcategory of $\mathbf{PC}(A)$ consisting of objects of *finite length*. For every object M of $\mathbf{PC}(A)$, denote by h_c^M the functor

$$\mathbf{LF}(A) \longrightarrow (\mathbf{Ab}), \quad N \longmapsto \mathrm{Hom}_c(M, N).$$

By [CA], § II.4, Th. 1, Lemma 4 and Cor. 1, one has the following results.⁽¹¹⁾

Proposition 0.2.E. *The functor $M \mapsto h_c^M$ is an anti-equivalence from $\mathbf{PC}(A)$ to the category $\mathbf{Lex}(\mathbf{LF}(A), (\mathbf{Ab}))$ of left exact functors $\mathbf{LF}(A) \rightarrow (\mathbf{Ab})$.*

Corollary 0.2.F.

- (i) *An object P of $\mathbf{PC}(A)$ is projective if and only if the functor h_c^P is exact (i.e. if and only if the functor $\mathrm{Hom}_c(P, -)$ is exact on $\mathbf{LF}(A)$).*
- (ii) *Let (M_i) be a filtered⁽¹²⁾ projective system of objects of $\mathbf{PC}(A)$. For every object N of $\mathbf{LF}(A)$, one has an isomorphism functorial in N :*

$$\mathrm{Hom}_c(\varprojlim M_i, N) \simeq \varinjlim \mathrm{Hom}_c(M_i, N).$$

- (iii) *Every filtered projective limit and every product⁽¹²⁾ of projective objects of $\mathbf{PC}(A)$ is a projective object of $\mathbf{PC}(A)$.*

Finally, one deduces from 0.2.F the following

Corollary 0.2.G. *Let $(M_i)_{i \in I}$ be a family of objects of $\mathbf{PC}(A)$. Then $\prod_{i \in I} M_i$ is projective if and only if every M_i is projective.*

Indeed, for every $N \in \mathrm{Ob} \mathbf{LF}(A)$, one has $\mathrm{Hom}_c(\prod_i M_i, N) \simeq \bigoplus_i \mathrm{Hom}_c(M_i, N)$.

0.2.1. — Each local component $A_{\mathfrak{m}}$ of A is a direct factor of A , hence a projective object of $\mathbf{PC}(A)$ (A is manifestly projective). Moreover, $A_{\mathfrak{m}}$ has $S_{\mathfrak{m}} = A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ as its unique simple quotient, hence is indecomposable. On the other hand, every simple object of $\mathbf{PC}(A)$ is isomorphic to a unique $S_{\mathfrak{m}}$. By [CA], IV § 3, Cor. 1 of Th. 3,⁽¹³⁾ one therefore has:

Proposition.

- (i) *Every projective object of $\mathbf{PC}(A)$ is a direct product of indecomposable projective objects, uniquely determined (up to isomorphism).*
- (ii) *Every indecomposable projective object is isomorphic to a unique $A_{\mathfrak{m}}$ ($\mathfrak{m} \in \Upsilon(A)$).*

Definition. A pseudocompact A -module M is said to be *topologically free* if it is isomorphic to the product of a family (A_i) of copies of A . In this case, a family (m_i) of elements of M is called a *pseudobasis* of M if the A -linear maps from A_i to M which send the identity element of A_i to m_i extend to an isomorphism from $\prod_i A_i$ to M .

⁽¹⁰⁾**N.D.E.** We have inserted in this paragraph Proposition 0.2.E and Corollary 0.2.F, which will be useful in 0.2.2. (In the original, these results appeared in 0.3.)

⁽¹¹⁾**N.D.E.** Indeed, $\mathbf{PC}(A)$ has a set of artinian cogenerators, filtered projective limits are exact in it, and $\mathbf{LF}(A)$ is the subcategory of artinian objects. The dual category $\mathbf{PC}(A)^0$ therefore has a set of noetherian generators, and filtered inductive limits are exact in it. By the proof of [CA] § II.4, Th. 1, the functor $M \mapsto \mathrm{Hom}_c(M, -)$ is an anti-equivalence from $\mathbf{PC}(A)$ to $\mathbf{Lex}(\mathbf{LF}(A), (\mathbf{Ab}))$. Similarly, Lemma 4 and Cor. 1 of *loc. cit.* state results “dual” to those of Corollary 0.2.F. For a “direct” proof, see [DG70], § V.2, Th. 3.1, Lemma 3.5, Cor. 3.3 & 3.4.

⁽¹²⁾**N.D.E.** Since every infinite product is a filtered projective limit of finite products, every additive functor which commutes with filtered projective limits also commutes with infinite products.

⁽¹³⁾**N.D.E.** This refers to the “dual” statements established in *loc. cit.*, § 2, Th. 2 and § 1, Prop. 2; for a “direct” proof, see [DG70], V, § 2, Th. 4.5 and Example 4.6 b).

0.2.2. — ⁽¹⁴⁾If M is a pseudocompact A -module, the A -module (without topology)

$$\mathrm{Hom}_c(M, A)$$

will be denoted by $M^\dagger$. Conversely, if N is an A -module, denote by $N^* = \mathrm{Hom}_A(N, A)$ its dual, equipped with the topology of pointwise convergence, i.e. a base of neighborhoods of 0 in N^* consists of the following submodules, where $x \in N$ and $\mathfrak{l}$ is an open ideal of A :

$$\mathcal{V}(x, \mathfrak{l}) = \{f \in N^* \mid f(x) \in \mathfrak{l}\}.$$

This makes N^* a pseudocompact A -module. Indeed, one first sees that if $N = A$, then $N^* = A$, equipped with its pseudocompact ring topology, and if N is a free module $A^{(I)}$, then N^* is the product A^I , equipped with the product topology. On the other hand, for every morphism $\phi : N_1 \rightarrow N_2$, the transpose morphism ${}^t\phi : N_2^* \rightarrow N_1^*$ is continuous, since the inverse image under ${}^t\phi$ of a submodule $\mathcal{V}(x_1, \mathfrak{l})$ of N_1^* is precisely the submodule $\mathcal{V}(\phi(x_1), \mathfrak{l})$ of N_2^* . Then, for arbitrary N , taking a presentation

$$A^{(J)} \xrightarrow{\phi} A^{(I)} \xrightarrow{\pi} N \rightarrow 0,$$

one sees that N^* is the kernel of the continuous morphism ${}^t\phi : A^I \rightarrow A^J$, so N^* is pseudocompact. When A is artinian (in which case one can take $\mathfrak{l} = 0$ above), one deduces from 0.2.F:

Proposition. *When A is artinian, the functors*

$$P \mapsto P^\dagger \quad \text{and} \quad Q \mapsto Q^*,$$

where P (resp. Q) is a projective object of $\mathbf{PC}(A)$ (resp. a projective A -module), establish an anti-equivalence between the category of projective pseudocompact A -modules and that of projective A -modules.⁽¹⁵⁾

In particular, when A is a field k , $P \mapsto P^\dagger$ is an anti-equivalence from the category of all pseudocompact k -modules (also called *linearly compact k -vector spaces*) to that of k -vector spaces.⁽¹⁶⁾

0.3. — ⁽¹⁷⁾Let L and M be two pseudocompact A -modules. The functor

$$\mathbf{LF}(A) \rightarrow (\mathrm{Ab}), \quad N \mapsto \mathrm{Bil}_c(L \times M, N),$$

where $\mathrm{Bil}_c(L \times M, N)$ denotes the abelian group of continuous A -bilinear maps from $L \times M$ to N , is left exact. By 0.2.E, there therefore exists a pseudocompact A -module $L \hat{\otimes}_A M$, unique up to unique isomorphism, which represents this functor, i.e. such that one has a functorial isomorphism, for every object N of $\mathbf{LF}(A)$:

$$\mathrm{Hom}_c(L \hat{\otimes}_A M, N) \simeq \mathrm{Bil}_c(L \times M, N).$$

Moreover, $L \hat{\otimes}_A M$ is identified with the projective limit $P = \widehat{L \otimes_A M}$ of the (discrete) A -modules $(L/L') \otimes_A (M/M')$, where L' and M' run respectively through the open submodules of L and M . Indeed, let $\varphi : L \times M \rightarrow N$ be a continuous bilinear map from $L \times M$ to a (discrete) A -module N of finite length. By Lemma 0.3.1 below, there exist open submodules L' and M' of L and M such that $\varphi(L' \times M) = \varphi(L \times M') = \{0\}$. This means that the map

⁽¹⁴⁾**N.D.E.** We have detailed the results of this paragraph, which play an important role in what follows (cf. 1.2.3, 1.3.5, 2.2.1, etc.).

⁽¹⁵⁾**N.D.E.** Recall also that, over an *artinian* ring, a module is projective if and only if it is flat; see, for example, VI B, N.D.E. (54) or 0.3.7 below.

⁽¹⁶⁾**N.D.E.** Note that the *direct sum in $\mathbf{PC}(k)$* of a family $(V_i)_{i \in I}$ of linearly compact k -vector spaces is $(\prod_{i \in I} V_i^\dagger)^*$.

⁽¹⁷⁾**N.D.E.** We have modified this paragraph, taking into account the additions made in 0.2.

$\bar{\varphi} : L \otimes_A M \longrightarrow N$ induced by φ has the form $\varphi' \circ p$, where p is the canonical projection from $L \otimes_A M$ to $(L/L') \otimes_A (M/M')$. If one denotes by $\hat{\varphi}$ the composite

$$P \longrightarrow (L/L') \otimes_A (M/M') \xrightarrow{\varphi'} N,$$

one sees that the map $\varphi \mapsto \hat{\varphi}$ is a functorial bijection from $\text{Bil}_c(L \times M, N)$ to $\text{Hom}_c(P, N)$, whence $P \simeq L \hat{\otimes}_A M$. The pseudocompact module $L \hat{\otimes}_A M$ is therefore the separated completion of $L \otimes_A M$, for the linear topology defined by the kernels of the canonical projections from $L \otimes_A M$ to $(L/L') \otimes_A (M/M')$, and will be called the *completed tensor product* of L and M . If x and y belong to L and M , the image of $x \otimes_A y$ in $L \hat{\otimes}_A M$ will be denoted by $x \hat{\otimes}_A y$.

Lemma 0.3.1. *Let L , M , and N be pseudocompact A -modules, N being of finite length. If $\varphi : L \times M \longrightarrow N$ is a continuous A -bilinear map, there exist open submodules L' and M' of L and M such that $\varphi(L' \times M) = \varphi(L \times M') = \{0\}$.*

Indeed, $\varphi^{-1}(0)$ is an open neighborhood of $(0, 0)$, hence contains an open set of the form $L_1 \times M_1$, where L_1 and M_1 are open submodules of L and M . Since L/L_1 has finite length, there exist elements $x_1, \dots, x_r$ of L such that $L_1 + Ax_1 + \dots + Ax_r = L$. If $M' \subset M_1$ is “sufficiently small”, one also has $\varphi(x_i, M') = 0$ for every i , because the map $y \mapsto \varphi(x_i, y)$ is continuous; from this one obtains $\varphi(L, M') = \{0\}$; similarly, $\varphi(L', M) = \{0\}$ if L' is sufficiently small.

Corollary 0.3.1.1. ⁽¹⁸⁾ *Let M be a pseudocompact A -module.*

- (i) *For every open submodule M' , there exists an open ideal $\mathfrak{l}$ of A such that $\overline{\mathfrak{l}M} \subset M'$.*
- (ii) *Consequently, $M \simeq \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$, where $\mathfrak{l}$ runs through the filtered projective system of open ideals of A and each $M/\overline{\mathfrak{l}M}$ is equipped with the quotient topology (which makes it a pseudocompact module, cf. 0.2.D).*

Indeed, consider the map $\varphi : A \times M \longrightarrow M/M'$, $(a, m) \mapsto am + M'$; by 0.3.1 there exists an open ideal $\mathfrak{l}$ of A such that $\mathfrak{l}M \subset M'$, and since M' is also closed, it also contains $\overline{\mathfrak{l}M}$. Since the intersection of the open submodules of M is zero, one therefore has $\bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0)$. Moreover, by 0.2.D, the map $\phi : M \longrightarrow \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$ is surjective; by (the proof of) 0.2.B, ϕ therefore induces an isomorphism $M/\text{Ker}(\phi) \xrightarrow{\sim} \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$, and we have just seen that $\text{Ker}(\phi) = \bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M}$ is zero.

Remark 0.3.1.2. ⁽¹⁹⁾ The completed tensor product satisfies the usual associativity condition: if L, M, P are pseudocompact A -modules, one has a canonical isomorphism

$$(L \hat{\otimes} M) \hat{\otimes} P \simeq L \hat{\otimes} (M \hat{\otimes} P);$$

indeed, these two objects represent the functor which associates to every object N of $\mathbf{LF}(A)$ the abelian group of continuous A -trilinear maps from $L \times M \times P$ to N .

0.3.2. — Let $L' \xrightarrow{f} L \xrightarrow{g} L'' \longrightarrow 0$ be an exact sequence and M an object of $\mathbf{PC}(A)$. It is clear that, for every object N of $\mathbf{LF}(A)$, the induced sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Bil}_c(L'' \times M, N) & \longrightarrow & \text{Bil}_c(L \times M, N) & \longrightarrow & \text{Bil}_c(L' \times M, N) \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & \text{Hom}_c(L'' \hat{\otimes}_A M, N) & \longrightarrow & \text{Hom}_c(L \hat{\otimes}_A M, N) & \longrightarrow & \text{Hom}_c(L' \hat{\otimes}_A M, N) \end{array}$$

are exact. By 0.2.E, this is equivalent to saying that the sequence

$$(*) \quad L' \hat{\otimes}_A M \xrightarrow{f \hat{\otimes} M} L \hat{\otimes}_A M \xrightarrow{g \hat{\otimes} M} L'' \hat{\otimes}_A M \longrightarrow 0$$

⁽¹⁸⁾ **N.D.E.** We have added this corollary.

⁽¹⁹⁾ **N.D.E.** We have added this remark.

is exact. Thus:

Corollary. *For every pseudocompact A -module M , the functor $-\widehat{\otimes}_A M$ is right exact.*

Take, in particular, the ring A for L , the inclusion of a closed ideal $\mathfrak{a}$ in A for f , and the canonical projection from A to $A/\mathfrak{a}$ for g . One can then identify $A\widehat{\otimes}_A M$ with M by means of the map $x\widehat{\otimes}_A m \mapsto xm$. Since the image of $\mathfrak{a}\widehat{\otimes}_A M$ is closed in M (cf. 0.2.B), and the image of $\mathfrak{a} \otimes_A M$ is everywhere dense in $\mathfrak{a}\widehat{\otimes}_A M$, the image of $f\widehat{\otimes}_A M$ is precisely the closure $\overline{\mathfrak{a}M}$ of $\mathfrak{a}M$ in M . The exact sequence (*) therefore yields the isomorphism

$$(A/\mathfrak{a})\widehat{\otimes}_A M \xrightarrow{\sim} M/\overline{\mathfrak{a}M}.$$

0.3.3. Nakayama's lemma. *Let A be a pseudocompact ring, M a pseudocompact A -module, and $\mathfrak{a}$ an ideal of A contained in the radical $\mathfrak{r}(A)$. The equality $\overline{\mathfrak{a}M} = M$ then implies $M = 0$.*

Indeed, suppose that $\overline{\mathfrak{a}M} = M$.⁽²⁰⁾ Let M' be an open submodule of M and M'' the quotient M/M' . Since M'' is discrete, $\mathfrak{a}M''$ is closed in M'' , hence equal to $\overline{\mathfrak{a}M''}$. By 0.3.2, the canonical map from $M/\overline{\mathfrak{a}M}$ to $M''/\overline{\mathfrak{a}M''}$ is surjective, so one has $\mathfrak{a}M'' = \overline{\mathfrak{a}M''} = M''$. Since M'' is a finitely generated A -module and $\mathfrak{a} \subset \mathfrak{r}(A)$, this implies $M'' = 0$ by the usual Nakayama lemma. Consequently, every open submodule M' of M equals M , so M is zero.⁽²¹⁾

0.3.4. — From Nakayama's lemma one obtains the usual consequences:

Corollary. *Let $\mathfrak{a}$ be a closed ideal contained in $\mathfrak{r}(A)$ and $f : M \rightarrow N$ a morphism of pseudocompact A -modules.*

- (i) *f is surjective if the induced map $f' : M/\overline{\mathfrak{a}M} \rightarrow N/\overline{\mathfrak{a}N}$ is surjective.*⁽²²⁾
- (ii) *If N is projective, f is invertible if f' is invertible.*

Indeed, (i) follows from Lemma 0.3.3 applied to $\text{Coker } f$. For (ii), suppose that f' is invertible. Then, by (i), f is surjective, hence has a section; one then applies 0.3.3 to $\text{Ker } f$. When A is local with maximal ideal $\mathfrak{m}$, one can also deduce from 0.3.3 the following *exchange theorem*:

Theorem. *Let A be a local pseudocompact ring, $\mathfrak{m}$ its maximal ideal, M a topologically free A -module with pseudobasis $(m_i)_{i \in I}$ (0.2.1), and N a (closed) direct factor of M . There exists a pseudobasis of M consisting of elements of N and of certain m_i .*

Indeed, this is clear when A is a field (use the duality of 0.2.2 and apply the usual exchange theorem); consequently, $N/\overline{\mathfrak{m}N}$ ⁽²³⁾ has as a complement a topologically free module over $A/\mathfrak{m}$ with pseudobasis $(\overline{m}_i)_{i \in J}$, where $\overline{m}_i$ is the image of m_i in $M/\overline{\mathfrak{m}M}$ and J is a subset of I . If P denotes the direct product $\prod_{i \in J} Am_i$, the canonical map from $N \oplus P$ to M is “bijective modulo $\mathfrak{m}$ ”; it is therefore bijective by the foregoing (for another proof see [CA], § IV.2, Prop. 8).

0.3.5. — Now consider three pseudocompact A -modules L, M, N , where N has finite length. Equipping the A -module $\text{Hom}_c(M, N)$ with the discrete topology, every element ψ of $\text{Hom}_c(L, \text{Hom}_c(M, N))$ defines a continuous bilinear map $\psi' : (\ell, m) \mapsto \psi(\ell)(m)$ from $L \times M$ to N . In this way one obtains a natural isomorphism

$$(1) \quad \text{Hom}_c(L, \text{Hom}_c(M, N)) \xrightarrow{\sim} \text{Hom}_c(L\widehat{\otimes}_A M, N),$$

hence another characterization of $\text{Hom}_c(L\widehat{\otimes}_A M, N)$, which we shall use when M is the projective limit of a filtered projective system of pseudocompact A -modules M_i . Then, by (1)

⁽²⁰⁾**N.D.E.** Replacing $\mathfrak{a}$ by its closure if necessary, one can suppose that $\mathfrak{a}$ is *closed*.

⁽²¹⁾**N.D.E.** since it equals the projective limit of the $M/M' = 0$.

⁽²²⁾**N.D.E.** Consequently, every pseudocompact A -module is a quotient of a topologically free A -module (one first reduces to the case where A is local); cf. the proof of 0.3.7.

⁽²³⁾**N.D.E.** We have corrected $N/\mathfrak{m}N$ to $N/\overline{\mathfrak{m}N}$, and likewise for $M/\overline{\mathfrak{m}M}$ below.

and 0.2.F(ii), one has natural isomorphisms

$$(2) \quad \begin{aligned} \mathrm{Hom}_c(L \widehat{\otimes}_A \varprojlim M_i, N) &\simeq \mathrm{Hom}_c(L, \mathrm{Hom}_c(\varprojlim M_i, N)) \\ &\simeq \mathrm{Hom}_c(L, \varinjlim \mathrm{Hom}_c(M_i, N)). \end{aligned}$$

Moreover, since the module $\varinjlim \mathrm{Hom}_c(M_i, N)$ is discrete, every continuous map with source L factors through a quotient of L of finite length. Consequently, the natural map below is an isomorphism:

$$(3) \quad \varinjlim \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) \longrightarrow \mathrm{Hom}_c(L, \varinjlim \mathrm{Hom}_c(M_i, N)).$$

Finally, by (1) and 0.2.F(ii) again, one has natural isomorphisms

$$(4) \quad \begin{aligned} \varinjlim \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) &\simeq \varinjlim \mathrm{Hom}_c(L \widehat{\otimes}_A M_i, N) \\ &\simeq \mathrm{Hom}_c(\varprojlim (L \widehat{\otimes}_A M_i), N). \end{aligned}$$

Combining isomorphisms (2), (3), (4), one obtains:

Proposition. *Let (M_i) be a filtered projective system of objects of $\mathbf{PC}(A)$, and let L (resp. N) be an object of $\mathbf{PC}(A)$ (resp. $\mathbf{LF}(A)$). One has an isomorphism functorial in N :*

$$\mathrm{Hom}_c(L \widehat{\otimes}_A \varprojlim M_i, N) \simeq \mathrm{Hom}_c(\varprojlim (L \widehat{\otimes}_A M_i), N),$$

and hence an isomorphism

$$L \widehat{\otimes}_A \varprojlim M_i \simeq \varprojlim (L \widehat{\otimes}_A M_i).$$

Thus the completed tensor product commutes with filtered projective limits.⁽²⁴⁾

0.3.6. — In particular,⁽²⁵⁾ the completed tensor product commutes with infinite products. For example, since the ring A is the product of its local components $A_{\mathfrak{m}}$ (0.1.1), every pseudocompact A -module M ($\simeq A \widehat{\otimes}_A M$) is identified with the product of the modules $M_{\mathfrak{m}} = A_{\mathfrak{m}} \widehat{\otimes}_A M$ (the *local components* of M). Similarly, let M and N be two pseudocompact A -modules. Recall (cf. 0.2.2) that $M^{\dagger}$ denotes $\mathrm{Hom}_c(M, A)$. Consider the map

$$\varphi : M^{\dagger} \otimes_A N^{\dagger} \longrightarrow (M \widehat{\otimes}_A N)^{\dagger}$$

which associates to an element $f \otimes g$ of $M^{\dagger} \otimes_A N^{\dagger}$ the map $m \widehat{\otimes} n \mapsto f(m)g(n)$ from $M \widehat{\otimes}_A N$ to A . This map φ is bijective when M is isomorphic to A .

Corollary. *When A is artinian, φ is an isomorphism whenever M is topologically free (or, more generally, projective).*

Indeed, for fixed N , the functor $M \mapsto (M \widehat{\otimes}_A N)^{\dagger}$ (resp. $M \mapsto M^{\dagger} \otimes_A N^{\dagger}$) transforms every direct product into a direct sum, by the foregoing and 0.2.F.

Remark 0.3.6.A. ⁽²⁶⁾Using 0.2.F in a similar way, one also obtains the following result: Let A be an artinian ring, M, Q objects of $\mathbf{PC}(A)$, and N an object of $\mathbf{LF}(A)$. Suppose that Q is projective; then one has natural isomorphisms

$$\mathrm{Hom}_c(M, Q) \xrightarrow{\sim} \mathrm{Hom}_A(Q^{\dagger}, M^{\dagger}) \quad \text{and} \quad Q^{\dagger} \otimes_A N \xrightarrow{\sim} \mathrm{Hom}_c(Q, N).$$

0.3.7. — For every $\mathfrak{m} \in \Upsilon(A)$, the functor $M \mapsto A_{\mathfrak{m}} \widehat{\otimes}_A M$ is evidently exact. Since every projective pseudocompact A -module P is a product of modules of the form $A_{\mathfrak{m}}$, it follows that the functor $M \mapsto P \widehat{\otimes}_A M$ is exact when P is projective. The converse is true:

Proposition. *Let A be a pseudocompact ring and P a pseudocompact A -module. The following conditions are equivalent:*

⁽²⁴⁾**N.D.E.** In this way one recovers 0.3.1.1: $L = L \widehat{\otimes}_A A = L \widehat{\otimes}_A \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} L/\overline{\mathfrak{l}}L$, where $\mathfrak{l}$ runs through the open ideals of A .

⁽²⁵⁾**N.D.E.** Cf. N.D.E. (12).

⁽²⁶⁾**N.D.E.** We have added this remark, which will be useful in 2.3.1.

- (i) P is a projective object of $\mathbf{PC}(A)$.
- (ii) Each local component $P_{\mathfrak{m}}$ is a topologically free $A_{\mathfrak{m}}$ -module.
- (iii) The functor $M \mapsto P \hat{\otimes}_A M$ is exact.

Indeed, the equivalence of (i) and (ii) follows from 0.2.F(iii) and 0.2.1, and we have just seen that (ii) $\Rightarrow$ (iii). Suppose that the functor $M \mapsto P \hat{\otimes}_A M$ is exact. Since $P \hat{\otimes}_A M$ is the product of its local components

$$(P \hat{\otimes}_A M)_{\mathfrak{m}} \simeq P_{\mathfrak{m}} \hat{\otimes}_{A_{\mathfrak{m}}} M_{\mathfrak{m}},$$

one is reduced to the case where the ring A is local. We shall then prove that P is *topologically free*. Let $\mathfrak{m}$ be the maximal ideal of A ; then $P/\mathfrak{m}P$ is a linearly compact vector space over $A/\mathfrak{m}$, hence a product of copies of $A/\mathfrak{m}$ (cf. 0.2.2). There is therefore a family $(A_i)_{i \in I}$ of copies of A and an isomorphism $\varphi : \prod_{i \in I} (A_i/\mathfrak{m}) \xrightarrow{\sim} P/\mathfrak{m}P$. Since $\prod_{i \in I} A_i$ is projective, there is a commutative square

$$\begin{array}{ccc} \prod_{i \in I} A_i & \xrightarrow{\psi} & P \\ \downarrow p & & \downarrow q \\ \prod_{i \in I} (A_i/\mathfrak{m}) & \xrightarrow[\sim]{\varphi} & P/\mathfrak{m}P \end{array}$$

where p and q denote the canonical projections. Applying Nakayama's lemma to Coker ψ , and noting that $(A/\mathfrak{m}) \hat{\otimes}_A \psi$ is precisely φ , one sees that ψ is surjective.⁽²⁷⁾ Putting $B = \prod_{i \in I} A_i$ and $N = \text{Ker } \psi$, one then has the following commutative exact diagram:

$$\begin{array}{ccccccc} & & & & 0 & \text{---} & \text{---} \\ & & & & \downarrow & & \uparrow \\ & & \mathfrak{m} \hat{\otimes}_A N & \longrightarrow & \mathfrak{m} \hat{\otimes}_A B & \longrightarrow & \mathfrak{m} \hat{\otimes}_A P \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A \hat{\otimes}_A N & \longrightarrow & A \hat{\otimes}_A B & \xrightarrow{\psi} & A \hat{\otimes}_A P \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & \text{---} & \longrightarrow & (A/\mathfrak{m}) \hat{\otimes}_A N & \longrightarrow & (A/\mathfrak{m}) \hat{\otimes}_A B & \xrightarrow{\varphi} (A/\mathfrak{m}) \hat{\otimes}_A P \longrightarrow 0. \end{array}$$

The “snake lemma” applied to the first two rows then shows that, in the bottom row, the morphism $(A/\mathfrak{m}) \hat{\otimes}_A N \longrightarrow (A/\mathfrak{m}) \hat{\otimes}_A B$ is a monomorphism. But then, since φ is an isomorphism, $(A/\mathfrak{m}) \hat{\otimes}_A N$ is zero; hence $N = 0$ (0.3.3) and ψ is an isomorphism.⁽²⁸⁾

Corollary 0.3.8. *Let A be a complete noetherian local ring and P a pseudocompact A -module. Then P is topologically free if and only if P is flat over A .*

Indeed, the canonical map from $M \otimes_A P$ to $M \hat{\otimes}_A P$ is bijective when M equals A , hence also when M is noetherian (take a finite presentation of M and use right exactness of the tensor product and the completed tensor product). Now P is flat if and only if the functor $M \mapsto M \otimes_A P$ is exact when M runs through the noetherian modules. Similarly, in the proof of Proposition 0.3.7 one has seen that P is topologically free if the sequence

$$0 \longrightarrow \mathfrak{m} \hat{\otimes}_A P \longrightarrow A \hat{\otimes}_A P \longrightarrow (A/\mathfrak{m}) \hat{\otimes}_A P \longrightarrow 0$$

⁽²⁷⁾**N.D.E.** This shows the result announced in N.D.E. (22): every pseudocompact A -module M is a quotient of a topologically free A -module. (Since products are exact in $\mathbf{PC}(A)$, one reduces to the case where A is local, treated above.)

⁽²⁸⁾**N.D.E.** An entirely analogous proof, using the “nilpotent Nakayama lemma”, shows that if A is artinian and P is a flat A -module, every local component of P is a free A -module (cf. [BAC], II § 3.2, Cor. 2 of Prop. 5).

is exact. Thus P is topologically free if and only if the functor $M \mapsto M \widehat{\otimes}_A P$ is exact when M runs through the noetherian modules. The corollary therefore follows from the equality $M \otimes_A P = M \widehat{\otimes}_A P$ established above.

0.4. — Let k be a pseudocompact ring; a *topological k -algebra* is a (commutative) topological ring A equipped with a morphism of topological rings $k \rightarrow A$. One says that A is a *profinite k -algebra* if the underlying topological k -module is pseudocompact. In this case, let $\mathfrak{l}$ be an open k -submodule of A . The composite map

$$\varphi : A \times A \xrightarrow{\text{mult.}} A \xrightarrow{\text{can.}} A/\mathfrak{l}$$

is continuous, so, by Lemma 0.3.1, there exists an open k -submodule $\mathfrak{n}$ of A such that $\varphi(A \times \mathfrak{n}) = 0$. This means that $\mathfrak{l}$ contains the open ideal $A\mathfrak{n}$ and implies that A is a *pseudocompact ring*. Similarly, let M be a topological A -module whose underlying k -module is pseudocompact. If M' is an open k -submodule of M , Lemma 0.3.1 applied to the map

$$A \times M \xrightarrow{\text{mult.}} M \xrightarrow{\text{can.}} M/M'$$

shows that M' contains an open A -submodule, hence that M is also a *pseudocompact A -module*.⁽²⁹⁾ Conversely:

Lemma. *Let A be a profinite k -algebra and M a pseudocompact A -module. Then the k -module $M|_k$ obtained by restriction of scalars is pseudocompact.*

Indeed, every pseudocompact A -module of finite length is isomorphic to a quotient A^n/L (where L is an open submodule of A^n), hence is a pseudocompact k -module. Since $M|_k$ is a projective limit of such modules, it is a pseudocompact k -module.

0.4.1. — If A and B are two profinite k -algebras, a *morphism* from A to B is, by definition, a continuous homomorphism of k -algebras. The category of profinite k -algebras will be denoted by $\mathbf{Alp}/k$. It evidently has projective limits: the algebra underlying a projective limit is the projective limit of the underlying algebras; the topology is the projective limit topology. It also has *finite inductive limits*:⁽³⁰⁾ for example, if $f : A \rightarrow B$ and $g : A \rightarrow C$ are two morphisms of profinite k -algebras, the amalgamated sum of B and C under A has $B \widehat{\otimes}_A C$ as its underlying topological A -module (by 0.4, f and g equip B and C with pseudocompact A -module structures); the multiplication on $B \widehat{\otimes}_A C$ evidently satisfies $(b \widehat{\otimes} c)(b' \widehat{\otimes} c') = (bb') \widehat{\otimes} (cc')$ if $b, b' \in B$ and $c, c' \in C$.

Definition 0.4.2. A profinite k -algebra C will be said to have *finite length* if the underlying k -module has finite length (hence is discrete); we shall denote by $\mathbf{Alf}/k$ the full subcategory of $\mathbf{Alp}/k$ consisting of k -algebras of finite length.⁽³¹⁾

For every profinite k -algebra A , denote by h^A the functor

$$\mathbf{Alf}/k \rightarrow (\text{Ens}), \quad C \mapsto \text{Hom}_{\mathbf{Alp}/k}(A, C).$$

It is clear that h^A is a left exact functor.⁽³²⁾ Moreover, the canonical projections $A \rightarrow A/\mathfrak{l}$ (where $\mathfrak{l}$ runs through the open ideals of A) induce, for every object C of $\mathbf{Alf}/k$, a canonical isomorphism

$$\varinjlim_{\mathfrak{l}} \text{Hom}_{\mathbf{Alf}/k}(A/\mathfrak{l}, C) \xrightarrow{\sim} \text{Hom}_{\mathbf{Alp}/k}(A, C),$$

⁽²⁹⁾**N.D.E.** We have added the following lemma; cf. N.D.E. (36) in Proposition 0.5.

⁽³⁰⁾**N.D.E.** We shall see in 0.4.2 that it has arbitrary inductive limits.

⁽³¹⁾**N.D.E.** Beware: k is an object of $\mathbf{Alf}/k$ only if k is artinian; cf. 1.2.2 below.

⁽³²⁾**N.D.E.** i.e. one which commutes with finite projective limits.

functorial in C . This means that h^A is the inductive limit of the representable functors $h^{A/\mathfrak{l}}$, i.e.

$$(*) \quad h^A \simeq \varinjlim_{\mathfrak{l}} h^{A/\mathfrak{l}}.$$

If B is another profinite k -algebra, the general properties of the bifunctor Hom and the isomorphism $\text{Hom}(h^C, h^B) = h^B(C)$ for C of finite length give isomorphisms

$$\begin{aligned} \text{Hom}_{\mathbf{Alp}/k}(B, A) &\simeq \varprojlim \text{Hom}_{\mathbf{Alp}/k}(B, A/\mathfrak{l}) \\ &\simeq \varprojlim \text{Hom}(h^{A/\mathfrak{l}}, h^B) \\ &\simeq \text{Hom}(\varinjlim h^{A/\mathfrak{l}}, h^B); \end{aligned}$$

combined with $(*)$, this shows that the contravariant functor $A \mapsto h^A$ is *fully faithful*. In fact:

Proposition. *The functor $A \mapsto h^A$ is an anti-equivalence from $\mathbf{Alp}/k$ to the category of left exact functors from $\mathbf{Alf}/k$ to (Ens) .*

Indeed, by the foregoing it suffices to show that every left exact functor $\mathbf{F} : \mathbf{Alf}/k \rightarrow (\text{Ens})$ is isomorphic to a functor of the form h^A ; for this, one can construct A as follows (cf. TDTE II, § 3). Since $\mathbf{F}$ is left exact, for every k -algebra C of finite length and every element ξ of $\mathbf{F}(C)$, there is a least subalgebra C' of C such that ξ belongs to the image of $\mathbf{F}(C')$ in $\mathbf{F}(C)$. If $C' = C$, one says that the pair (C, ξ) is *minimal*. The minimal pairs form a category if one takes as morphisms from (C, ξ) to (D, η) the homomorphisms φ from C to D such that $(\mathbf{F}\varphi)(\xi) = \eta$; it is clear that such a φ is a surjection and that the category of minimal pairs is “left filtered”. Moreover, one can restrict to pairs (C, ξ) such that C belongs to a set containing k -algebras of finite length of each isomorphism type.⁽³³⁾ Thus the functor $(C, \xi) \mapsto C$, whose source category is that of minimal pairs and whose target category is that of profinite k -algebras, has a projective limit; one takes this projective limit for A .

Corollary. *The category $\mathbf{Alp}/k$ has infinite inductive limits.*

Indeed, the category of left exact functors from $\mathbf{Alf}/k$ to (Ens) has projective limits, defined “argument by argument”, i.e. if $(\mathbf{F}_i)$ is a projective system of such functors, one has, for every object C of $\mathbf{Alf}/k$:

$$(34) \quad (\varprojlim \mathbf{F}_i)(C) = \varprojlim \mathbf{F}_i(C).$$

0.5. — ⁽³⁵⁾Let $\varphi : k \rightarrow \ell$ be a homomorphism of pseudocompact rings (cf. 0.1.3). One can generalize the construction of 0.3 as follows.

Definition 0.5.A. For every object M of $\mathbf{PC}(k)$ (resp. N of $\mathbf{PC}(\ell)$), we shall denote by $M \widehat{\otimes}_k N$ the separated completion of $M \otimes_k N$ for the linear topology defined by the kernels of the projections

$$M \otimes_k \ell \rightarrow (M/M') \otimes_k (N/N'),$$

⁽³³⁾**N.D.E.** Indeed, every discrete k -module of length n is isomorphic to k^n/L , where L is an open submodule of k^n of colength n . One can then consider the *set* of (isomorphism classes of) profinite k -algebra structures on each such module.

⁽³⁴⁾**N.D.E.** If $(A_i)_{i \in I}$ is an inductive system in $\mathbf{Alp}/k$, its inductive limit in $\mathbf{Alp}/k$ is the separated completion of the k -algebra B , the inductive limit of the underlying k -algebras, for the topology defined by the ideals I such that $I \cap A_i$ is an open ideal of A_i for every i , and $\text{length}_k(B/I) < \infty$. Note that if, for example, K/k is an algebraic field extension of infinite degree, and (k_i) denotes the filtered inductive system of subextensions of finite degree, then the inductive limit of the system (k_i) in $\mathbf{Alp}/k$ is the zero ring!

⁽³⁵⁾**N.D.E.** We have detailed this paragraph.

where M' (resp. N') is an open submodule of M (resp. N). Then $M\widehat{\otimes}_k N$ is a *pseudocompact ℓ -module*. If $m \in M$ and $x \in N$, we shall denote by $m\widehat{\otimes}_k x$ the canonical image of $m \otimes_k x$ in $M\widehat{\otimes}_k N$. This applies in particular when $N = \ell$; in this case one says that $M\widehat{\otimes}_k \ell$ is the pseudocompact ℓ -module *deduced from M by the base change $k \rightarrow \ell$* .

Remarks 0.5.B.

- (i) When considering such a base change, ℓ will not in general be a profinite k -algebra: a typical example is the case where k is a field and ℓ is an arbitrary extension K of k .
- (ii) However, if the k -module underlying N is pseudocompact (for example if ℓ is a profinite k -algebra), then, by 0.4, every open k -submodule of N contains an open ℓ -submodule of N ; consequently, $M\widehat{\otimes}_k N$ in this case coincides with the completed tensor product (cf. 0.3) of the pseudocompact k -modules M and N , and the notation therefore entails no ambiguity.

The k -module $N|_k$ obtained by restriction of scalars is in any case a topological k -module, i.e. the map $k \times N \rightarrow N$, $(t, n) \mapsto \varphi(t)n$ is continuous. Denote by $\text{Hom}_c(M, N|_k)$ the abelian group of continuous homomorphisms of k -modules from M to $N|_k$.

Proposition 0.5.C. *For every $M \in \text{Ob } \mathbf{PC}(k)$ and $N \in \text{Ob } \mathbf{PC}(\ell)$, one has a functorial isomorphism*

$$\text{Hom}_{\mathbf{PC}(\ell)}(M\widehat{\otimes}_k \ell, N) \simeq \text{Hom}_c(M, N|_k).$$

(36)

Indeed, let ϕ be a continuous homomorphism $M \rightarrow N|_k$; then the map $\phi' : M \times \ell \rightarrow N$, $(m, \lambda) \mapsto \lambda\phi(m)$ is continuous and “bilinear” (i.e. k -linear in the first factor and ℓ -linear in the second). If N' is an open ℓ -submodule of N , one shows as in Lemma 0.3.1 that there exist an open submodule M' of M and an open ideal ℓ' of ℓ such that $\phi'(M' \times \ell)$ and $\phi'(M \times \ell')$ are contained in N' . It follows that ϕ induces a continuous homomorphism of ℓ -modules $\Phi : M\widehat{\otimes}_k \ell \rightarrow N$ such that $\Phi(m\widehat{\otimes}\lambda) = \lambda\phi(m)$ for every $m \in M$ and $\lambda \in \ell$. Conversely, to every morphism $f : M\widehat{\otimes}_k \ell \rightarrow N$ one associates the morphism $f' : m \mapsto f(m\widehat{\otimes}_k 1)$ from M to $N|_k$. One then obtains, as in 0.3.2, 0.3.5, and 0.3.1.2, the following:

Corollary 0.5.D. *The functor $\mathbf{PC}(k) \rightarrow \mathbf{PC}(\ell)$, $M \mapsto M\widehat{\otimes}_k \ell$, is right exact and commutes with filtered projective limits, i.e. if (M_i) is a filtered projective system of objects of $\mathbf{PC}(k)$, one has a canonical isomorphism*

$$(\varprojlim M_i)\widehat{\otimes}_k \ell \simeq \varprojlim (M_i\widehat{\otimes}_k \ell).$$

Moreover, if M, N are pseudocompact k -modules, one has a canonical isomorphism

$$(M\widehat{\otimes}_k N)\widehat{\otimes}_k \ell \simeq (M\widehat{\otimes}_k \ell)\widehat{\otimes}_\ell (N\widehat{\otimes}_k \ell).$$

Definition 0.5.E. Finally, if A is a profinite k -algebra, there is one and only one profinite ℓ -algebra structure on $A\widehat{\otimes}_k \ell$ such that, if $a, b \in A$ and $\lambda, \mu \in \ell$,

$$(a\widehat{\otimes}_k \lambda)(b\widehat{\otimes}_k \mu) = (ab)\widehat{\otimes}_k (\lambda\mu).$$

One says that $A\widehat{\otimes}_k \ell$ is the profinite ℓ -algebra deduced from A by the *extension of scalars* (or “base change”) $k \rightarrow \ell$.

1. Formal varieties over a pseudocompact ring

1.1. — To every pseudocompact ring A one can associate a formal scheme (EGA I, 10.4.2) as follows: the underlying topological space is the set $\Upsilon(A)$ of open prime (hence maximal)

(36)**N.D.E.** Consequently, if ℓ is a profinite k -algebra, the functor $M \mapsto M\widehat{\otimes}_k \ell$ is left adjoint to the restriction-of-scalars functor $\mathbf{PC}(\ell) \rightarrow \mathbf{PC}(k)$ (cf. Lemma 0.4).

ideals of A , equipped with the discrete topology; the structure sheaf has the cartesian product $\prod_{\mathfrak{m} \in E} A_{\mathfrak{m}}$ as its space of sections over a subset E of $\Upsilon(A)$. The formal scheme thus obtained is denoted by $\mathrm{Spf}(A)$ (the *formal spectrum* of A).⁽³⁷⁾ If A and B are two pseudocompact rings, a *morphism* from $\mathrm{Spf}(B)$ to $\mathrm{Spf}(A)$ consists of a map f from $\Upsilon(B)$ to $\Upsilon(A)$ and a family of continuous homomorphisms $f_y^\sharp : A_{f(y)} \rightarrow B_y$, for $y \in \Upsilon(B)$. Such a morphism induces a continuous homomorphism $f^\sharp$ from $A = \prod_{x \in \Upsilon(A)} A_x$ to $B = \prod_{y \in \Upsilon(B)} B_y$. The converse is true:

Proposition. *The contravariant functor $A \mapsto \mathrm{Spf}(A)$ is fully faithful.*

Indeed, if $\varphi : A \rightarrow B$ is a continuous homomorphism of algebras, the inverse image $\varphi^{-1}(\mathfrak{n})$ of an open maximal ideal of B is an open prime ideal of A , hence maximal in A . One therefore obtains a map $\mathfrak{n} \mapsto \varphi^{-1}(\mathfrak{n})$ from $\Upsilon(B)$ to $\Upsilon(A)$, and φ induces a continuous homomorphism $A_{\varphi^{-1}(\mathfrak{n})} \rightarrow B_{\mathfrak{n}}$. Thus φ induces a morphism of formal schemes $\mathrm{Spf}(\varphi) : \mathrm{Spf}(B) \rightarrow \mathrm{Spf}(A)$. One easily verifies that $\mathrm{Spf}(\varphi)^\sharp = \varphi$, and that $\mathrm{Spf}(f^\sharp) = f$ for every morphism $f : \mathrm{Spf}(B) \rightarrow \mathrm{Spf}(A)$, whence the proposition. Although here we speak only of formal schemes of the form $\mathrm{Spf}(A)$, we shall use the language of formal schemes rather than that of pseudocompact rings in order to support our assertions by geometric intuition.

1.2. — Let k be a pseudocompact ring.

Definition 1.2.A. ⁽³⁸⁾ We shall call a *formal variety over k* any formal scheme X over $\mathrm{Spf}(k)$ which is isomorphic to a formal k -scheme $\mathrm{Spf}(A)$ for some profinite k -algebra A . The algebra A is then isomorphic to the *affine algebra of X* , that is, the algebra of sections of the structure sheaf $\mathcal{O}_X$ of X . Denote by $\mathbf{Vaf}/k$ the full subcategory of the category of formal schemes over $\mathrm{Spf}(k)$ whose objects are formal k -varieties.⁽³⁹⁾

By 1.1, the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from $\mathbf{Alp}/k$ (0.4.1) to $\mathbf{Vaf}/k$. Thus, by Corollary 0.4.2:

Proposition 1.2.B. *The category $\mathbf{Vaf}/k$ has projective and inductive limits.*⁽⁴⁰⁾

For example, let $X \in \mathrm{Ob} \mathbf{Vaf}/k$ and let $f : Y \rightarrow X$ and $g : Z \rightarrow X$ be two formal k -varieties over X , and let A, B, C be the affine algebras of X, Y, Z ; by 0.4.1, the affine algebra of the fibered product $Y \times_X Z$ is identified with $B \hat{\otimes}_A C$,⁽⁴¹⁾ so that the inclusion of $\mathbf{Vaf}/k$ in the category of all formal k -schemes commutes with finite projective limits (cf. EGA I, 10.7). Inductive limits of formal k -varieties correspond to projective limits of their affine algebras.

Example 1.2.C. (Cokernels). Let, for example, $d, e : X \rightrightarrows Y$ be a double arrow in $\mathbf{Vaf}/k$; the affine algebra of $\mathrm{Coker}(d, e)$ is isomorphic to the kernel of the homomorphisms induced on the affine algebras of X and Y , but one can also give the following construction of $\mathrm{Coker}(d, e)$: the topological space underlying $\mathrm{Coker}(d, e)$ is the cokernel of the underlying spaces;⁽⁴²⁾ if p is the canonical projection from the set underlying Y to the cokernel, and z belongs to the

⁽³⁷⁾**N.D.E.** Cf. Remarks 0.1.2.

⁽³⁸⁾**N.D.E.** We have added the numbering 1.2.A to 1.2.E for later references.

⁽³⁹⁾**N.D.E.** Note that, by the definition of morphisms (1.1), every $X \in \mathrm{Ob} \mathbf{Vaf}/k$ is the direct sum in $\mathbf{Vaf}/k$ of the one-point formal varieties $\mathrm{Spf}(\mathcal{O}_{X,x})$, for $x \in X$.

⁽⁴⁰⁾**N.D.E.** In particular, cokernels exist in $\mathbf{Vaf}/k$; see below. Observe, moreover, that a filtered projective limit in $\mathbf{Vaf}/k$ whose transition morphisms are all surjective may be empty; cf. N.D.E. (34).

⁽⁴¹⁾**N.D.E.** Note that $Y \times_X Z$ is the direct sum, for $x \in X$, of the formal varieties $B_x \hat{\otimes}_{\mathcal{O}_{X,x}} C_x$, where B_x is the product of the $\mathcal{O}_{Y,y}$ for the $y \in Y$ such that $f(y) = x$, and C_x is defined analogously. This will be used in 2.5.1.

⁽⁴²⁾**N.D.E.** i.e. the quotient set of Y by the identifications $d(x) = e(x)$, for every $x \in X$, equipped with the quotient topology, which here is the discrete topology.

cokernel, the local algebra of $\text{Coker}(d, e)$ at z is the kernel of the double arrow

$$d^{\natural}, e^{\natural} : \prod_{p(y)=z} \mathcal{O}_{Y,y} \rightrightarrows \prod_{q(x)=z} \mathcal{O}_{X,x},$$

where one has put $q = p \circ d = p \circ e$ and where $d^{\natural}$ and $e^{\natural}$ are induced by the homomorphisms $d_x^{\natural}$ and $e_x^{\natural}$ (notation of 1.1).

Definition 1.2.D. If $\varphi : k \rightarrow \ell$ is a homomorphism of pseudocompact rings and X a formal k -variety with affine algebra A , the formal scheme $X \times_{\text{Spf}(k)} \text{Spf}(\ell)$ obtained by base change is a formal ℓ -variety, which we shall also denote by $X \widehat{\otimes}_k \ell$, and whose affine algebra is the completed tensor product $A \widehat{\otimes}_k \ell$ (cf. 0.5 and EGA I, § 10).

Remark 1.2.E. Since every formal variety over k decomposes into formal varieties over the local components of k , *in certain proofs we shall suppose that k is a local pseudocompact ring.*

Let us now give some examples while fixing our terminology.

1.2.1. — A k -functor will, by definition, be a *covariant* functor from $\mathbf{Alf}/k$ to $(\mathbf{Ens})$. By 1.1 and 0.4.2, one can identify $\mathbf{Vaf}/k \simeq (\mathbf{Alp}/k)^0$ with a full subcategory of the category of k -functors, by associating to every formal k -variety X the functor

$$\mathbf{Alf}/k \rightarrow (\mathbf{Ens}), \quad C \mapsto X(C) = \text{Hom}_{\mathbf{Vaf}/k}(\text{Spf}(C), X).$$

Later we shall encounter k -functors $\mathbf{F}$ which associate to every object C of $\mathbf{Alf}/k$ a module $\mathbf{F}(C)$ over C , and to every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$ a k -linear map $\mathbf{F}(\varphi) : \mathbf{F}(C) \rightarrow \mathbf{F}(D)$ such that, if $x \in \mathbf{F}(C)$ and $\lambda \in C$,

$$\mathbf{F}(\varphi)(\lambda x) = \varphi(\lambda) \mathbf{F}(\varphi)(x).$$

By Exp. I, 3.1, such an $\mathbf{F}$ is equipped with an $\mathbf{O}_k$ -module structure, if one denotes by $\mathbf{O}_k$ the ring-valued k -functor which associates to every object C of $\mathbf{Alf}/k$ the ring underlying C .

Definitions.

- (i) An $\mathbf{O}_k$ -module $\mathbf{F}$ will be said to be *admissible* if every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$ induces a bijection from $D \otimes_C \mathbf{F}(C)$ to $\mathbf{F}(D)$.⁽⁴³⁾
- (ii) One says that $\mathbf{F}$ is *flat* if it is admissible and if, for every object C of $\mathbf{Alf}/k$, $\mathbf{F}(C)$ is a flat C -module.⁽⁴⁴⁾

For example, if M is a k -module (*not necessarily pseudocompact*), we shall denote (as in Exp. I, 4.6.1) by $\mathbf{W}(M)$ the $\mathbf{O}_k$ -module $C \mapsto C \otimes_k M$; then $\mathbf{W}(M)$ is flat when M is flat over k . Moreover, the functor $M \mapsto \mathbf{W}(M)$ has as its right adjoint the functor which associates to every $\mathbf{O}_k$ -module $\mathbf{F}$ the k -module $\varprojlim_{\mathfrak{l}} \mathbf{F}(k/\mathfrak{l})$, where $\mathfrak{l}$ runs through the open ideals of k .

1.2.2. — In what follows, an $\mathbf{O}_k$ -module will always be denoted by a boldface letter such as $\mathbf{F}$; when k is artinian, we shall then write simply F instead of $\mathbf{F}(k)$. In this case, it goes without saying that the functor $\mathbf{F} \mapsto F$ induces an equivalence from the category of flat $\mathbf{O}_k$ -modules to that of flat k -modules! Thus the terminology we have adopted has only the purpose of allowing us to reason “as if k were always artinian”. In accordance with Exposé I § 3.1, we shall use analogous conventions for other algebraic structures: thus an $\mathbf{O}_k$ -algebra (resp. an $\mathbf{O}_k$ -coalgebra, resp. an $\mathbf{O}_k$ -Lie algebra, resp. an $\mathbf{O}_k$ -Lie p -algebra) will consist of an $\mathbf{O}_k$ -module $\mathbf{M}$ and, for every object C of $\mathbf{Alf}/k$, a C -algebra structure (resp. a C -coalgebra, resp. a C -Lie algebra, resp. a C -Lie p -algebra structure) on $\mathbf{M}(C)$; one supposes further that, for every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$, the map from $D \otimes_C \mathbf{M}(C)$ to $\mathbf{M}(D)$ induced by

⁽⁴³⁾**N.D.E.** We have introduced this terminology; cf. N.D.E. (47) in 1.2.3.

⁽⁴⁴⁾**N.D.E.** and hence projective, since C is artinian.

$\mathbf{M}(\varphi)$ is a homomorphism of D -algebras (resp. of D -coalgebras, etc.). Finally, note that if $\mathbf{F}$ and $\mathbf{G}$ are two $\mathbf{O}_k$ -modules, $\mathbf{F} \otimes_k \mathbf{G}$ will denote the $\mathbf{O}_k$ -module $C \mapsto \mathbf{F}(C) \otimes_C \mathbf{G}(C)$.

1.2.3. — ⁽⁴⁵⁾Let us begin with the following lemma.

Lemma 1.2.3.A. *Let $k \rightarrow K$ be a morphism of pseudocompact rings, B a pseudocompact K -module, and M a projective k -module (without topology). One has a canonical isomorphism of pseudocompact K -modules*

$$(2) \quad \theta : \mathrm{Hom}_k(M, k) \widehat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_k(M, B).$$

Here $M^* = \mathrm{Hom}_k(M, k)$ is equipped with the topology defined in 0.2.2, which makes it a pseudocompact k -module, and the topology of $\mathrm{Hom}_k(M, B)$ is defined analogously: a base of neighborhoods of 0 consists of the following K -submodules, where $x \in M$ and B' is an open K -submodule of B :

$$\mathcal{H}(x, B') = \{f \in \mathrm{Hom}_k(M, B) \mid f(x) \in B'\}.$$

Finally, $M^* \widehat{\otimes}_k B$ is the pseudocompact K -module deduced from M^* by base change; cf. 0.5.A. This being so, θ is evidently an isomorphism when $M = k$; moreover, both sides of (2), considered as functors in M , transform direct sums into products (in particular, commute with finite direct sums). One therefore obtains that (2) is an isomorphism when M is a free k -module, and then when M is projective.

Definition 1.2.3.B. Let N be a pseudocompact k -module. We denote by $\mathbf{V}_k^f(N)$ or $\mathbf{N}^\dagger$ ⁽⁴⁶⁾ the $\mathbf{O}_k$ -module which associates to every $C \in \mathrm{Ob} \mathbf{A}lf/k$ the C -module $(C \widehat{\otimes}_k N)^\dagger$ dual to $C \widehat{\otimes}_k N$ (0.2.2), i.e. the C -module

$$\mathrm{Hom}_c(C \widehat{\otimes}_k N, C) \simeq \mathrm{Hom}_c(N, C|_k),$$

where $C|_k$ denotes the k -module C obtained by restriction of scalars. This $\mathbf{O}_k$ -module $\mathbf{V}_k^f(N)$ will be called the *dual* of N .

If $k_{\mathfrak{m}}$ is a local component of k , then, for every object C of $\mathbf{A}lf/k$,

$$\mathrm{Hom}_c(k_{\mathfrak{m}}, C|_k) = C_{\mathfrak{m}} = C \otimes_k k_{\mathfrak{m}}$$

is a projective C -module, and moreover one has $D \otimes_C C_{\mathfrak{m}} = D_{\mathfrak{m}}$ for every morphism $C \rightarrow D$ of $\mathbf{A}lf/k$; thus $\mathbf{V}_k^f(k_{\mathfrak{m}})$ is a flat $\mathbf{O}_k$ -module (cf. 1.2.1). Since every projective pseudocompact k -module is a product of modules $k_{\mathfrak{m}}$ (cf. Prop. 0.2.1), from Corollary 0.2.F one deduces:

Corollary 1.2.3.C. $\mathbf{V}_k^f(N)$ is a flat $\mathbf{O}_k$ -module when N is a projective pseudocompact k -module.

Definition 1.2.3.D. Conversely, if $\mathbf{M}$ is an *admissible* $\mathbf{O}_k$ -module (cf. 1.2.1),⁽⁴⁷⁾ we call the following pseudocompact k -module the *dual* of $\mathbf{M}$ and denote it by $\Gamma^*(\mathbf{M})$. For $\mathfrak{l}$ running through the open ideals of k , equip each $k/\mathfrak{l}$ -module

$$\mathbf{M}(k/\mathfrak{l})^* = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l})$$

with the topology described in 0.2.2, which makes it a pseudocompact k -module. Since $\mathbf{M}(k/\mathfrak{l}) \simeq \mathbf{M}(k/\mathfrak{l}') \otimes (k/\mathfrak{l})$ for $\mathfrak{l} \supset \mathfrak{l}'$, one has transition morphisms

$$\begin{aligned} \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}') &\longrightarrow \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}) \\ &= \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l}), \end{aligned}$$

⁽⁴⁵⁾**N.D.E.** We have added the following lemma and introduced the numbering 1.2.3.A to 1.2.3.F for later references.

⁽⁴⁶⁾**N.D.E.** In fact, we shall not use this second notation; see N.D.E. (47).

⁽⁴⁷⁾**N.D.E.** The editors do not understand the following definition unless $\mathbf{M}$ is assumed admissible, hence the addition of this hypothesis. Moreover, we have written $\Gamma^*(\mathbf{M})$ instead of M^* in order to avoid confusion between M^* , the module dual to the functor $\mathbf{M}$, and $\mathbf{N}^\dagger$, the functor dual to the module N ; cf. N.D.E. (46). Finally, we have detailed the construction of $\Gamma^*(\mathbf{M})$.

and by definition $\Gamma^*(\mathbf{M})$ is the projective limit of this (filtered) projective system of pseudocompact k -modules.

From now on, suppose further that $\mathbf{M}$ is a *flat* $\mathbf{O}_k$ -module (cf. 1.2.1); then each $\mathbf{M}(k/\mathfrak{l}')$ is a projective $(k/\mathfrak{l}')$ -module, so the transition morphisms above are surjective, and hence so are the projections $\Gamma^*(\mathbf{M}) \rightarrow \mathbf{M}(k/\mathfrak{l})^*$, since filtered projective limits are exact in $\mathbf{PC}(k)$. If $\tau : k \rightarrow K$ is a morphism of pseudocompact rings, denote by $\mathbf{M} \otimes_k K$ or simply $\mathbf{M}_K$ the $\mathbf{O}_K$ -functor defined as follows. If C is a K -algebra of finite length, the kernel of $k \rightarrow C$ is an open ideal $\mathfrak{l}$, and one puts $\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}) \otimes_k C$; one then has $\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}') \otimes_k C$ for every open ideal $\mathfrak{l}'$ contained in $\mathfrak{l}$. One then defines $\Gamma_K^*(\mathbf{M}_K)$ as the projective limit, for I running through the open ideals of K , of the pseudocompact K -modules

$$\mathrm{Hom}_{K/I}(\mathbf{M}_K(K/I), K/I) = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), K/I),$$

where in the right-hand term $\mathfrak{l}$ is any open ideal of k such that $\tau(\mathfrak{l}) \subset I$. Moreover, by 1.2.3.A, the right-hand term is identified with $\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k (K/I)$. Since the projections $\Gamma^*(\mathbf{M}) \rightarrow \mathbf{M}(k/\mathfrak{l})^*$ are surjective, one sees that the projective limit of the $\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k (K/I)$ is precisely the pseudocompact K -module $\Gamma^*(\mathbf{M}) \hat{\otimes}_k K$ (cf. 0.5.A). Thus one has obtained that, for every flat $\mathbf{O}_k$ -module $\mathbf{M}$, formation of $\Gamma^*(\mathbf{M})$ *commutes with extension of the base*, i.e. one has

$$(\star) \quad \Gamma_K^*(\mathbf{M} \otimes_k K) \simeq \Gamma^*(\mathbf{M}) \hat{\otimes}_k K.$$

Proposition 1.2.3.E.

- (i) *The functors $N \mapsto \mathbf{V}_k^f(N)$ and $\mathbf{M} \mapsto \Gamma^*(\mathbf{M})$ define an anti-equivalence between the category of projective pseudocompact k -modules and that of flat $\mathbf{O}_k$ -modules.*⁽⁴⁸⁾
- (ii) *Moreover, if $k \rightarrow K$ is a morphism of pseudocompact rings, the preceding anti-equivalence “commutes with base change” in the following sense: if $N \simeq \Gamma^*(\mathbf{M})$, then $N \hat{\otimes}_k K \simeq \Gamma_K^*(\mathbf{M} \otimes_k K)$.*

PROOF. On the one hand, one has a natural transformation $\phi_N : N \rightarrow \Gamma^*(\mathbf{V}_k^f(N))$. Since the functor $\Gamma^* \circ \mathbf{V}_k^f$ commutes with products, by 0.3.5 and 0.2.F, it suffices to verify that ϕ_N is an isomorphism when N is a local component $k_{\mathfrak{m}}$ of k . In this case, for every open ideal $\mathfrak{l}$ of k contained in $\mathfrak{m}$, the natural morphism

$$(k/\mathfrak{l})_{\mathfrak{m}} \rightarrow \mathrm{Hom}_{k/\mathfrak{l}}(\mathrm{Hom}_c(k_{\mathfrak{m}} \hat{\otimes} k/\mathfrak{l}, k/\mathfrak{l}), k/\mathfrak{l})$$

is an isomorphism, whence the result. On the other hand, let $\mathbf{M}$ be a flat $\mathbf{O}_k$ -module. Let us show that $\Gamma^*(\mathbf{M}) = \varprojlim \mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathbf{PC}(k)$. Let $N \rightarrow N'$ be a surjective morphism between objects of $\mathbf{LF}(k)$. By 0.2.F(i) and (ii), it suffices to show that the natural map

$$\varinjlim \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N) \rightarrow \varinjlim \mathrm{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N')$$

is surjective. But this is clear, since N and N' are $k/\mathfrak{l}_0$ -modules for some open ideal $\mathfrak{l}_0$; hence, if $\mathfrak{l} \subset \mathfrak{l}_0$, every morphism $\mathbf{M}(k/\mathfrak{l})^* \rightarrow N'$ lifts to a morphism $\mathbf{M}(k/\mathfrak{l})^* \rightarrow N$, since $\mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathbf{PC}(k/\mathfrak{l})$. One has a morphism $\psi : \mathbf{M} \rightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))$ of functors from $\mathbf{A}f/k$ to $(\mathbf{Ens})$. Let B be an object of $\mathbf{A}f/k$; let us show that

$$(1) \quad \begin{aligned} \psi(B) : \mathbf{M}(B) &\rightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))(B) \\ &= \mathrm{Hom}_c(\varprojlim_{\mathfrak{l}} (\mathbf{M}(k/\mathfrak{l})^* \hat{\otimes}_k B), B) \end{aligned}$$

is a bijection (in the equality above, we have used the fact that $\hat{\otimes}$ commutes with filtered projective limits). Let $\mathfrak{l}_0$ be an open ideal of k contained in the kernel of $k \rightarrow B$. By Lemma

⁽⁴⁸⁾**N.D.E.** We have added part (ii) below, as well as the following proof. The original indicated: “If $\mathbf{M}$ is a flat $\mathbf{O}_k$ -module, it is clear that $\mathbf{M}$ is precisely the dual of M^* , so the functors $N \mapsto N^*$ and $\mathbf{M} \mapsto M^*$ define an anti-equivalence...”

1.2.3.A, for every $\mathfrak{l} \subset \mathfrak{l}_0$, one has a canonical isomorphism of pseudocompact B -modules

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), B),$$

and, since $\mathbf{M}(B) = \mathbf{M}(k/\mathfrak{l}) \otimes_{k/\mathfrak{l}} B$, the right-hand term equals $\mathrm{Hom}_B(\mathbf{M}(B), B)$. Thus the projective system in (1) is constant for $\mathfrak{l} \subset \mathfrak{l}_0$, and (1) reduces to the canonical morphism

$$\mathbf{M}(B) \longrightarrow \mathrm{Hom}_c(\mathrm{Hom}_B(\mathbf{M}(B), B), B),$$

which is an isomorphism by 0.2.2, since B is artinian and $\mathbf{M}(B)$ is a projective B -module. This proves part (i) of the proposition, and part (ii) follows from isomorphism $(\star)$ established above. $\square$

Remark 1.2.3.F. Let us return to the statement of the proposition, and suppose further that N is a topologically free pseudocompact k -module. In this case one can choose bases for the C -modules $\mathbf{V}_k^f(N)(C)$ “in a coherent manner”. Indeed, let (n_i) be a pseudobasis of N (0.2.1) and n_i^C the canonical image of n_i in $C \widehat{\otimes}_k N$, for $C \in \mathbf{AIf}/k$. If one defines the element δ_i^C of $(C \widehat{\otimes}_k N)^\dagger$ by the equalities $\delta_i^C(n_i^C) = 1$ and $\delta_i^C(n_j^C) = 0$ for $i \neq j$, the family (δ_i^C) is a basis of $\mathbf{V}_k^f(N)(C)$; moreover, for every morphism $\varphi : C \longrightarrow D$ of $\mathbf{AIf}/k$, $\mathbf{V}_k^f(N)(\varphi)$ sends δ_i^C to δ_i^D .

1.2.4. — ⁽⁴⁹⁾For every pseudocompact k -module E , let $\widehat{S}_k(E)$ be the completed symmetric algebra of E , defined in the following manner. Let $T_k(E)$ be the direct sum of the pseudocompact k -modules

$$\widehat{\bigotimes}_k^n E = E \widehat{\otimes}_k \cdots \widehat{\otimes}_k E \quad (n \geq 0; \text{ if } n = 0, \widehat{\bigotimes}_k^0 E = k);$$

one makes $T_k(E)$ a graded k -algebra by defining multiplication in the usual manner; let $S_k(E)$ be the quotient of $T_k(E)$ by the homogeneous ideal whose n th component is the closed k -submodule of $\widehat{\bigotimes}_k^n E$ generated by the elements

$$\begin{aligned} & x_1 \widehat{\otimes} \cdots \widehat{\otimes} x_i \widehat{\otimes} x_{i+1} \widehat{\otimes} \cdots \widehat{\otimes} x_n \\ & - x_1 \widehat{\otimes} \cdots \widehat{\otimes} x_{i+1} \widehat{\otimes} x_i \widehat{\otimes} \cdots \widehat{\otimes} x_n. \end{aligned}$$

If one denotes this quotient by $S_k^n(E)$, $S_k(E)$ is evidently a graded k -algebra with n th component $S_k^n(E)$. Equip $S_k(E)$ with the linear topology defined by the set Υ of ideals $\mathfrak{l}$ such that $S_k(E)/\mathfrak{l}$ is a k -module of finite length and $S_k^n(E) \cap \mathfrak{l}$ is an open submodule of $S_k^n(E)$ for every n . Then the profinite algebra $\widehat{S}_k(E)$ is the separated completion of $S_k(E)$ for this topology.⁽⁵⁰⁾ Denote by $\mathbf{V}_k^f(E)$ the formal k -variety $\mathrm{Spf}(\widehat{S}_k(E))$. It represents the k -functor $\mathbf{V}_k^f(E)$, i.e. for every object C of $\mathbf{AIf}/k$, one has a canonical isomorphism

$$\begin{aligned} \mathbf{V}_k^f(E)(C) &= \mathrm{Hom}_c(E, C|_k) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{AIf}/k}(\widehat{S}_k(E), C) \\ &= \mathrm{Hom}_{\mathbf{Vaf}/k}(\mathrm{Spf}(C), \mathbf{V}_k^f(E)). \end{aligned}$$

Throughout what follows, we identify $\mathbf{V}_k^f(E)$ with the k -functor $\mathbf{V}_k^f(E)$.

⁽⁴⁹⁾**N.D.E.** In this paragraph we have changed the order, first defining $\widehat{S}_k(E)$ and then stating that $\mathbf{V}_k^f(E)$ represents $\mathbf{V}_k^f(E)$.

⁽⁵⁰⁾**N.D.E.** For example, let k be a field, E a pseudocompact k -vector space, and $V = \mathrm{Hom}_c(E, k)$; one has $E \simeq V^*$ (cf. 0.2.2). For every finite-dimensional subspace W of V , let $\mathcal{F}(W)$ be the set of closed points of the k -scheme $\mathbb{V}(W) = \mathrm{Spec} S_k(W^*)$. Denote by $\mathcal{F}(V)$ the inductive limit of the $\mathcal{F}(W)$. Then $\widehat{S}_k(E)$ is the product, for $x \in \mathcal{F}(V)$, of the pseudocompact k -algebras

$$\widehat{S}_k(E)_x = \varprojlim_W \widehat{\mathcal{O}}_{W,x},$$

where W runs through the finite-dimensional subspaces of V such that $x \in \mathcal{F}(W)$, and $\widehat{\mathcal{O}}_{W,x}$ denotes the separated completion of the local ring $\mathcal{O}_{\mathbb{V}(W),x}$ for the topology defined by the ideals of finite codimension (which here coincides with the $\mathfrak{m}$ -adic topology). If k is algebraically closed and $(v_i)_{i \in I}$ is a basis of V , so that E has a pseudobasis $(e_i)_{i \in I}$, each local component $\widehat{S}_k(E)_x$ is isomorphic to the formal power series ring $k[[e_i, i \in I]]$, equipped with the topology defined by the ideals of finite codimension.

1.2.5. — By 1.2.4, the zero morphism from E to k is associated to a morphism of profinite algebras $\pi : \widehat{S}_k(E) \rightarrow k$; this morphism π induces the zero map on $S_k^n(E)$ for $n \geq 1$ and defines a section of the structural morphism $\mathbb{V}_k^f(E) \rightarrow \mathrm{Spf}(k)$. We shall denote by $\mathbb{V}_k^{f,0}(E)$ the formal variety whose points are the images of the points of $\mathrm{Spf}(k)$ under the section $\mathrm{Spf}(\pi)$ and which has the same local algebras as $\mathbb{V}_k^f(E)$ at these points.⁽⁵¹⁾ Then the affine algebra of $\mathbb{V}_k^{f,0}(E)$ is the separated completion of $S_k(E)$ for the topology defined by the ideals $\mathfrak{l} \in \Upsilon$ (cf. 1.2.4) which contain $S_k^n(E)$ for sufficiently large n . One deduces that it is the infinite product

$$k[[E]] = k \times E \times S_k^2(E) \times S_k^3(E) \times \cdots.$$

On the other hand, let C be an object of $\mathbf{AIf}/k$, u an element of $\mathbb{V}_k^f(C) = \mathrm{Hom}_c(E, C)$, and $\phi : \widehat{S}_k(E) \rightarrow C$ the morphism of profinite k -algebras corresponding to u . Then $\mathrm{Ker}(\phi)$ is an open ideal (i.e. ϕ belongs to $\mathbb{V}_k^{f,0}(C)$) if and only if $\mathrm{Ker}(\phi)$ contains $S_k^n(E)$ for sufficiently large n , i.e. if and only if $u(E)$ is contained in the radical $\mathfrak{r}(C)$ of C . Thus: *for every object C of $\mathbf{AIf}/k$, one has canonical isomorphisms*

$$\mathbb{V}_k^{f,0}(E)(C) \simeq \mathrm{Hom}_{\mathbf{AIf}/k}(k[[E]], C) \simeq \mathrm{Hom}_c(E, \mathfrak{r}(C)).$$

1.2.6. — A formal k -variety V is said to have *finite length* if its affine algebra does. Similarly, if S is a scheme, an S -scheme X is said to have *finite length* if X is finite over S and the direct image of $\mathcal{O}_X$ on S is an $\mathcal{O}_S$ -module of finite length.⁽⁵²⁾ Thus giving an S -scheme X of finite length “is the same thing” as giving a finite set $\{s_1, \dots, s_n\}$ of closed points of S and an $\mathcal{O}_{S,s_i}$ -algebra of finite length at each of these points. One therefore sees that S -schemes of finite length are identified with formal varieties of finite length over the following formal scheme $\widehat{S}$. The topological space underlying $\widehat{S}$ is the set of closed points of S equipped with the discrete topology; if s is such a closed point, the stalk $\mathcal{O}_{\widehat{S},s}$ of the structure sheaf $\mathcal{O}_{\widehat{S}}$ at s is the separated completion $\widehat{\mathcal{O}_{S,s}}$ of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength; one therefore has $\widehat{S} = \mathrm{Spf} \mathcal{A}(\widehat{S})$, where $\mathcal{A}(\widehat{S})$ is the product of the $\widehat{\mathcal{O}_{S,s}}$, for s running through the closed points of S , equipped with the product topology.

Definition. If X is an S -scheme, denote by $\widehat{X}/\widehat{S}$ the formal variety over $k = \mathcal{A}(\widehat{S})$ defined as follows. The underlying topological space consists of the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S ; the local ring $\mathcal{O}_{\widehat{X}/\widehat{S},x}$ is the separated completion of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I of $\mathcal{O}_{X,x}$ such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{S,s}$ -module (N.B. since $[\kappa(x) : \kappa(s)] < \infty$, this is equivalent to saying that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$ -module).

Let $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ be the category of formal varieties of finite length over $\widehat{S}$ (identified with that of S -schemes of finite length). By 1.1 and 1.2.1, the category $\mathbf{Vaf}/\widehat{S}$ of formal varieties over $\widehat{S}$ is equivalent to that of contravariant left exact functors from $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ to (Ens) . In particular, for every S -scheme X , the functor $T \mapsto \mathrm{Hom}_{((\mathrm{Sch})/S)}(T, X)$, from $\mathbf{Vaf}_{/\widehat{S}}^{\mathrm{eff}}$ to (Ens) , is such a left exact functor and therefore corresponds to a formal variety over $\widehat{S}$. This is precisely $\widehat{X}/\widehat{S}$.⁽⁵³⁾

Proposition. *For every S -scheme X , the functors*

$$\mathrm{Hom}_{\mathbf{Vaf}/\widehat{S}}(-, \widehat{X}/\widehat{S}) \quad \text{and} \quad \mathrm{Hom}_{((\mathrm{Sch})/S)}(-, X)$$

⁽⁵¹⁾**N.D.E.** We have detailed the remainder of this paragraph.

⁽⁵²⁾**N.D.E.** We have added the following sentence.

⁽⁵³⁾**N.D.E.** We have expanded the following proposition by inserting the fact that the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with finite projective limits (in the original, this appeared in the proof of 1.3.4—the proof given here is more direct than the original). Moreover, this result will be useful in Section 2 and in 3.3.1.

have the same restriction to $\mathbf{Vaf}_{\widehat{S}}^{\text{eff}}$. In this way one obtains a functor $X \mapsto \widehat{X}/\widehat{S}$ from $((\text{Sch})/S)$ to $\mathbf{Vaf}/\widehat{S}$ which commutes with finite projective limits.

Indeed, one easily sees that the formal variety $\widehat{X}/\widehat{S}$ has the required property and that the correspondence $X \mapsto \widehat{X}/\widehat{S}$ is functorial. Let us prove the second assertion. Put $\mathcal{S} = ((\text{Sch})/S)$ and $\mathcal{V} = \mathbf{Vaf}/\widehat{S}$, and write $\widehat{X}$ instead of $\widehat{X}/\widehat{S}$. One knows (1.2.B) that $\mathcal{V}$ has arbitrary projective limits. Let $(X_i)_{i \in I}$ be a projective system in $\mathcal{S}$, and suppose that $X = \varprojlim X_i$ exists in $\mathcal{S}$ (which is the case if I is finite). Since the functor which associates to every object Y of $\mathcal{S}$ (resp. V of $\mathcal{V}$) the functor $h_Y = \text{Hom}_{\mathcal{S}}(-, Y)$ (resp. $h_V = \text{Hom}_{\mathcal{V}}(-, V)$) commutes with projective limits, for every S -scheme T of finite length one has functorial isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{S}}(T, X) &\simeq \varprojlim \text{Hom}_{\mathcal{S}}(T, X_i) \\ &\simeq \varprojlim \text{Hom}_{\mathcal{V}}(T, \widehat{X}_i) \simeq \text{Hom}_{\mathcal{V}}(T, \varprojlim \widehat{X}_i). \end{aligned}$$

Consequently, the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with projective limits when they exist in $((\text{Sch})/S)$; in particular, it commutes with finite projective limits.

1.3. Proposition. *Let $f : X \rightarrow Y$ be a morphism of formal varieties over k , A and B the affine algebras of X and Y , and $g : B \rightarrow A$ the morphism induced by f . Then f is a monomorphism in $\mathbf{Vaf}/k$ if and only if g is a surjection.*

⁽⁵⁴⁾By 1.1, $A \mapsto \text{Spf}(A)$ is an anti-equivalence from $\mathbf{Alp}/k$ to $\mathbf{Vaf}/k$, so f is a monomorphism if and only if g is an epimorphism, and this is indeed the case if g is surjective. Conversely, suppose that g is an epimorphism and let us show that it is surjective. For every $\mathfrak{n} \in \Upsilon(B)$, put $A_{\mathfrak{n}} = A \widehat{\otimes}_B B_{\mathfrak{n}}$; by 0.4, A is a pseudocompact B -module, hence is the product of the $A_{\mathfrak{n}}$ (cf. 0.3.6). Then g is the product of the morphisms $g_{\mathfrak{n}} : B_{\mathfrak{n}} \rightarrow A_{\mathfrak{n}}$ deduced from g by base change. These are still epimorphisms, which reduces us to proving the result when B is local, with maximal ideal $\mathfrak{n}$. Put $K = B/\mathfrak{n}$. By Nakayama's lemma 0.3.3, it suffices to show that the morphism $g \widehat{\otimes}_B K$ is surjective; it is deduced from g by base change, hence is an epimorphism in $\mathbf{Alp}/K$. One can therefore suppose that $B = K$ is a field. Now f is a monomorphism if and only if the diagonal morphism $X \rightarrow X \times_Y X$ is an isomorphism, that is, if the homomorphism $x \widehat{\otimes}_K x' \mapsto xx'$ is an isomorphism from $A \widehat{\otimes}_K A$ to A . Since K is a field, this implies $A = K$.

Remark. ⁽⁵⁵⁾ It follows from the proposition that every monomorphism $f : X \rightarrow Y$ of formal varieties is an isomorphism from X to a formal subvariety (necessarily closed!) of Y .

1.3.1. — The preceding proposition implies, in particular, that every monomorphism $f : X \rightarrow Y$ in $\mathbf{Vaf}/k$ is effective (cf. IV 1.3).⁽⁵⁶⁾ The same does not hold for epimorphisms, as one easily sees by slightly modifying the counterexample of Exp. V, § 2.c);⁽⁵⁷⁾ this is why we shall consider a pleasant class of effective epimorphisms.

⁽⁵⁴⁾**N.D.E.** We have detailed the beginning of the proof.

⁽⁵⁵⁾**N.D.E.** We have added this remark.

⁽⁵⁶⁾**N.D.E.** i.e. in the opposite category $(\mathbf{Vaf}/k)^0 = \mathbf{Alp}/k$, the morphism $g : B \rightarrow A$ corresponding to f is an effective epimorphism. This is indeed the case, since g is surjective and hence induces (cf. the proof of 0.2.B) an isomorphism of profinite k -algebras $B/I \xrightarrow{\sim} A$, where $I = \text{Ker } g$. Consequently, every morphism $\phi : B \rightarrow C$ of $\mathbf{Alp}/k$ which vanishes on I descends to a morphism $\bar{\phi} : A \rightarrow C$ such that $\phi = \bar{\phi} \circ g$.

⁽⁵⁷⁾**N.D.E.** i.e. let k be a field, $X = \text{Spf}(k[[T]])$, and $Y = \text{Spf}(B)$, where B is the k -subalgebra of $A = k[[T]]$ topologically generated by T^3 and T^4 (i.e. B consists of the formal power series $\sum a_n T^n$ such that $a_n = 0$ for $n = 1, 2, 5$). Then $X \rightarrow Y$ is an epimorphism which is not effective; indeed, the cokernel of $X \times_Y X \rightarrow X$ is $\text{Spf}(B')$, where B' is the subalgebra of A consisting of the a such that $a \widehat{\otimes} 1 = 1 \widehat{\otimes} a$, and B' contains T^5 .

Lemma. ⁽⁵⁸⁾ *Let $f : X \rightarrow Y$ be a morphism of formal k -varieties, let A, B be the affine algebras of X, Y , and let $f^\sharp : B \rightarrow A$ be the morphism induced by f . The following conditions are equivalent:*

- (i) *For every $x \in X$, the homomorphism $f_x^\sharp : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$ makes $\mathcal{O}_{X, x}$ a topologically free pseudocompact $\mathcal{O}_{Y, f(x)}$ -module.*
- (ii) *For every $y \in Y$, the local component $A_y = \prod_{f(x)=y} \mathcal{O}_{X, x}$ is a topologically free pseudocompact B_y -module.*
- (iii) *$f^\sharp : B \rightarrow A$ makes A a projective pseudocompact B -module.*
- (iv) *The functor $\mathbf{PC}(B) \rightarrow \mathbf{PC}(A)$, $M \mapsto M \hat{\otimes}_B A$, is exact.*

If these conditions hold, one says that f is topologically flat.

The implications (i) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) follow from 0.2.F(iii) and 0.3.7. Conversely, suppose that (ii) holds, and let $x \in X$ and $y = f(x)$. Since $\mathcal{O}_{X, x}$ is a direct factor of A_y , it is a projective pseudocompact B_y -module, hence topologically free by 0.2.1 (since B_y is local). On the other hand, a morphism $f : X \rightarrow Y$ of formal k -varieties is said to be *surjective* if it induces a surjection of the underlying sets.

Proposition. *Let $f : X \rightarrow Y$ be a surjective and topologically flat morphism of formal k -varieties. Then f is an effective epimorphism (cf. IV 1.3).*

Indeed, let A, B be the affine algebras of X, Y , and $g : B \rightarrow A$ the morphism induced by f . One must show that Y is identified with the cokernel of $X \times_Y X \rightrightarrows X$, i.e. that, for every $\mathfrak{n} \in \Upsilon(B)$, $B_{\mathfrak{n}}$ is identified with the subring of $A_{\mathfrak{n}} = A \hat{\otimes}_B B_{\mathfrak{n}}$ consisting of the a such that $a \hat{\otimes} 1 = 1 \hat{\otimes} a$. One can therefore suppose that B is local, with maximal ideal $\mathfrak{n}$. Our hypothesis then means that g makes A a nonzero topologically free B -module. By Nakayama's lemma 0.3.3, $A/\mathfrak{n}A$ is nonzero, so the morphism $g' : B/\mathfrak{n} \rightarrow A/\mathfrak{n}A$ deduced from g is injective. By Lemma 1.3.2 below, B is a direct factor of A as a B -module, say $A = B \oplus A'$; it follows that B is identified with the part of A consisting of the a such that $a \hat{\otimes}_B 1 = 1 \hat{\otimes}_B a$.

1.3.2. Lemma. *Let B be a local pseudocompact ring, $\mathfrak{n}$ its maximal ideal, M, N two projective pseudocompact B -modules, and g a morphism $M \rightarrow N$. If $(B/\mathfrak{n}) \hat{\otimes}_B g$ is injective, g is an isomorphism from M to a direct factor of N .*

Indeed, suppose that $g' = (B/\mathfrak{n}) \hat{\otimes}_B g$ is injective. Since $B/\mathfrak{n}$ is a field, g' then has a retraction r' . Denote by p and q the canonical projections from M and N to $M/\mathfrak{n}M$ and $N/\mathfrak{n}N$; since N is projective, there exists a morphism $r : N \rightarrow M$ such that $p \circ r = r' \circ q$; consequently, r' is deduced from r by passage to the quotient. Then, since $r' \circ g'$ is an isomorphism, so is $r \circ g$, by 0.3.4 (since M is projective). Let s be the inverse isomorphism of $r \circ g$; then $s \circ r$ is a retraction of g .

1.3.3. Proposition. *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms of formal k -varieties.*

- (i) *If f and g are topologically flat, so is $g \circ f$.*
- (ii) *If f and $g \circ f$ are topologically flat and f is surjective, g is topologically flat.*
- (iii) *If f is topologically flat, so is $f \times_Y Y'$ for every base change $Y' \rightarrow Y$.*

Assertions (i) and (iii) are clear. To prove (ii), call A, B, C the affine algebras of X, Y, Z , and $f' : B \rightarrow A$, $g' : C \rightarrow B$ the morphisms induced by f and g . Since $g \circ f$ is topologically flat, $f' \circ g'$ makes A a projective pseudocompact C -module; similarly, f' makes A a faithful projective pseudocompact B -module. When P runs through the pseudocompact C -modules and N through the pseudocompact B -modules, the functors $P \mapsto P \hat{\otimes}_C A$ and $N \mapsto N \hat{\otimes}_B A$ are therefore exact; since the second is moreover faithful, the functor $P \mapsto P \hat{\otimes}_C B$ is exact; by 0.3.7, B is therefore a projective pseudocompact C -module.

⁽⁵⁸⁾ **N.D.E.** We have added this lemma, which explains the terminology “topologically flat”.

1.3.4. Proposition. *Let S be a scheme, Y a locally noetherian S -scheme, and $f : X \rightarrow Y$ a faithfully flat S -morphism locally of finite type, so that f is an effective epimorphism, i.e. the sequence below is exact (cf. IV 6.3.1(iv) and IV 1.3):*

$$(*) \quad X \times_Y X \xrightarrow[\text{pr}_2]{\text{pr}_1} X \xrightarrow{f} Y.$$

Then the morphism of formal $\widehat{S}$ -varieties $\widehat{f} : \widehat{X}/\widehat{S} \rightarrow \widehat{Y}/\widehat{S}$ (cf. 1.2.6) is surjective and topologically flat, and the sequence below deduced from $()$ is exact:*

$$(\widehat{*}) \quad \widehat{X \times_Y X} / \widehat{S} \xrightarrow[\widehat{\text{pr}_2}]{\widehat{\text{pr}_1}} \widehat{X}/\widehat{S} \xrightarrow{\widehat{f}} \widehat{Y}/\widehat{S}.$$

Indeed, let y be a point of Y with projection s on S such that $\kappa(y)$ is a finite extension of the residue field $\kappa(s)$ of s . Since f is surjective and locally of finite type, $f^{-1}(y)$ is nonempty and locally of finite type over $\kappa(y)$; the closed points of $f^{-1}(y)$ are then the points of $\widehat{X}/\widehat{S}$ projecting to y . This shows that $\widehat{f}$ is surjective. ⁽⁵⁹⁾Let x be a closed point of $f^{-1}(y)$. Since Y is locally noetherian and f locally of finite type, the local ring $\mathcal{O}_{Y,y}$ (resp. $\mathcal{O}_{X,x}$) is noetherian, so the powers of the maximal ideal have finite colength, and thus the local ring of $\widehat{Y}/\widehat{S}$ at y (resp. of $\widehat{X}/\widehat{S}$ at x) is the completion $\widehat{\mathcal{O}}_{Y,y}$ of $\mathcal{O}_{Y,y}$ (resp. $\widehat{\mathcal{O}}_{X,x}$ of $\mathcal{O}_{X,x}$) for the $\mathfrak{m}$ -adic topology. Then, since f is flat, $\widehat{\mathcal{O}}_{X,x}$ is flat over $\widehat{\mathcal{O}}_{Y,y}$ by SGA 1, IV 5.8. Thus, by 0.3.8, $\widehat{\mathcal{O}}_{X,x}$ is a topologically free $\widehat{\mathcal{O}}_{Y,y}$ -module. This shows that $\widehat{f}$ is topologically flat. Thus, by Proposition 1.3.1, $\widehat{f}$ is an effective epimorphism, i.e. the sequence below (where we have written $\widehat{X}$ instead of $\widehat{X}/\widehat{S}$) is exact:

$$\widehat{X} \times_{\widehat{Y}} \widehat{X} \xrightarrow[\widehat{\text{pr}_2}]{\widehat{\text{pr}_1}} \widehat{X} \xrightarrow{\widehat{f}} \widehat{Y}.$$

Moreover, by 1.2.6, one has a natural isomorphism (in particular, commuting with the projections to $\widehat{X}$)

$$\widehat{X \times_Y X} \simeq \widehat{X} \times_{\widehat{Y}} \widehat{X}.$$

Consequently, sequence $(\widehat{*})$ is exact.

1.3.5. — Let k be a pseudocompact ring. A formal variety X over k is said to be *topologically flat* if its affine algebra A is a *projective* pseudocompact k -module, i.e. if the structural morphism $X \rightarrow \text{Spf}(k)$ is topologically flat. ⁽⁶⁰⁾First note that 0.2.2 and 0.3.6 imply the following result (analogous to VII A, 3.1.1).

Lemma 1.3.5.A. *Suppose that k is artinian. The functors $A \mapsto A^\dagger = \text{Hom}_c(A, k)$ and $C \mapsto C^* = \text{Hom}_k(C, k)$ define an anti-equivalence between the category of topologically flat profinite k -algebras and that of flat k -coalgebras.*

Indeed, if A is a topologically flat profinite k -algebra, then, by 0.3.6, one has an isomorphism of k -modules

$$A^\dagger \otimes_k A^\dagger \xrightarrow{\sim} (A \widehat{\otimes} A)^\dagger,$$

so multiplication $A \widehat{\otimes} A \rightarrow A$ induces by duality a k -coalgebra structure on $A^\dagger$. The rest then follows from Proposition 0.2.2. Let us return to the case of an arbitrary pseudocompact ring k .

Definition 1.3.5.B. To every formal k -variety X whose affine ring A is a *projective* pseudocompact k -module, one associates a flat $\mathbf{O}_k$ -coalgebra $\mathbf{H}(X)$, defined as follows. Denote by

⁽⁵⁹⁾**N.D.E.** We have detailed what follows; we have then taken advantage of the addition made in 1.2.6.

⁽⁶⁰⁾**N.D.E.** We have added the following lemma, used implicitly in the original; moreover, we have introduced the numbering 1.3.5.A to 1.3.5.D for later references.

$\mathbf{H}(X)$ the $\mathbf{O}_k$ -module $\mathbf{V}_k^f(A)$ “dual to A ”; it is a flat $\mathbf{O}_k$ -module, since the pseudocompact k -module underlying A is projective (cf. 1.2.3.C). Moreover, by 0.3.6, one has

$$\mathbf{V}_k^f(A \hat{\otimes} A) \simeq \mathbf{V}_k^f(A) \otimes \mathbf{V}_k^f(A),$$

and thus multiplication on A induces by transposition a diagonal morphism

$$\mathbf{H}(X) = \mathbf{V}_k^f(A) \longrightarrow \mathbf{V}_k^f(A \hat{\otimes} A) = \mathbf{H}(X) \otimes \mathbf{H}(X)$$

which makes $\mathbf{H}(X)$ a flat $\mathbf{O}_k$ -coalgebra. We shall say that $\mathbf{H}(X)$ is *the coalgebra of X over $\mathbf{O}_k$* .

Definition 1.3.5.C. Conversely, to every $\mathbf{O}_k$ -coalgebra $\mathbf{C}$ one can associate a k -functor (cf. 1.2.1) $\mathrm{Spf}^*(\mathbf{C})$, defined as follows. For every object B of $\mathbf{AIf}/k$, put (with the notation of VII A 3.1)

$$(1) \quad \begin{aligned} \mathrm{Spf}^*(\mathbf{C})(B) &= \mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \\ &= \{x \in \mathbf{C}(B) \mid \varepsilon_{\mathbf{C}(B)}(x) = 1 \text{ and } \Delta_{\mathbf{C}(B)}(x) = x \otimes x\}. \end{aligned}$$

⁽⁶¹⁾ Suppose further that the $\mathbf{O}_k$ -module underlying $\mathbf{C}$ is *admissible* (cf. 1.2.1), and put

$$(2) \quad A = \Gamma^*(\mathbf{C}) = \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^*.$$

Then the algebra structures on each $\mathbf{C}(k/\mathfrak{l})^*$ equip A with a profinite k -algebra structure. For every object B of $\mathbf{AIf}/k$, one has

$$(3) \quad \begin{aligned} \mathrm{Hom}_{\mathbf{Vaf}/k}(\mathrm{Spf}(B), \mathrm{Spf}(A)) &= \mathrm{Hom}_{\mathbf{AIf}/k}(A, B) \\ &= \mathrm{Hom}_{\mathbf{AIf}/B}(A \hat{\otimes} B, B). \end{aligned}$$

Finally, suppose that $\mathbf{C}$ is a flat $\mathbf{O}_k$ -module. Then, by 1.2.3.E, $A = \Gamma^*(\mathbf{C})$ is a *projective* pseudocompact k -module. Moreover, in the proof of *loc. cit.* one has seen that if $\mathfrak{l}_0$ is an open ideal of k contained in the kernel of $k \longrightarrow B$, one has isomorphisms

$$(4) \quad \begin{aligned} A \hat{\otimes} B &= \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^* \hat{\otimes}_k B \simeq \mathrm{Hom}_{k/\mathfrak{l}_0}(\mathbf{C}(k/\mathfrak{l}_0), B) \\ &\simeq \mathrm{Hom}_B(\mathbf{C}(B), B), \end{aligned}$$

and we shall denote the right-hand term by $\mathbf{C}(B)^*$. Finally, by Lemma 1.3.5.A applied to the artinian ring B , one has a natural isomorphism

$$(5) \quad \mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{AIf}/B}(\mathbf{C}(B)^*, B).$$

Then, combining (1), (5), (4), (3), and (2), when $\mathbf{C}$ is a flat $\mathbf{O}_k$ -coalgebra one obtains an isomorphism of functors

$$(\star) \quad \mathrm{Spf}^*(\mathbf{C}) \simeq \mathrm{Spf}(A) = \mathrm{Spf}(\Gamma^*(\mathbf{C})).$$

Consequently, if one denotes by $\mathscr{A}(X)$ the affine algebra of a formal k -variety X , taking 1.2.3.E into account one obtains:

Proposition 1.3.5.D.

- (i) *The functors $X \longmapsto \mathbf{H}(X) = \mathbf{V}_k^f(\mathscr{A}(X))$ and $\mathbf{C} \longmapsto \mathrm{Spf}^*(\mathbf{C}) = \mathrm{Spf}(\Gamma^*(\mathbf{C}))$ define an equivalence between the category of topologically flat formal k -varieties and that of flat $\mathbf{O}_k$ -coalgebras.*
- (ii) *Moreover, this equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of pseudocompact rings, $X \hat{\otimes}_k K$ corresponds to $\mathbf{H}(X) \otimes_k K$.*

⁽⁶¹⁾ **N.D.E.** We have detailed what follows.

1.3.6. — In the remainder of this exposé, we shall repeatedly define a topologically flat formal k -variety X by exhibiting the coalgebra $\mathbf{H}(X)$. We shall then have to interpret certain geometric properties of X by means of $\mathbf{H}(X)$. Here we give an example of this situation: suppose that a section σ of the structural morphism $X \rightarrow \mathrm{Spf}(k)$ is given, and ask under what condition σ induces an isomorphism of the underlying topological spaces.⁽⁶²⁾ To begin with, suppose that k is artinian. Let (H, Δ, ε) be a flat k -coalgebra, $H^+ = \mathrm{Ker}(\varepsilon)$, and $A = H^*$ the profinite k -algebra dual to H . Suppose that a morphism of k -coalgebras $k \rightarrow H$ is given, i.e. an element φ of H such that $\varepsilon(\varphi) = 1$ and $\Delta(\varphi) = \varphi \otimes \varphi$. On the one hand, φ defines a continuous morphism of algebras $\Phi : A \rightarrow k$, and hence a section $\sigma : \mathrm{Spf}(k) \rightarrow \mathrm{Spf}(A)$ of the projection $\mathrm{Spf}(A) \rightarrow \mathrm{Spf}(k)$. On the other hand, one defines k -submodules of H by putting $H_0 = k\varphi$ and, for $n \geq 1$,

$$H_n = \{x \in H \mid \Delta(x) - x \otimes \varphi \in H_{n-1} \otimes H^+\};$$

this also holds for $n = 0$ if one puts $H_{-1} = (0)$. One sees by induction on n that $H_{n-1} \subset H_n$. One calls $H_0 \subset H_1 \subset \dots$ the *filtration of H defined by φ* .

Remark. Since $\Delta(H_n) \subset H_n \otimes H_0 \oplus H_{n-1} \otimes H^+$, one has $\Delta(H_n) \subset H_n \otimes H$. Since Δ is cocommutative (i.e. $\sigma \circ \Delta = \Delta$, where $\sigma(a \otimes b) = b \otimes a$), one also has $\Delta(H_n) \subset H \otimes H_n$. When H/H_n is flat over k , it follows that H_n is a subcoalgebra of H (see also 1.3.6.A(iii) below). But in general $\Delta : H_n \rightarrow H_n \otimes H$ does not factor through $H_n \otimes H_n$.⁽⁶³⁾

Lemma 1.3.6.A. *Let k be an artinian ring, H a flat k -coalgebra, $A = H^*$ the dual profinite k -algebra, and φ an element of H such that $\varepsilon(\varphi) = 1$ and $\Delta(\varphi) = \varphi \otimes \varphi$. Let $\Phi : A \rightarrow k$ be the continuous morphism of algebras, $\sigma : \mathrm{Spf}(k) \rightarrow \mathrm{Spf}(A)$ the section of $\mathrm{Spf}(A) \rightarrow \mathrm{Spf}(k)$, and (H_n) the filtration of H defined by φ . Denote by I the kernel of Φ .*

- (i) *For every $n \geq 1$, H_{n-1} is the orthogonal in H of the closure $\overline{I^n}$ of I^n .*
- (ii) *Consequently, σ induces a bijection of the underlying sets if and only if $H = \bigcup_n H_n$.⁽⁶⁴⁾*
- (iii) *If, moreover, each H/H_n is flat over k , then, for every $n \geq 0$,*

$$(*) \quad \Delta(H_n) \subset \sum_{i=0}^n H_i \otimes H_{n-i};$$

in particular, each H_n is then a subcoalgebra of H .

PROOF. Note that, for every $x \in H$, the map $A \rightarrow k$, $f \mapsto f(x)$ is continuous, so if I^n is annihilated by x , the same holds for its closure $\overline{I^n}$. For every $n \geq 1$, one then puts

$$(\overline{I^n})^\perp = \{x \in H \mid f(x) = 0 \text{ for every } f \in \overline{I^n}\}.$$

Suppose that $\sigma : \mathfrak{m} \mapsto \Phi^{-1}(\mathfrak{m})$ is a bijection from $\mathrm{Spf}(k)$ to $\mathrm{Spf}(A)$. Since I is contained in the intersection of the $\Phi^{-1}(\mathfrak{m})$, it follows from 0.1.2 that the sequence of ideals (I^n) tends to 0. Let $x \in H$; since $J(x) = \{f \in A \mid f(x) = 0\}$ is an open and closed k -submodule of A , it contains I^n for sufficiently large n ; in other words, $x \in (\overline{I^n})^\perp$ for sufficiently large n . Conversely, suppose that $H = \bigcup_n (\overline{I^n})^\perp$. Let $\mathfrak{p}$ be an open prime ideal of A ; by the definition of the topology of A (0.2.2), $\mathfrak{p}$ contains an open k -submodule of the form

$$\mathcal{V}(x_1, \dots, x_s) = \{f \in A \mid f(x_1) = \dots = f(x_s) = 0\}.$$

⁽⁶²⁾**N.D.E.** In Lemma 1.3.6.A which follows, we have detailed the proof of parts (i) and (ii), and have added part (iii).

⁽⁶³⁾**N.D.E.** For example, let k_0 be a field, $k = k_0[T]/(T^n)$, where $n \geq 4$, and $H = k\varphi \oplus kx$, with $\varepsilon(x) = 0$ and $\Delta(x) = x \otimes \varphi + \varphi \otimes x + tx \otimes x$, where t is the image of T in k . Then $H_i = k\varphi \oplus t^{n-i}x$ for $i = 0, \dots, n$, but for $2 \leq i \leq n-2$, $\Delta(t^{n-i}x)$ does not belong to the image of $H_i \otimes H_i$ in $H \otimes H$.

⁽⁶⁴⁾**N.D.E.** In this case one says that the coalgebra H is connected; cf. addition 2.9.

By hypothesis, there exists an integer n such that $x_1, \dots, x_s \in (\overline{I^n})^\perp$, hence $I^n \subset \mathfrak{p}$. Moreover, since k is artinian, $\mathrm{Spf}(k)$ is a finite set $\{\mathfrak{m}_1, \dots, \mathfrak{m}_r\}$ and there exists an integer $t \geq 1$ such that $(\prod_i \mathfrak{m}_i)^t = 0$, whence

$$\prod_i \Phi^{-1}(\mathfrak{m}_i)^t \subset I.$$

Thus $\mathfrak{p}$ contains the product of the $\Phi^{-1}(\mathfrak{m}_i)^{tn}$; since $\mathfrak{p}$ is prime, it follows that $\mathfrak{p}$ contains, hence equals, one of the $\Phi^{-1}(\mathfrak{m}_i)$. One has thus proved that

$$\sigma \text{ is a bijection} \iff H = \bigcup_n (\overline{I^n})^\perp.$$

(65) On the other hand, one has $H = k\varphi \oplus H^+$; denote by π the projection $H \rightarrow H^+$ with kernel $k\varphi$. For every $n \geq 0$, let Δ^n be the “iterated” comultiplication $H \rightarrow H^{\otimes(n+1)}$, $\overline{\Delta}^n$ the composite of Δ^n with the projection $\pi^{\otimes(n+1)} : H^{\otimes(n+1)} \rightarrow (H^+)^{\otimes(n+1)}$, and

$$H'_n = \mathrm{Ker}(\overline{\Delta}^n) = \{x \in H \mid \Delta^n(x) \in \sum_{i=0}^n H^{\otimes(n-i)} \otimes H_0 \otimes H^{\otimes i}\}.$$

(One puts $\Delta^0 = \mathrm{id}_H$, whence $H'_0 = H_0$.) One easily sees, by induction on n , that

$$(*) \quad H_n \subset H'_n \subset (\overline{I^{(n+1)}})^\perp.$$

Until now, one has not used the hypothesis that H is flat over k . Now suppose that H is flat, hence projective over k , so that $A^\dagger = H$ by 0.2.2, and let us show that $H_n = (\overline{I^{(n+1)}})^\perp$. This is clear for $n = 0$. Suppose therefore that it holds for $n < r$. Then H_{r-1} is the kernel of the morphism $H \rightarrow (\overline{I^r})^\dagger$, and thus, since H^+ is flat, the morphism

$$(H/H_{r-1}) \otimes H^+ \rightarrow (\overline{I^r})^\dagger \otimes H^+$$

is injective. On the other hand, the hypothesis implies that I is a projective pseudocompact k -module (since it is a direct factor of $A = H^*$), whence, by 0.3.6,

$$(\overline{I^r} \widehat{\otimes} I)^\dagger \simeq (\overline{I^r})^\dagger \otimes I^\dagger = (\overline{I^r})^\dagger \otimes H^+.$$

Then the exact sequence

$$\overline{I^r} \widehat{\otimes} I \rightarrow A \rightarrow A/\overline{I^{r+1}} \rightarrow 0$$

gives by duality the exact sequence

$$(1) \quad (\overline{I^r})^\dagger \otimes H^+ \xleftarrow{\delta} H \leftarrow (A/\overline{I^{r+1}})^\dagger \leftarrow 0,$$

where δ is obtained by composing Δ_H with the projection

$$(2) \quad H \otimes H \rightarrow H \otimes H^+ \rightarrow (H/H_{r-1}) \otimes H^+ \hookrightarrow (\overline{I^r})^\dagger \otimes H^+.$$

Now, for every $u \in H$, the projection of $\Delta(u)$ to $H \otimes H^+$ is $\Delta(u) - u \otimes \varphi$. Then (1) and (2) show that, if u belongs to $(A/\overline{I^{r+1}})^\dagger = (\overline{I^{r+1}})^\perp$, $\Delta(u) - u \otimes \varphi$ belongs to the kernel of the map $H \otimes H^+ \rightarrow (H/H_{r-1}) \otimes H^+$, that is, to $H_{r-1} \otimes H^+$, and hence $u \in H_r$. This completes the proof of parts (i) and (ii), and also shows that $H_n = \mathrm{Ker}(\overline{\Delta}^n)$. Let us prove (iii). For every $i \geq 0$, put $H_i^+ = H_i \cap H^+$. Let $n \geq 1$. For every $x \in H_n$, $\bar{x} = x - \varepsilon(x)\varphi$ belongs to H_n^+ , and one has

$$\Delta(x) = \varepsilon(x)\varphi \otimes \varphi + \bar{x} \otimes \varphi + \varphi \otimes \bar{x} + \overline{\Delta}(\bar{x}).$$

Thus it suffices to show that

$$\overline{\Delta}(H_n^+) \subset \sum_{i=1}^{n-1} H_i^+ \otimes H_{n-i}^+.$$

(65) **N.D.E.** This also holds without assuming H cocommutative (k remaining a commutative artinian ring): in this case, a base of neighborhoods of 0 in $A = H^*$ is given by the two-sided ideals J such that A/J has finite length over k , and the preceding proof shows that $H = \bigcup_n (\overline{I^n})^\perp$ if and only if the $\Phi^{-1}(\mathfrak{m}_i)$ are the only open prime ideals of A .

For every $i = 0, \dots, n-1$, $\overline{\Delta}^n$ factors as

$$\begin{array}{ccc} H^+ & \xrightarrow{\overline{\Delta}} & H^+ \otimes H^+ \xrightarrow{\overline{\Delta}^i \otimes \overline{\Delta}^{n-i-1}} (H^+)^{\otimes(i+1)} \otimes (H^+)^{\otimes(n-i)} \\ & \downarrow & \uparrow g \\ \frac{H^+}{H_i^+} \otimes \frac{H^+}{H_{n-i-1}^+} & \xrightarrow{f} & \frac{H^+}{H_i^+} \otimes (H^+)^{\otimes(n-i)}. \end{array}$$

Moreover, since H^+/H_i^+ and $(H^+)^{\otimes(n-i)}$ are flat, the maps f and g above are injective. It follows that $\overline{\Delta}(H_n^+)$ is contained in $H_i^+ \otimes H^+ + H^+ \otimes H_{n-i-1}^+$ for every $i = 0, \dots, n-1$. Part (iii) then follows from the lemma below applied to $M = H^+$ and $E_i = H_{i-1}^+$. $\square$

Lemma 1.3.6.B. *Let k be a ring and $0 = E_0 \subset E_1 \subset \dots \subset E_n \subset M$ be k -modules. Suppose that M/E_i is flat for every i . Then one has the equality*

$$\bigcap_{i=0}^n (E_i \otimes M + M \otimes E_i) = \sum_{i=1}^n E_i \otimes E_{n-i+1}.$$

Denote by K (resp. S) the left-hand (resp. right-hand) term. One easily sees that $S \subset K$; let us prove the converse. For $i = 0, \dots, n$, put $K_i = K \cap (E_i \otimes M)$. For every $i = 1, \dots, n$, since M/E_{n-i+1} and E_i/E_{i-1} are flat, the map τ_i below is injective, and the composite map

$$\begin{aligned} (E_i/E_{i-1}) \otimes M &\longrightarrow (E_i/E_{i-1}) \otimes (M/E_{n-i+1}) \\ &\xrightarrow{\tau_i} (M/E_{i-1}) \otimes (M/E_{n-i+1}) \end{aligned}$$

has kernel $(E_i/E_{i-1}) \otimes E_{n-i+1}$. Since the image of K_i in $(E_i \otimes M)/(E_{i-1} \otimes M)$ is contained in, and contains, this kernel, one deduces that

$$K_i = K_{i-1} + E_i \otimes E_{n-i+1},$$

whence the lemma. ⁽⁶⁶⁾To finish this paragraph, let us return to an arbitrary pseudocompact ring k . Let $(\mathbf{H}, \Delta, \varepsilon)$ be a flat $\mathbf{O}_k$ -coalgebra, $\mathbf{H}^+ = \text{Ker}(\varepsilon)$, $A = \Gamma^*(\mathbf{H})$ the dual profinite k -algebra, and $X = \text{Spf}(A)$, so that $\mathbf{H} = \mathbf{H}(X)$ (cf. 1.3.5). Suppose that a morphism of $\mathbf{O}_k$ -coalgebras $\varphi : \mathbf{O}_k \rightarrow \mathbf{H}$ is given; it defines a continuous morphism of k -algebras $A \rightarrow k$, and hence a section σ of the structural morphism $X \rightarrow \text{Spf}(k)$. For every object B of $\mathbf{Alf}/k$, write $\mathbf{H}_0(B) = \varphi(B) = B\varphi_B$, where φ_B is the element $\varphi(1_B)$ of $\mathbf{H}(B)$, and define $\mathbf{O}_k$ -submodules $\mathbf{H}_n$ of $\mathbf{H}$ by putting, for $n \geq 1$,

$$\mathbf{H}_n(B) = \{u \in \mathbf{H}(B) \mid \Delta(u) - u \otimes \varphi_B \in \mathbf{H}_{n-1}(B) \otimes \mathbf{H}^+(B)\}.$$

In this way one obtains a filtration $\mathbf{H}_0 \subset \mathbf{H}_1 \subset \dots$ of $\mathbf{H}(X)$. By the foregoing, one has:

Proposition. *For σ to induce an isomorphism on the underlying topological spaces, it is necessary and sufficient that $\mathbf{H}(X)$ be the union of the $\mathbf{H}_n$.*

1.4. Theorem. *Let k be a pseudocompact ring and $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence pair in $\mathbf{Vaf}/k$ (cf. Exp. V, § 2.b) such that d_1 is topologically flat.*

- (i) *The canonical projection from X to X/X_1 ($= \text{Coker}(d_0, d_1)$) is surjective and topologically flat, and the morphism $X_1 \rightarrow X \times_{X/X_1} X$ with components d_0 and d_1 is an isomorphism.*
- (ii) *If X is topologically flat over k , so is X/X_1 .*

First note that (ii) follows from (i), by 1.3.3(ii). The proof of (i) occupies paragraphs 1.4.1, 1.4.2, and 1.4.3.

1.4.1. — Let us first show that one can reduce to the case where X has a single point. Since we are dealing with an equivalence pair, one sees as in Exp. V, § 3.e) that one defines an

⁽⁶⁶⁾**N.D.E.** We have added what follows; the original was restricted to the case where k is artinian.

equivalence relation on the set underlying X by declaring two points x, y equivalent if there exists a point z of X_1 such that $d_0(z) = y$ and $d_1(z) = x$. One can evidently suppose without loss that X contains a single equivalence class for this relation, in other words that X/X_1 has a single point (see the construction of X/X_1 given in 1.2). In this case, let x be any point of X and U the formal variety which has x as its sole point and has the same local ring as X at x . One then sees as in Exp. V, § 6 that the equivalence relation induced by (d_0, d_1) on U still satisfies the hypotheses of the theorem, and that it suffices to prove it for this latter equivalence relation (U is a “quasi-section”). Let us briefly recall the principle of the proof given in Exp. V, § 6. Put $V = d_0^{-1}(U) = U \times_{i, d_0} X_1$, where i is the inclusion of U in X ; let u and v be the morphisms with source V induced respectively by d_0 and d_1 :

$$X \xleftarrow{v} V \xrightarrow{u} U.$$

It is clear that u and v are topologically flat and u is surjective; since X contains a single equivalence class, v is surjective. If (v_0, v_1) is the inverse image of the equivalence pair (d_0, d_1) under v (cf. V, 3.a)), it follows from V, 3.c) and 3.d) that X/X_1 and the quotient of U by the equivalence relation induced by (d_0, d_1) are both identified with $\text{Coker}(v_0, v_1)$. One then sees, as in the proof of V, 6.1, that if the conclusion of Theorem 1.4 holds for U , it also holds for X .

1.4.2. — One is thus reduced to the case where X has a single point.⁽⁶⁷⁾ Consider then the following commutative diagram (cf. V, § 1, (0,1,2)):

$$\begin{array}{ccccc} X_2 & \xrightarrow{d'_1} & X_1 & \xrightarrow{d_0} & X \\ d'_2 \downarrow & & d'_0 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X & \longrightarrow & X/X_1 \\ & & d_0 \downarrow & & \end{array}$$

where X_2 is the fibered product $X_1 \times_{d_1, d_0} X_1$, and d'_0, d'_1, d'_2 are respectively the morphisms “ $(x, y, z) \mapsto (x, y)$ ”, “ $(x, y, z) \mapsto (x, z)$ ”, and “ $(x, y, z) \mapsto (y, z)$ ”.⁽⁶⁸⁾ If B, A, A_1, A_2 denote respectively the affine algebras of $X/X_1, X, X_1, X_2$, the preceding diagram induces a commutative diagram

$$\begin{array}{ccccc} A_2 & \xleftarrow{j_1} & A_1 & \xleftarrow{i_0} & A \\ j_2 \uparrow & & j_0 \uparrow & & \uparrow i \\ A_1 & \xleftarrow{i_1} & A & \xleftarrow{i} & B \\ & & i_0 \downarrow & & \end{array}$$

in which both rows are exact and the squares determined by (i_0, j_0) and (i_1, j_1) are cocartesian. Since the morphism $X_1 \rightarrow X \times X$ with components d_0, d_1 is a monomorphism by hypothesis, the morphism $A \hat{\otimes}_k A \rightarrow A_1$ with components i_0, i_1 is surjective, by Proposition 1.3. This means that i_1 makes A_1 a pseudocompact A -module (assumed topologically free) generated by $i_0(A)$. Since A is local, Lemma 1.4.3 below implies the existence of a topologically free k -module V and a morphism of pseudocompact k -modules $f : V \rightarrow A$ such that the morphism

$$\alpha_1 : A \hat{\otimes}_k V \rightarrow A_1, \quad a \hat{\otimes} v \mapsto i_1(a) \cdot i_0(f(v))$$

is invertible. Let $\alpha : B \hat{\otimes}_k V \rightarrow A$ and $\alpha_2 : A_1 \hat{\otimes}_k V \rightarrow A_2$ be the morphisms

$$b \hat{\otimes} v \mapsto i(b) \cdot f(v) \quad \text{and} \quad a_1 \hat{\otimes} v \mapsto j_2(a_1) \cdot j_0 i_0(f(v)).$$

⁽⁶⁷⁾**N.D.E.** One can therefore suppose that k is local.

⁽⁶⁸⁾**N.D.E.** Cf. Exp. V, § 2.b).

In the following commutative diagram, both rows are therefore exact and the two left-hand squares are cocartesian. Since α_1 is invertible, so is α_2 , hence so is α .

$$\begin{array}{ccccc}
 A_2 & \xleftarrow{j_1} & A_1 & \xleftarrow{i_0} & A \\
 \alpha_2 \uparrow & & \alpha_1 \uparrow & & \alpha \uparrow \\
 A_1 \widehat{\otimes}_k V & \xleftarrow{i_1 \widehat{\otimes} V} & A \widehat{\otimes}_k V & \xleftarrow{i \widehat{\otimes} V} & B \widehat{\otimes}_k V \\
 & i_0 \widehat{\otimes} V & & &
 \end{array}$$

This shows, on the one hand, that A is topologically free over B with pseudobasis $f(V)$ (cf. 0.2.1), and that one obtains a pseudobasis of A_1 over A (where A_1 is considered as an A -module by means of i_1) by taking the image of $f(V)$ under i_0 ; this implies that the morphism $A \widehat{\otimes}_B A \rightarrow A_1$ with components i_1, i_0 is invertible:

$$A \widehat{\otimes}_B A \simeq A \widehat{\otimes}_B B \widehat{\otimes}_k V \simeq A \widehat{\otimes}_k V \simeq A_1.$$

This proves Theorem 1.4, modulo the following Lemma 1.4.3.

1.4.3. Lemma. *Let k be a pseudocompact ring, A a local profinite k -algebra, A_1 a topologically free A -module, and $i_0 : M \rightarrow A_1$ a morphism of pseudocompact k -modules. Suppose that the map*

$$A \widehat{\otimes}_k M \rightarrow A_1, \quad a \widehat{\otimes} m \mapsto a \cdot i_0(m)$$

is surjective. There then exist a topologically free k -module V and a morphism of pseudocompact k -modules $f : V \rightarrow M$ such that the map

$$A \widehat{\otimes}_k V \rightarrow A_1, \quad a \widehat{\otimes} v \mapsto a \cdot i_0(f(v))$$

is bijective.

Since every pseudocompact k -module is a quotient of a topologically free k -module (cf. N.D.E. (27)), one can suppose without loss that M is topologically free; take therefore for M the direct product of a family $(M_i)_{i \in I}$ of copies of k . In this case $A \widehat{\otimes}_k M$ is precisely the product $\prod_{i \in I} A \widehat{\otimes}_k M_i$. Since the map $a \widehat{\otimes} m \mapsto a \cdot i_0(m)$ is surjective and A_1 is projective, the kernel of this map is a direct factor of $A \widehat{\otimes}_k M$; since A is local, it follows from the exchange theorem (0.3.4) that this kernel has as a complement a partial product $\prod_{i \in J} A \widehat{\otimes}_k M_i$, where J denotes some subset of I . One can therefore take $V = \prod_{i \in J} M_i$.

1.5. — Let k be a pseudocompact ring.

Definition. We shall say that a family of morphisms $f_i : X_i \rightarrow X$ of $\mathbf{Vaf}/k$ is a *topologically flat surjective family* if the morphism $\coprod_i X_i \rightarrow X$ induced by the f_i is surjective and topologically flat; this means that each f_i is topologically flat and every point of X belongs to the image of at least one of the X_i .

It follows from 1.3.3 that topologically flat surjective families define a pretopology on $\mathbf{Vaf}/k$ (IV 4.2.5); the corresponding topology will be called the *flat topology* on $\mathbf{Vaf}/k$. By IV, 4.3.5, a functor $\mathbf{F} : (\mathbf{Vaf}/k)^0 \rightarrow (\mathbf{Ens})$ is a sheaf for the flat topology if and only if $\mathbf{F}$ transforms every direct sum into a direct product and the sequence

$$\mathbf{F}(Y) \xrightarrow{\mathbf{F}(f)} \mathbf{F}(X) \xrightarrow[\mathbf{F}(\text{pr}_2)]{\mathbf{F}(\text{pr}_1)} \mathbf{F}(X \times_Y X)$$

is exact for every surjective topologically flat morphism $f : X \rightarrow Y$. ⁽⁶⁹⁾By IV, 4.5, Proposition 1.3.1 implies that the flat topology is less fine than the canonical topology, i.e. for every object X of $\mathbf{Vaf}/k$, the functor $h_X : T \mapsto \text{Hom}_{\mathbf{Vaf}/k}(T, X)$ is a sheaf for the flat

⁽⁶⁹⁾**N.D.E.** We have modified the original in what follows. In particular, we have replaced the statement “if $d_0, d_1 : X_1 \rightrightarrows X$ is an equivalence relation such that d_1 is topologically flat, formation of the quotient commutes with h ” by the theorem below.

topology. (In what follows, one identifies X with h_X as usual (cf. Exp. I).) By IV, 4.6.5, one can reformulate Theorem 1.4 as follows.

Theorem. *Let k be a pseudocompact ring, $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence pair in $\mathbf{Vaf}/k$, and X/X_1 the quotient formal variety (i.e. $\text{Coker}(d_0, d_1)$, cf. 1.2). If d_1 is topologically flat, X/X_1 represents the quotient sheaf for the flat topology.*

1.6. — To conclude these generalities on formal varieties, it remains to define briefly formal varieties étale over k .⁽⁷⁰⁾

Recollections 1.6.A.

- (i) First recall (cf. EGA 0_{IV}, 19.10.2) that a topological k -algebra A is said to be *formally étale over k* (for the given topologies on k and A , resp. for the discrete topologies) if, for every discrete topological k -algebra C (not necessarily artinian), and every nilpotent ideal J of C , every continuous morphism of k -algebras $A \rightarrow C/J$ lifts uniquely to a continuous morphism $A \rightarrow C$ (A being equipped with the given topology, resp. the discrete topology). One sees at once that this property is preserved by base change, i.e. for every morphism $k \rightarrow k'$ of pseudocompact rings, $A \widehat{\otimes}_k k'$ is then formally étale over k' . Moreover, one easily sees that it suffices to verify the lifting condition for every square-zero ideal J ; cf. EGA IV₄, 17.1.2(ii). One says that A is *étale over k* if it is formally étale over k for the discrete topologies and if, in addition, A is a finitely presented k -algebra (cf. EGA IV₄, 17.3.2(ii)). In what follows, k being a pseudocompact ring and A a profinite k -algebra, “formally étale” will be used in the sense of the given topologies (unless otherwise specified).
- (ii) Recall also that if $F \in k[T]$ is a monic polynomial of degree $d \geq 1$ which is separable (i.e. such that the ideal generated by F and its derivative F' is $k[T]$), then the k -algebra $B = k[T]/(F)$ (which is free of rank d over k , and equipped with the product topology) is formally étale over k . Indeed, let C be a discrete k -algebra (so that the kernel of $k \rightarrow C$ is an open ideal $\mathfrak{l}$ of k), J a square-zero ideal of C , and $\phi : B \rightarrow C/J$ a continuous morphism of k -algebras. Note that, since B is a free k -module of finite rank, $\mathfrak{l}B$ is an open ideal of B , so every lifting $\Phi : B \rightarrow C$ of ϕ is automatically continuous. Let t be the image of T in B and u_0 any lifting of $\phi(t)$ in C ; then $F(u_0) \in J$ (since $\phi(t)$ is a root of F); moreover, there exist $G, H \in k[T]$ such that $GF + HF' = 1$, whence $H(u_0)F'(u_0) = 1 - G(u_0)F(u_0)$, and the right-hand term is invertible, since $F(u_0)$ has square zero, so $F'(u_0)$ is invertible. Seek $h \in J$ such that $x = x_0 + h$ is a root of F ; this is equivalent to $0 = F(x) = F(u_0) + F'(u_0)h$, and since $F'(u_0)$ is invertible, this has the unique solution $h = -F'(u_0)^{-1}F(u_0) \in J$. Of course, the same proof (without making continuity assumptions for the morphisms $k \rightarrow C$, ϕ , and Φ) shows that B is also an étale k -algebra.
- (iii) Finally, recall that if A is a finite product $A_1 \times \cdots \times A_n$, then A is formally étale over k if and only if the A_i are.⁽⁷¹⁾ Indeed, it suffices to see this for $n = 2$; in this case let $e = 1_{A_1}$ be the idempotent such that $A_1 = Ae$ and $A_2 = A(1 - e)$, and suppose that a continuous morphism $A \rightarrow C/J$ is given, where J is a square-zero ideal. Since the polynomial $F = X^2 - X$ is separable (one has $F' = 2X - 1$ and $(F')^2 - 4F = 1$), the idempotent $\phi(e)$ of C/J lifts uniquely to an idempotent f of C ,

⁽⁷⁰⁾**N.D.E.** The original continued as follows: “A formal variety X over k is said to be étale if the diagonal morphism $\Delta_X : X \rightarrow X \times X$ is a local isomorphism, that is, if Δ_X induces an isomorphism from $\mathcal{O}_{X \times X, \Delta_X(x)}$ to $\mathcal{O}_{X, x}$ for every point x of X . One easily sees with the aid of SGA I that this formulation is equivalent to the following two: the formal variety X is topologically flat, and, for every point $x \in X$ projecting to $s \in \text{Spf}(k)$, $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is a finite separable extension of the residue field $\kappa(s)$ of s ; or again, if A denotes the affine algebra of X , the local components (0.1) of $A \widehat{\otimes}_k (k/\mathfrak{l})$ are finite étale algebras over $k/\mathfrak{l}$, for every open ideal $\mathfrak{l}$ of k .” In what follows, we have rectified the omission of the flatness hypothesis in the first condition above, and detailed the equivalence of these conditions.

⁽⁷¹⁾**N.D.E.** Let us point out in passing that the proof given in EGA 0_{IV}, 19.3.5(v) is erroneous.

whence $C = Cf \oplus C(1 - f)$, and then giving a lifting of Φ is equivalent to giving two morphisms $\Phi_1 : A_1 \rightarrow Cf$ and $\Phi_2 : A_2 \rightarrow C(1 - f)$ lifting the restrictions of ϕ to A_1 and A_2 . The same argument shows that if e is an idempotent of k such that $A = Ae$, then A is formally étale over k if and only if it is so over the localization k_e (which is identified with ke).

Now let X be a formal k -variety and A its affine algebra. If x is a point of X (i.e. an open maximal ideal $\mathfrak{m}$ of A) with image s in $\mathrm{Spf}(k)$, the local component of k corresponding to s will be denoted by $k_{\mathfrak{m}}$ or k_s .

Definition 1.6.B. The following conditions are equivalent:

- (a) A is *formally étale over k* .
- (b) For every $\mathfrak{m} \in \Upsilon(A)$, $A_{\mathfrak{m}}$ is formally étale over k (or over $k_{\mathfrak{m}}$).
- (c) For every open ideal $\mathfrak{l}$ of k , $A \widehat{\otimes}_k (k/\mathfrak{l}) = A/\overline{A\mathfrak{l}}$ is formally étale over $k/\mathfrak{l}$.
- (d) For every $\mathfrak{m} \in \Upsilon(A)$ and every open ideal $\mathfrak{l}$ of $k_{\mathfrak{m}}$, $A_{\mathfrak{m}}/\overline{A_{\mathfrak{m}}\mathfrak{l}}$ is formally étale over $k_{\mathfrak{m}}/\mathfrak{l}$.

One says that X is *étale over k* if it satisfies these conditions, and denotes by $\mathbf{Vaf}_{/k}^{\text{ét}}$ the full subcategory of $\mathbf{Vaf}/k$ consisting of formal varieties étale over k .⁽⁷²⁾

Observe that if $\phi : A \rightarrow C/J$ is a continuous morphism of k -algebras, where C is a discrete k -algebra and J a square-zero ideal, then $I = \mathrm{Ker}(\phi)$ is an open ideal of A , hence A/I is artinian, so I is contained in only finitely many open maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$, and therefore contains the product of the components $A_{\mathfrak{m}}$ for $\mathfrak{m} \neq \mathfrak{m}_i$, which equals $A(1 - e)$, where e denotes the idempotent of A such that $Ae = \prod_{i=1}^r A_{\mathfrak{m}_i}$. Thus $\phi(e) = 1_{C/J}$, and giving a continuous lifting of ϕ is equivalent to giving one of the morphism from $Ae \simeq A/A(1 - e)$ to C/J induced by ϕ . Moreover, one knows (cf. N.D.E. (24)) that $A \simeq \varprojlim_{\mathfrak{l}} A/\overline{A\mathfrak{l}}$. In view of these remarks and the preceding recollections, one easily obtains the equivalence of the indicated conditions.

Definitions 1.6.C. Write $\kappa(k) = \prod_{s \in \mathrm{Spf}(k)} \kappa(s)$, equipped with the product topology, i.e. the formal variety $\mathrm{Spf}(\kappa(k))$ is the direct sum of the $\mathrm{Spec} \kappa(s)$, for $s \in \mathrm{Spf}(k)$. Moreover, denote by $S_{\kappa(k)}$ the scheme which is the direct sum of the $\mathrm{Spec} \kappa(s)$, for $s \in \mathrm{Spf}(k)$. For every formal variety X over k , denote by $X_{\kappa} = X \widehat{\otimes}_k \kappa(k)$ the formal variety over $\kappa(k)$ obtained by base change, i.e. X_{κ} has the same points as X and for every $x \in X$ projecting to s on $\mathrm{Spf}(k)$, one has $\mathcal{O}_{X_{\kappa}, x} = \mathcal{O}_{X, x} \otimes_k \kappa(s)$. This base-change functor $\mathbf{Vaf}/k \rightarrow \mathbf{Vaf}/\kappa(k)$ sends $\mathbf{Vaf}_{/k}^{\text{ét}}$ into $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ (cf. 1.6.A(i)).

One then has (cf. SGA 1, I 6.2):

Lemma 1.6.D. For every $Y \in \mathrm{Ob} \mathbf{Vaf}_{/k}^{\text{ét}}$ and $X \in \mathrm{Ob} \mathbf{Vaf}/k$, the canonical map

$$\mathrm{Hom}_{\mathbf{Vaf}/k}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{Vaf}/\kappa(k)}(X_{\kappa}, Y_{\kappa})$$

is bijective. In particular, the functor $\mathbf{Vaf}_{/k}^{\text{ét}} \rightarrow \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is fully faithful (and we shall see below that it is an equivalence).

Indeed, denote by A (resp. B) the affine algebra of X (resp. Y), and by $\mathfrak{r}$ the radical of k , and suppose that a morphism $B \widehat{\otimes}_k \kappa(k) \rightarrow A \widehat{\otimes}_k \kappa(k)$ is given, or, equivalently, a morphism $\phi : B \rightarrow A/\mathfrak{r}A$. For every open ideal $\mathfrak{l}$ of k , there exists $n \in \mathbf{N}^*$ such that $\mathfrak{r}^n \subset \mathfrak{l}$, whence $(\mathfrak{r}A)^n \subset \mathfrak{l}A \subset \overline{\mathfrak{l}A}$, and since the multiplication map $m_{n-1} : A^n \rightarrow A$ is continuous, one therefore also has $(\mathfrak{r}A)^n \subset \overline{\mathfrak{l}A}$, i.e. $\overline{\mathfrak{r}A}/\overline{\mathfrak{l}A}$ is a nilpotent ideal of $A/\overline{\mathfrak{l}A}$. Consequently, ϕ lifts uniquely to a morphism $\phi_{\mathfrak{l}} : B \rightarrow A/\overline{\mathfrak{l}A}$. By uniqueness, these morphisms form a projective

⁽⁷²⁾N.D.E. Note that $\mathbf{Vaf}_{/k}^{\text{ét}}$ is stable under finite projective limits.

system, hence give a continuous morphism $\Phi : B \longrightarrow \varprojlim_{\mathfrak{l}} A/\overline{\mathfrak{l}A} = A$. Moreover, Φ is unique, since if Φ' is a second lifting of ϕ , Φ' and Φ agree modulo $\overline{\mathfrak{l}A}$ for every $\mathfrak{l}$, hence are equal.

Proposition 1.6.E. (a) *Let X be a formal variety over k and A its affine algebra. The following conditions are equivalent:*

- (i) *X is étale over k .*
- (ii) *X is topologically flat over k and the diagonal morphism $\Delta : X \longrightarrow X \times X$ is a local isomorphism, i.e. Δ_X induces an isomorphism $\mathcal{O}_{X \times X, \Delta(x)} \xrightarrow{\sim} \mathcal{O}_{X, x}$ for every point x of X .*
- (iii) *For every $x \in X$ projecting to s on $\mathrm{Spf}(k)$, $\mathcal{O}_{X, x}$ is isomorphic to $k_s[T]/(F)$, where $F \in k_s[T]$ is a monic separable polynomial (cf. 1.6.A(ii)).*
- (iv) *X is topologically flat over k , and for every point $x \in X$ projecting to s on $\mathrm{Spf}(k)$, $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is a finite separable extension of $\kappa(s)$.*
- (v) *For every open ideal $\mathfrak{l}$ of k , each local component of $A \widehat{\otimes}_k (k/\mathfrak{l})$ is a finite étale algebra over the artinian ring $k/\mathfrak{l}$.*

(b) $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is identified with the category of étale schemes over $S_{\kappa(k)}$ (cf. 1.6.C), and the functor $X \longmapsto X_{\kappa}$ induces an equivalence of categories $\mathbf{Vaf}_{/k}^{\text{ét}} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ (cf. SGA 1, I 6.1).

PROOF. (a) Denote by I the kernel of the multiplication morphism $m : A \widehat{\otimes}_k A \longrightarrow A$. Suppose that X is étale over k , i.e. A is formally étale over k . Then, by EGA 0_{IV}, 20.7.4, the separated completion of I/I^2 for the quotient topology of that of I is zero, i.e. one has $I = \overline{I^2}$. Now, for every $x \in X$, I is contained in the maximal ideal $\mathfrak{m}_{\Delta(x)}$ of $A \widehat{\otimes}_k A$, so the localization $I_{\mathfrak{m}_{\Delta(x)}}$ is contained in the maximal ideal of $\mathcal{O}_{X \times X, \Delta(x)}$, and hence, by Nakayama's lemma 0.3, one has $I_{\mathfrak{m}_{\Delta(x)}} = 0$, and thus $\Delta : X \longrightarrow X \times X$ is a local isomorphism. Now suppose that Δ is a local isomorphism and that k is a field κ , and let us show that each $\mathcal{O}_{X, x}$ is a finite-dimensional étale κ -algebra. Replacing X by $\mathrm{Spf}(\mathcal{O}_{X, x})$, one can suppose that $A = \mathcal{O}_{X, x}$ is local. One then proceeds as in the proof of EGA IV₄, 17.4.1, (b) $\Rightarrow$ (d''). Let K be a finite normal extension of κ containing the residue field $\kappa(x)$, and let $B = A \otimes_{\kappa} K = A \widehat{\otimes}_{\kappa} K$ (since $[K : \kappa] < \infty$, $-\widehat{\otimes}_{\kappa} K$ and $-\otimes_{\kappa} K$ coincide) and $X_K = \mathrm{Spf}(B) = X \widehat{\otimes}_{\kappa} K$. Then $\Delta_K : X_K \longrightarrow X_K \widehat{\otimes}_K X_K$ is still a local isomorphism, so for every $y \in X_K$, multiplication $B_y \widehat{\otimes}_K B_y \longrightarrow B_y$ induces an isomorphism $(B_y \widehat{\otimes}_K B_y)_{\mathfrak{m}_{\Delta_K(y)}} \xrightarrow{\sim} B_y$. Now, since the residue field of B_y is K (cf. for example VI A, 1.1.1, N.D.E. (11)), $C = B_y \widehat{\otimes}_K B_y$ is already a local ring (indeed, $\mathfrak{n} = \mathfrak{m}_y \widehat{\otimes}_K B_y + B_y \widehat{\otimes}_K \mathfrak{m}_y$ consists of topologically nilpotent elements, hence is contained in the radical of C , and $C/\mathfrak{n} = K \widehat{\otimes}_K K = K$ is a field), so one obtains that the multiplication morphism $B_y \widehat{\otimes}_K B_y \longrightarrow B_y$ is an isomorphism. Taking a pseudobasis of B_y over K containing the identity element 1, one deduces that $B_y = K$. Since, moreover, B is finite over A , X_K is a finite set, so $B = A \otimes_{\kappa} K$ is the product of finitely many copies of K , and this implies that A is a finite étale κ -algebra. One thus obtains that, if κ is a field, every profinite κ -algebra A étale over κ is the product of finite separable extensions K_i of κ , equipped with the product topology, so the formal variety $\mathrm{Spf}(A)$ is the direct sum of the $\mathrm{Spf}(K_i) = \mathrm{Spec}(K_i)$, and one deduces that $\mathbf{Vaf}_{/\kappa}^{\text{ét}}$ is identified with the category of étale κ -schemes. The foregoing shows implication (ii) $\Rightarrow$ (iv) (since $\mathcal{O}_{X, x} \widehat{\otimes}_k \kappa(s)$ is formally étale over $\kappa(s)$), and implies part (b) of 1.6.E. Indeed, let k again be an arbitrary pseudocompact ring. By the foregoing, $\mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$ is identified with the category of étale schemes over $S_{\kappa(k)}$. Let us show that $X \longmapsto X_{\kappa}$ induces an equivalence $\mathbf{Vaf}_{/k}^{\text{ét}} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa(k)}^{\text{ét}}$. In view of 1.6.D, it suffices to show that, for every $s \in \mathrm{Spf}(k)$ and every finite separable extension K of $\kappa(s)$, there exists an étale k_s -algebra A such that $A \widehat{\otimes}_k \kappa(s) \simeq K$. Let ξ be a primitive element of the extension $K/\kappa(s)$, n its degree, and $F \in k_s[T]$ a monic polynomial of degree n whose image $\overline{F}$ in $\kappa(s)[T]$ is the minimal polynomial of ξ . Since $\overline{F}'$ is invertible in $\kappa(s)[T]/(\overline{F})$, it follows from Nakayama's lemma that F' is invertible in $k_s[T]/(F)$, so F is a separable polynomial,

and hence, by 1.6.A(ii), $A = k_s[T]/(F)$ is an étale k_s -algebra such that $A \hat{\otimes}_k \kappa(s) \simeq K$. One thus obtains that every local étale profinite k_s -algebra is free of finite rank over k_s (hence a fortiori topologically free over k), and therefore every étale profinite k -algebra is topologically free over k . Moreover, condition (v) implies condition (d) of 1.6.B, hence implies (i). One has therefore obtained that (i), (iii), and (v) are equivalent, and imply (ii), which implies (iv). Finally, let A be a profinite k -algebra satisfying (iv); let us show that A is formally étale over k . For this one can suppose that A and k are local; denote by κ the residue field of k ; by hypothesis, $K = A \hat{\otimes}_k \kappa$ is a finite separable extension of κ , say of degree n . By what has been seen above (and taking Lemma 1.6.D into account), there then exist a k -algebra B , free of rank n , formally étale over k , and a continuous morphism $\phi : B \rightarrow A$ such that $\phi \hat{\otimes}_k \kappa$ is an isomorphism. Since A is topologically flat over k , this implies $\text{Coker}(\phi) \hat{\otimes}_k \kappa = 0$ and also $\text{Ker}(\phi) \hat{\otimes}_k \kappa = 0$; by Nakayama's lemma 0.3, one therefore has $\text{Coker}(\phi) = 0 = \text{Ker}(\phi)$, so ϕ is an isomorphism (cf. the proof of 0.2.B). This completes the proof of 1.6.E. $\square$

⁽⁷³⁾Let X be a formal variety over k . For every $x \in X$ projecting to s on $\text{Spf}(k)$, the residue field $\kappa(x)$ is a finite extension of $\kappa(s)$, and one denotes by $\kappa_e(x)$ the separable closure of $\kappa(s)$ in $\kappa(x)$.

Proposition 1.6.F.

- (i) *The inclusion of $\mathbf{Vaf}_{/k}^{\text{ét}}$ in $\mathbf{Vaf}/k$ has a left adjoint $X \mapsto X_e$: the variety X_e has the same points as X ; for every $x \in X$ projecting to s on $\text{Spf}(k)$, let ξ be a primitive element of $\kappa_e(x)$, n its degree, u an arbitrary lifting of ξ in $\mathcal{O}_{X,x}$, and $F \in k[T]$ monic of degree n annihilating u ; put $\mathcal{O}_{X_e,x} = k[T]/(F)$. Then for every $Y \in \text{Ob } \mathbf{Vaf}_{/k}^{\text{ét}}$, the canonical maps below are bijective:*

$$\begin{array}{ccc} \text{Hom}_{\mathbf{Vaf}/k}(X, Y) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{Vaf}/\kappa(k)}(X_\kappa, Y_\kappa) \\ & & \downarrow \sim \\ \text{Hom}_{\mathbf{Vaf}/k}(X_e, Y) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{Vaf}/\kappa(k)}((X_e)_\kappa, Y_\kappa), \end{array}$$

the vertical arrow being induced by the inclusions $\kappa_e(x) \hookrightarrow \kappa(x)$ for every $x \in X$. This defines, in particular, a morphism $p : X \rightarrow X_e$, and every morphism of formal k -varieties $\phi : X \rightarrow Y$, with Y étale over k , factors uniquely through p .

- (ii) *The functor $X \mapsto X_e$ commutes with finite products.*

Indeed, (i) follows from 1.6.E(b) and 1.6.D. Let us prove (ii). In view of the equivalence of categories 1.6.E(b), one can suppose that k is a field. In this case, one easily sees that if X is a semilocal formal k -variety, i.e. whose affine algebra A is semilocal, the affine algebra of X_e is the largest subalgebra of A which is étale over k ; denote it by A_e . One is thus reduced to seeing that if K, L are two extensions of k of finite degree, the inclusion $K_e \otimes L_e \subset (K \otimes L)_e$ is an equality. Let p be the characteristic exponent of k (i.e. $p = 1$ if $\text{char}(k) = 0$ and $p = \text{char}(k)$ otherwise); then for every $x \in K \otimes L$, there exists $n \in \mathbf{N}^*$ such that $x^{p^n} \in K_e \otimes L_e$, so every subalgebra B of $K \otimes L$ is radicial over $K_e \otimes L_e$, and it follows that $K_e \otimes L_e = (K \otimes L)_e$. Observe, retaining the preceding notation, that $\mathcal{O}_{X_e,x}$ is not necessarily a subalgebra of $\mathcal{O}_{X,x}$, but this is the case when X is topologically flat over k , by the following proposition.

1.6.1. — Let Y be an étale formal k -variety and $f : X \rightarrow Y$ a morphism in $\mathbf{Vaf}/k$. One then has the cartesian square below, where Γ_f is the graph morphism $X \rightarrow X \times Y$ with

⁽⁷³⁾**N.D.E.** In what follows, we have used the preceding additions to detail the construction of the functor $X \mapsto X_e$ and show that it commutes with finite products (this is used in 2.5.1).

components id_X and f :

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times Y \\ f \downarrow & & \downarrow f \boxtimes \text{id}_Y \\ Y & \xrightarrow{\Delta_Y} & Y \times Y. \end{array}$$

It follows that Γ_f is a local isomorphism, hence that $f = \text{pr}_Y \circ \Gamma_f$ is topologically flat if pr_Y is, for example if X is topologically flat over k . Conversely, since $Y \rightarrow k$ is topologically flat, $X \rightarrow k$ will also be so if f is (cf. 1.3.3). Taking, in particular, for f the canonical morphism $p : X \rightarrow X_e$ of 1.6, one obtains:

Proposition. *Let X be a formal variety over k . The morphism $X \rightarrow X_e$ is topologically flat if and only if X is topologically flat over k .*

Remark 1.6.2. ⁽⁷⁴⁾ When k is a perfect field, the functor $X \mapsto X_e$ is also right adjoint to the inclusion $\mathbf{Vaf}_{/k}^{\text{ét}} \hookrightarrow \mathbf{Vaf}/k$. Indeed, in this case one has $\mathcal{O}_{X_e, x} = \kappa(x)$ for every $x \in X$, and the canonical projections $\mathcal{O}_{X, x} \rightarrow \kappa(x)$ define a morphism $s : X_e \rightarrow X$ which is a section of $p : X \rightarrow X_e$. For every morphism $f : X \rightarrow Y$, the diagrams below commute:

$$\begin{array}{ccc} \mathcal{O}_{Y, f(x)} & \longrightarrow & \mathcal{O}_{X, x} \\ \downarrow & & \downarrow \\ \kappa(f(x)) & \longrightarrow & \kappa(x) \end{array} \quad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ s_X \uparrow & & \uparrow s_Y \\ X_e & \xrightarrow{f_e} & Y_e \end{array}$$

so s is functorial in X , and it follows that $X \mapsto X_e$ is right adjoint to the inclusion $\mathbf{Vaf}_{/k}^{\text{ét}} \hookrightarrow \mathbf{Vaf}/k$. Thus, being a right adjoint, $X \mapsto X_e$ commutes with projective limits when they exist in $\mathbf{Vaf}_{/k}^{\text{ét}}$, ⁽⁷⁵⁾ hence in particular with finite products. (This can also be verified directly: for every formal k -variety X with affine algebra A , the affine algebra of X_e is the quotient of A by its radical $\mathfrak{r}(A)$, and since k is perfect, the quotient of $\mathcal{O}_{X, x} \hat{\otimes} \mathcal{O}_{Y, y}$ by its radical is the algebra $\kappa(x) \otimes_k \kappa(y)$, since the latter is semisimple.)

2. Generalities on formal groups

2.1. — Let k be a pseudocompact ring and G a formal k -group, that is, a group in the category $\mathbf{Vaf}/k$ of formal varieties over k . Let A be the affine algebra of G . The composition law of G evidently defines a *diagonal morphism*, i.e. a homomorphism of profinite k -algebras $\Delta_A : A \rightarrow A \hat{\otimes}_k A$; this homomorphism satisfies the following conditions:

(i) The diagram

$$\begin{array}{ccc} A & \xrightarrow{\Delta_A} & A \hat{\otimes}_k A \\ \Delta_A \downarrow & & \downarrow \Delta_A \hat{\otimes} \text{id}_A \\ A \hat{\otimes}_k A & \xrightarrow{\text{id}_A \hat{\otimes} \Delta_A} & A \hat{\otimes}_k A \hat{\otimes}_k A \end{array}$$

commutes.

(ii) There exists a (necessarily unique) *augmentation*, that is, a homomorphism of profinite k -algebras $\varepsilon_A : A \rightarrow k$ such that the composite maps

$$\begin{aligned} A &\xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\varepsilon_A \hat{\otimes} \text{id}_A} k \hat{\otimes}_k A \simeq A \\ \text{and } A &\xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\text{id}_A \hat{\otimes} \varepsilon_A} A \hat{\otimes}_k k \simeq A \end{aligned}$$

⁽⁷⁴⁾**N.D.E.** We have inserted this remark here; in the original it appeared in 2.5.2.

⁽⁷⁵⁾**N.D.E.** This is the case for finite projective limits.

- are the identity maps of A .
- (iii) There exists a (necessarily unique) *antipode*, that is, a homomorphism of profinite k -algebras $c_A : A \rightarrow A$ such that the composite map

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{c_A \hat{\otimes} \text{id}_A} A \hat{\otimes}_k A \xrightarrow{m_A} A$$

equals $\eta_A \circ \varepsilon_A$, if m_A denotes the linear map which sends $a \hat{\otimes} b$ to ab , and η_A the map $\lambda \mapsto \lambda 1_A$ from k to A .

Conversely, the data $(\Delta_A, \varepsilon_A, c_A)$ satisfying (i)–(iii) equip G with a formal k -group structure.⁽⁷⁶⁾ Explicitly, for every profinite k -algebra B , the set $\text{Hom}_c(A, B)$ of continuous morphisms of k -modules $\phi : A \rightarrow B$ is equipped with a group structure, functorial in B , defined by

$$\phi \cdot \phi' = m_B \circ (\phi \hat{\otimes} \phi') \circ \Delta_A,$$

the identity element being $\eta_B \circ \varepsilon_A$ (where m_B is multiplication on B and η_B the map $\lambda \mapsto \lambda 1_B$ from k to B), and $\phi \circ c_A$ being the inverse of ϕ ; and the set $\text{Hom}_{\mathbf{Alf}/k}(A, B)$ of continuous morphisms of k -algebras $A \rightarrow B$ is a subgroup of it (since the algebra B is commutative).

Definition. A morphism of formal k -groups $\theta : K \rightarrow G$ is, by definition, a morphism of formal k -varieties which respects the group structures. If B (resp. A) is the affine algebra of K (resp. G), and $f : A \rightarrow B$ is the morphism corresponding to θ , this is equivalent to saying that f is compatible with comultiplications, i.e.

$$(f \hat{\otimes} f) \circ \Delta_A = \Delta_B \circ f$$

(the conditions $\varepsilon_B \circ f = \varepsilon_A$ and $c_B \circ f = f \circ c_A$ then being automatically satisfied). Denote by $\mathbf{Grf}/k$ the category of formal k -groups.

Notation. In what follows, we shall call the ideal $I_A = \text{Ker}(\varepsilon_A)$ the *augmentation ideal* of A , and denote by $\omega_{G/k}$ the pseudocompact k -module $I_A/\overline{I_A^2}$, that is, the quotient of I_A by the closed ideal generated by the products xy , for $x, y \in I_A$.

2.2. — Let $\mathbf{H}$ be a group in the category of coalgebras over $\mathbf{O}_k$, i.e. for every object C of $\mathbf{Alf}/k$, $\mathbf{H}(C)$ is equipped with a group-coalgebra structure over C (cf. VII A 3.2; following Manin, we shall say *bialgebra*⁽⁷⁷⁾ instead of group-coalgebra); moreover, if $\varphi : C \rightarrow D$ is a morphism of $\mathbf{Alf}/k$, the map $D \otimes_C \mathbf{H}(C) \rightarrow \mathbf{H}(D)$ is a homomorphism of D -bialgebras.

Definition. We shall summarize the above properties by saying that $\mathbf{H}$ is a *bialgebra over* $\mathbf{O}_k$.

It is clear that the functor $\mathbf{H} \mapsto \text{Spf}^*(\mathbf{H})$ of 1.3.5 commutes with finite products. It therefore transforms a bialgebra over $\mathbf{O}_k$ into a *group-valued k -functor*, that is, a (covariant) functor from $\mathbf{Alf}/k$ to the category of groups. Indeed, for every k -algebra C of finite length, the elements of

$$\text{Spf}^*(\mathbf{H})(C) = \text{Spf}^*(\mathbf{H}(C)) = \{x \in \mathbf{H}(C) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$$

form a group for multiplication in the algebra $\mathbf{H}(C)$ (cf. VII A 3.2.2). Note, moreover, that the condition $\Delta(x) = x \otimes x$ implies the equality $\varepsilon(x) = \varepsilon(x)^2$, hence also $\varepsilon(x) = 1$ if C is local and $x \neq 0$.⁽⁷⁸⁾

⁽⁷⁶⁾**N.D.E.** One also says that A is a cogroup in the category of profinite k -algebras. Moreover, we have detailed what follows; in particular, we have made explicit what a morphism of formal groups $K \rightarrow G$ is; cf. Proposition 2.3.1.

⁽⁷⁷⁾**N.D.E.** One also says “bigèbre” (cf. [BAlg], III § 11.4); recall (cf. VII A, 3.1, N.D.E. (26)) that all the “bialgebras” considered here are assumed cocommutative and equipped with an antipode, i.e. they are in fact cocommutative Hopf algebras.

⁽⁷⁸⁾**N.D.E.** Since $\Delta(x) = x \otimes x$, one has $x = \varepsilon(x)x$, so $\varepsilon(x) = 0$ can occur only if $x = 0$.

2.2.1. — A bialgebra $\mathbf{H}$ over $\mathbf{O}_k$ is said to be *flat* if the underlying module is flat (cf. 1.2.1).⁽⁷⁹⁾ If $\mathbf{H}$ is flat, then, by 1.3.5, $A = \Gamma^*(\mathbf{H})$ is a topologically flat profinite k -algebra, and $\mathrm{Spf}^*(\mathbf{H})$ is isomorphic, as a functor from $(\mathbf{A}lf/k)^0 = \mathbf{V}af/k$ to $(\mathbf{Ens})$, to the functor

$$\mathrm{Spf}(A) : C \longmapsto \mathrm{Hom}_{\mathbf{V}af/k}(\mathrm{Spf}(C), \mathrm{Spf}(A)).$$

The group structure of $\mathrm{Spf}^*(\mathbf{H})$ therefore equips $\mathcal{G}(\mathbf{H}) = \mathrm{Spf}(A)$ with a formal group structure, explicitly described as follows. For every object C of $\mathbf{A}lf/k$, since the C -module underlying $\mathbf{H}(C)$ is projective, one deduces from Lemma 1.2.3.A by induction on n natural isomorphisms

$$\begin{aligned} (\mathbf{H}(C)^*)^{\widehat{\otimes}(n+1)} &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^*)^{\widehat{\otimes}n}) \\ &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^{\otimes n})^*) \simeq (\mathbf{H}(C)^{\otimes(n+1)})^*. \end{aligned}$$

From this one deduces (for $n = 1, 2$) that the C -algebra structure of $\mathbf{H}(C)$ equips $\mathbf{H}(C)^*$ with a diagonal map satisfying conditions 2.1(i)–(iii), all this functorially in C . Consequently, $A = \Gamma^*(\mathbf{H}) = \varprojlim \mathbf{H}(k/l)^*$ is equipped with a cogroup structure in $\mathbf{A}lp/k$, which defines the announced formal group structure on $\mathcal{G}(\mathbf{H})$. Conversely, let G be a topologically flat formal k -group with affine algebra A , and denote by $\mathbf{H}(G)$ the $\mathbf{O}_k$ -coalgebra $\mathbf{V}_k^f(A)$ (cf. 1.2.3). The diagonal morphism $\Delta_A : A \longrightarrow A \widehat{\otimes}_k A$ then induces, for every k -algebra C of finite length, a C -linear map

$$\mathbf{V}_k^f(A)(C) \otimes_C \mathbf{V}_k^f(A)(C) \longrightarrow \mathbf{V}_k^f(A)(C)$$

which makes the coalgebra $\mathbf{V}_k^f(A)(C)$ a C -bialgebra. One says that $\mathbf{H}(G)$ is *the covariant bialgebra of the formal group G* .⁽⁸⁰⁾ Thus, by Proposition 1.3.5.D:

Proposition.

- (i) *The functors $G \longmapsto \mathbf{H}(G)$ and $\mathbf{H} \longmapsto \mathcal{G}(\mathbf{H})$ define an equivalence between the category of topologically flat formal k -groups and that of flat $\mathbf{O}_k$ -bialgebras.*⁽⁸¹⁾
- (ii) *This equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of pseudocompact rings, then $\mathbf{H}(G \widehat{\otimes}_k K) = \mathbf{H}(G) \otimes_k K$ and $\mathcal{G}(\mathbf{H} \otimes_k K) = \mathcal{G}(\mathbf{H}) \widehat{\otimes}_k K$.*

When k is an artinian ring and G a topologically flat formal k -group, the functor $\mathbf{H}(G)$ is evidently determined by its value $H(G) = \mathbf{H}(G)(k)$ at k . One also says that $H(G)$ is the (covariant) bialgebra of G .⁽⁸²⁾ Consequently, denoting by $\mathcal{H}_k$ the category of k -Hopf algebras and by $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$ the full subcategory consisting of k -Hopf algebras flat over k and cocommutative, one therefore obtains:

Corollary. *Let k be an artinian ring.*

- (i) *The functors $G \longmapsto H(G)$ and $H \longmapsto \mathcal{G}(H) = \mathrm{Spf}^* H(G)$ define an equivalence between the category of topologically flat formal k -groups and $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$.*
- (ii) *This equivalence “commutes with base change”: if $k \longrightarrow K$ is a morphism of artinian rings, then $H(G \widehat{\otimes}_k K) = H(G) \otimes_k K$ and $\mathcal{G}(H \otimes_k K) = \mathcal{G}(H) \widehat{\otimes}_k K$.*

Moreover, denote also by $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{com}}$ the full subcategory of $\mathcal{H}_k$ consisting of k -Hopf algebras flat over k and commutative, and recall that the functor $K \longmapsto \mathcal{O}(K)$ is an anti-equivalence from the category of affine group k -schemes to $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{com}}$.

2.2.2. — For simplicity suppose that k is artinian, and let G be a topologically flat formal k -group.⁽⁸³⁾ Then G is commutative if and only if its affine algebra $\mathcal{A}(G)$ has a cocommutative

⁽⁷⁹⁾**N.D.E.** We have detailed the original in what follows.

⁽⁸⁰⁾**N.D.E.** We have added the adjective “covariant” to make clear that the functor $G \longmapsto \mathbf{H}(G)$ is covariant; this terminology is used in [Di73], I § 2.14.

⁽⁸¹⁾**N.D.E.** Recall that in this exposé “bialgebra” means “cocommutative Hopf algebra”.

⁽⁸²⁾**N.D.E.** We have introduced here the notation $\mathcal{H}_{\mathrm{plat}/k}^{\mathrm{cocom}}$, which will be useful below.

⁽⁸³⁾**N.D.E.** We have detailed the original in what follows, in order to introduce the notation $D'(G)$ and $D(K)$; cf. [Ca62], § 14.

comultiplication, which is equivalent to saying that the bialgebra $H(G)$ has commutative multiplication. In this case $H(G)$ is a commutative and cocommutative Hopf algebra flat over k , so if one puts $D'(G) = \operatorname{Spec} H(G)$, $D'(G)$ is a *commutative group k -scheme, affine and flat* over k . Conversely, if T is such a group k -scheme, its affine algebra $\mathcal{O}(T)$ is a commutative group in the category of flat cocommutative k -coalgebras, and hence, by 1.3.5.C, one obtains a topologically flat formal k -group $D(T)$ by putting

$$D(T) = \operatorname{Spf}^* \mathcal{O}(T) = \operatorname{Spf}(\mathcal{A}), \quad \text{where } \mathcal{A} = \mathcal{O}(T)^*.$$

Since, by 1.3.5.D, one has canonical isomorphisms $G = \operatorname{Spf}^* H(G)$ and $H(D(T)) = \mathcal{O}(T)$, one obtains canonical isomorphisms

$$\begin{aligned} D(D'(G)) &= \operatorname{Spf}^* \mathcal{O}(D'(G)) = \operatorname{Spf}^* H(G) = G, \\ D'(D(T)) &= \operatorname{Spec} H(D(T)) = \operatorname{Spec} \mathcal{O}(T) = T. \end{aligned}$$

Moreover, denoting by $k\text{-Gr.}$ the category of group k -schemes, by Corollary 2.2.1 one has functorial isomorphisms

$$\operatorname{Hom}_{\mathbf{Grf}/k}(G, D(T)) \simeq \operatorname{Hom}_{\mathcal{H}_k}(H(G), \mathcal{O}(T)) \simeq \operatorname{Hom}_{k\text{-Gr.}}(T, D'(G)).$$

One therefore obtains the following:

Proposition. (*Cartier duality*). *Let k be an artinian ring.*

- (i) *The functors $G \mapsto D'(G) = \operatorname{Spec} H(G)$ and $T \mapsto D(T) = \operatorname{Spf}^* \mathcal{O}(T)$ induce an anti-equivalence between the category $\mathcal{FC}_k$ of topologically flat commutative formal k -groups and the category $\mathcal{AC}_k$ of commutative group k -schemes, affine and flat, i.e. G and $D'(G)$ (resp. T and $D(T)$) are related by the equalities⁽⁸⁴⁾*

$$H(G) = \mathcal{O}(D'(G)) \quad \text{and} \quad \mathcal{O}(T) = H(D(T)).$$

- (ii) *This anti-equivalence “commutes with base change”: if $k \rightarrow K$ is a morphism of artinian rings, then $D'(G \hat{\otimes}_k K) = D'(G) \otimes_k K$ and $D(T \otimes_k K) = D(T) \hat{\otimes}_k K$.*
- (iii) *In particular, if k is a field, one obtains an anti-equivalence between the category of commutative formal k -groups and that of affine commutative group k -schemes which commutes with extension of the base field.*

2.3. — Now consider an arbitrary⁽⁸⁵⁾ formal k -group G with affine algebra A . Continue to denote by $\mathbf{H}(G)$ the $\mathbf{O}_k$ -module $\mathbf{V}_k^f(A)$ dual to A , and denote by φ_G the functorial homomorphism

$$\varphi_G : \mathbf{H}(G) \otimes_k \mathbf{H}(G) \rightarrow \mathbf{H}(G \times G)$$

induced by the natural map (0.3.6), for every object C of $\mathbf{Alf}/k$:

$$(A \hat{\otimes}_k C)^\dagger \otimes_C (A \hat{\otimes}_k C)^\dagger \rightarrow ((A \hat{\otimes}_k C) \hat{\otimes}_C (A \hat{\otimes}_k C))^\dagger.$$

If $m : G \times G \rightarrow G$ is multiplication on G , the composite map

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G) \xrightarrow{\varphi_G} \mathbf{H}(G \times G) \xrightarrow{\mathbf{H}(m)} \mathbf{H}(G)$$

makes $\mathbf{H}(G)$ an algebra over $\mathbf{O}_k$; for every $C \in \mathbf{Alf}/k$, the identity element of $\mathbf{H}(G)(C) = (A \hat{\otimes}_k C)^\dagger$ is the augmentation of $A \hat{\otimes}_k C$ (cf. 2.1).⁽⁸⁶⁾ If G is not topologically flat over k , φ_G is not necessarily an isomorphism, and thus the morphism $\delta_G : \mathbf{H}(G) \rightarrow \mathbf{H}(G \times G)$ induced by the diagonal morphism “ $x \mapsto (x, x)$ ” from G to $G \times G$ does not necessarily factor

⁽⁸⁴⁾**N.D.E.** If one denotes by $\mathcal{FC}_k^f$ (resp. $\mathcal{AC}_k^f$) the full subcategory of $\mathcal{FC}_k$ (resp. $\mathcal{AC}_k$) consisting of objects G (resp. T) such that $H(G)$ (resp. $\mathcal{O}(T)$) is a finite (and hence finite locally free) k -module, then $\mathcal{FC}_k^f$ and $\mathcal{AC}_k^f$ both have the same objects as the category $\mathcal{C}$ of commutative and cocommutative Hopf k -algebras finite and flat over k , the correspondence $G \mapsto H(G)$ (resp. $T \mapsto \mathcal{O}(T)$) being covariant (resp. contravariant), and in this way one recovers the “Cartier duality” of the category $\mathcal{AC}_k$ already seen in VII A, 3.3.1.

⁽⁸⁵⁾**N.D.E.** i.e. not necessarily topologically flat over k .

⁽⁸⁶⁾**N.D.E.** We have changed the order of the sentences in what follows.

through $\mathbf{H}(G) \otimes_k \mathbf{H}(G)$, i.e. $\mathbf{H}(G)$ is not necessarily an $\mathbf{O}_k$ -bialgebra. For this reason, in the general case we shall simply say that $\mathbf{H}(G)$ is “the covariant algebra” of the formal group G . Of course, when G is topologically flat over k , φ_G is an isomorphism and one recovers the $\mathbf{O}_k$ -bialgebra structure on $\mathbf{H}(G)$ defined in 2.2.1.

2.3.1. Proposition. *Let K, G be two formal k -groups with affine algebras B, A . Suppose that K is topologically flat over k . There then exists a canonical bijection from $\mathrm{Hom}_{\mathbf{Grf}/k}(K, G)$ to the set of homomorphisms of unital $\mathbf{O}_k$ -algebras $h : \mathbf{H}(K) \rightarrow \mathbf{H}(G)$ such that the diagram*

$$(*) \quad \begin{array}{ccc} \mathbf{H}(K) \otimes_k \mathbf{H}(K) & \xrightarrow{h \otimes h} & \mathbf{H}(G) \otimes_k \mathbf{H}(G) \\ \uparrow \Delta_{\mathbf{H}(K)} & & \searrow \varphi_G \\ \mathbf{H}(K) & \xrightarrow{h} & \mathbf{H}(G) \end{array} \quad \begin{array}{c} \nearrow \delta_G \\ \mathbf{H}(G \times G) \end{array}$$

commutes.

Since K is topologically flat, $\mathbf{H}(K)$ is equipped with a bialgebra structure (cf. 2.2), and $\Delta_{\mathbf{H}(K)}$ is its diagonal morphism; in other words, with the notation of 2.3, one has $\Delta_{\mathbf{H}(K)} = \varphi_K^{-1} \circ \delta_K$. When G is also topologically flat over k , our proposition follows from the equivalence of categories established in 2.2.1. In the general case one can suppose that k is artinian and reason with the algebras $H(K) = B^\dagger$ and $H(G) = A^\dagger$. Let $\mathrm{Hom}_c(A, B)$ be the set of continuous k -linear maps from A to B , and $\mathrm{Hom}_k(B^\dagger, A^\dagger)$ the set of k -linear maps from $B^\dagger$ to $A^\dagger$.⁽⁸⁷⁾ By 0.3.6.A one knows that, if M, P are pseudocompact k -modules and P is projective, the canonical map

$$\mathrm{Hom}_c(M, P) \rightarrow \mathrm{Hom}_k(P^\dagger, M^\dagger), \quad f \mapsto {}^t f$$

(where ${}^t f$ denotes the transpose of f) is bijective. (One will apply this to $M = A \hat{\otimes} A$ and $P = B$, or to $M = A$ and $P = B \hat{\otimes} B$.) Let $f \in \mathrm{Hom}_c(A, B)$. Consider the diagrams below, in which the squares (0) commute and the two unnamed vertical arrows are ${}^t(f \hat{\otimes} f)$.

$$\begin{array}{ccccccc} A \hat{\otimes} A & \xrightarrow{m_A} & A & \xrightarrow{\Delta_A} & A \hat{\otimes} A \\ f \hat{\otimes} f \downarrow & (1) & \downarrow f & (2) & \downarrow f \hat{\otimes} f \\ B \hat{\otimes} B & \xrightarrow{m_B} & B & \xrightarrow{\Delta_B} & B \hat{\otimes} B \end{array}$$

$$\begin{array}{ccccccc} A^\dagger \otimes A^\dagger & \xrightarrow{\varphi_G} & (A \hat{\otimes} A)^\dagger & \xleftarrow{\delta_G = m_A^t} & A^\dagger & \xleftarrow{\Delta_A^t} & (A \hat{\otimes} A)^\dagger \xleftarrow{\varphi_G} A^\dagger \otimes A^\dagger \\ {}^{tf} \otimes {}^{tf} \uparrow & (0) & \uparrow & (1') & {}^{tf} \uparrow & (2') & \uparrow (0) \uparrow {}^{tf} \otimes {}^{tf} \\ B^\dagger \otimes B^\dagger & = & (B \hat{\otimes} B)^\dagger & \xleftarrow{\delta_K = m_B^t} & B^\dagger & \xleftarrow{\Delta_B^t} & (B \hat{\otimes} B)^\dagger = B^\dagger \otimes B^\dagger \end{array}$$

If $f : A \rightarrow B$ corresponds to a morphism of formal groups $K \rightarrow G$, the squares (1) and (2) commute, and $\varepsilon_B \circ f = \varepsilon_A$; consequently, the squares (1') and (2') commute and ${}^t f$ sends the identity of $B^\dagger = H(K)$ to that of $A^\dagger = H(G)$, i.e. ${}^t f$ is a morphism of unital k -algebras $H(K) \rightarrow H(G)$ such that diagram (*) of the proposition commutes. Conversely, if ${}^t f$ satisfies these conditions, $\varepsilon_B \circ f = \varepsilon_A$ and the squares (1') and (2') commute. Since, for $M = A \hat{\otimes} A$ and $P = B$ (resp. $M = A$ and $P = B \hat{\otimes} B$), the map $g \mapsto {}^t g$ is injective, one deduces that the squares (1) and (2) commute, so f is compatible with the multiplications and diagonal morphisms of A and B . It remains to see that $f(1_A) = 1_B$. Now it follows

⁽⁸⁷⁾ **N.D.E.** We have detailed the remainder of the proof.

from the foregoing that $\varepsilon_B f(1) = 1$, $\Delta_B f(1) = f(1) \hat{\otimes} f(1)$, and $f(1) \cdot f(1) = f(1)$. The first two conditions imply, by 2.1(iii), that $f(1)$ has $c_B f(1)$ as its inverse in B ; consequently, $f(1) \cdot f(1) = f(1)$ implies $f(1) = 1$. Thus $f : A \rightarrow B$ is a morphism in $\mathbf{Alp}/k$ compatible with the comultiplications of A and B .

2.3.2. — For simplicity, now suppose that the ring k is artinian. When G is topologically flat over k , the algebra $H(G) = \mathbf{H}(G)(k)$ can be characterized by a universal property (due to Cartier). Recall (cf. 1.2.1) that if U is a k -module, $\mathbf{W}(U)$ denotes the functor which associates to every k -algebra C of finite length the C -module $U \otimes_k C$.⁽⁸⁸⁾ If U is a k -algebra (associative, with identity), the same holds for $U \otimes_k C$; we shall denote by $\mathbf{W}(U)^\times$ the group-valued k -functor which associates to every $C \in \text{Ob } \mathbf{Alf}/k$ the multiplicative group of invertible elements of the algebra $U \otimes_k C$:

$$\mathbf{W}(U)^\times(C) = (U \otimes_k C)^\times.$$

Moreover, identify G with the group-valued k -functor $C \mapsto \text{Hom}_{\mathbf{Vaf}/k}(\text{Spf}(C), G)$ and denote by $\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times)$ the set of homomorphisms of group-valued k -functors from G to $\mathbf{W}(U)^\times$. One has the following

Proposition. *Let k be an artinian ring. For every formal group G topologically flat over k and every k -algebra U , there is a canonical isomorphism*

$$\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times) \xrightarrow{\sim} \text{Hom}_{k\text{-Alg.}}(H(G), U).$$

Denote by A the affine algebra of G ; by hypothesis it is a projective object of $\mathbf{PC}(k)$, and $H(G) = A^\dagger$. For every object P of $\mathbf{PC}(k)$, denote by $U \hat{\otimes}_k P$ the projective limit of the k -modules $U \otimes_k (P/N)$, where N runs through the open submodules of P . One has linear maps

$$U \otimes_k (P/N) \rightarrow \text{Hom}_k((P/N)^*, U)$$

which send $u \otimes x$ to the k -linear map $f \mapsto f(x)u$ and form a filtered projective system. One therefore obtains, by passage to the projective limit, a morphism

$$(1) \quad U \hat{\otimes}_k P \xrightarrow{\psi_P} \text{Hom}_k(P^\dagger, U).$$

When $P = k$, ψ_k is evidently an isomorphism; moreover, both sides of (1), considered as functors in P , commute with infinite products (every product being a filtered projective limit of finite products). One therefore obtains that (1) is an isomorphism when P is a product of copies of k , and then when P is a projective object of $\mathbf{PC}(k)$ (both sides of (1) commuting with finite direct sums). Now denote by $\text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ the set of morphisms of k -functors from G to $\mathbf{W}(U)$. Since $G = \text{Spf}(A) = \varprojlim \text{Spf}(A/\mathfrak{l})$, where $\mathfrak{l}$ runs through the open ideals of A , one has canonical isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) &= \varprojlim \text{Hom}_{\mathcal{F}}(\text{Spf}(A/\mathfrak{l}), \mathbf{W}(U)) \\ &= \varprojlim U \otimes_k (A/\mathfrak{l}) = U \hat{\otimes}_k A. \end{aligned}$$

By the foregoing, one therefore obtains a canonical isomorphism

$$(2) \quad \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) = U \hat{\otimes}_k A \xrightarrow{\psi_A} \text{Hom}_k(H(G), U).$$

For every k -algebra C of finite length, multiplication makes $U \otimes_k C$ a monoid with identity, and every morphism of monoids with identity $G(C) \rightarrow U \otimes_k C$ is necessarily a morphism of groups $G(C) \rightarrow (U \otimes_k C)^\times$. Consequently, one obtains that $\text{Hom}_{k\text{-Gr.}}(G, \mathbf{W}(U)^\times)$ is the subset of $\text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ consisting of morphisms of k -functors in monoids with identity. It remains to see that these morphisms correspond to the k -linear maps from

⁽⁸⁸⁾**N.D.E.** We have added the preceding sentence, and in what follows we have written $\mathbf{W}(U)^\times$ instead of U_m . Recall, moreover (cf. 1.2.1), that one calls a covariant functor $\mathbf{Alf}/k \rightarrow (\text{Ens})$ a “ k -functor”.

$H(G)$ to U which preserve multiplication and the identity elements.⁽⁸⁹⁾ To simplify the notation, $H(G) = A^\dagger$ will be denoted by H , and we shall write $\widehat{\otimes}$ instead of $\widehat{\otimes}_k$. Denote by $\Delta_A, m_A, \varepsilon_A$ (resp. $\Delta_H, m_H, \varepsilon_H$) the comultiplication, multiplication, and augmentation of A (resp. H). Let $\theta \in \text{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$; denote by γ its image in $U \widehat{\otimes} A$, and by $\phi : H \rightarrow U$ the associated k -linear map. Then θ sends the identity section $s \in G(k)$ to an element u of U , and since s corresponds to the augmentation $\varepsilon : A \rightarrow k$, which is the identity element 1_H of H , one sees that $\theta(s) = 1_U$ if and only if $\phi(1_H) = 1_U$. Moreover, the morphism $\theta \circ m_G : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the element $(\text{id}_U \widehat{\otimes} \Delta_A)(\gamma)$, and this corresponds, by duality, to the map $\phi \circ m_H : H \otimes H \rightarrow U$. On the other hand, the morphism $\theta \circ \text{pr}_1 : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the element $\gamma \widehat{\otimes} 1_A$ of $U \widehat{\otimes} A \widehat{\otimes} A$, which corresponds by duality to $\phi \circ (\text{id}_H \otimes \varepsilon) : H \otimes H \rightarrow U$. Similarly, $\theta \circ \text{pr}_2$ corresponds to the element $\tau(\gamma \widehat{\otimes} 1_A)$ of $U \widehat{\otimes} A \widehat{\otimes} A$ (where $\tau(u \widehat{\otimes} a \widehat{\otimes} b) = u \widehat{\otimes} b \widehat{\otimes} a$), which corresponds by duality to $\phi \circ (\varepsilon \otimes \text{id}_H) : H \otimes H \rightarrow U$. Finally, the multiplication map $\mu = m_U \widehat{\otimes} A \widehat{\otimes} A$ below,

$$\begin{aligned}
 & (U \widehat{\otimes} A \widehat{\otimes} A) \times (U \widehat{\otimes} A \widehat{\otimes} A), \\
 & (u \widehat{\otimes} a_1 \widehat{\otimes} a_2, u' \widehat{\otimes} a_3 \widehat{\otimes} a_4) \mapsto uu' \widehat{\otimes} a_1 a_3 \widehat{\otimes} a_2 a_4,
 \end{aligned}$$

can be viewed as the composite of the endomorphism σ_{23} of $(U \otimes U) \widehat{\otimes} A \widehat{\otimes} A$ which “interchanges the factors a_2 and a_3 ”, and the map

$$(U \otimes U) \widehat{\otimes} A \widehat{\otimes} A \xrightarrow{m_U \widehat{\otimes} m_A \widehat{\otimes} m_A} U \widehat{\otimes} A \widehat{\otimes} A.$$

One deduces that the map $\mu \circ (\phi \circ \text{pr}_1, \phi \circ \text{pr}_2) : G \times G \rightarrow \mathbf{W}(U)$ corresponds to the composite map β below:

$$\begin{array}{ccc}
 & H \otimes H & \xrightarrow{\beta} U \\
 \text{id}_H \circ \sigma_{23} \circ \text{id}_H \swarrow & & \uparrow m_U \\
 H \otimes H \otimes H & & U \otimes U \\
 \searrow (\text{id}_H \otimes \varepsilon_H) \otimes (\varepsilon_H \otimes \text{id}_H) & \nearrow \phi \otimes \phi & \\
 & H \otimes H &
 \end{array}$$

Finally, θ is compatible with the laws of G and $\mathbf{W}(U)^\times$ if and only if $\theta \circ m_G$ equals $\mu \circ (\phi \circ \text{pr}_1, \phi \circ \text{pr}_2)$, which is equivalent, by the foregoing, to $\phi \circ m_H = \beta$. Now it is clear that $(\phi \circ m_H)(x \otimes y) = \phi(xy)$, and one easily sees that $\beta(x \otimes y) = \phi(x)\phi(y)$.

2.4. — Let us now return to any pseudocompact ring k to apply to formal groups the results of 1.4–1.5 on passage to the quotient by a topologically flat equivalence relation.⁽⁹⁰⁾ Let $u : H \rightarrow G$ be a monomorphism of formal k -groups, $\mu : G \times G \rightarrow G$ the “multiplication” morphism of G , and λ the composite morphism

$$\lambda : G \times H \xrightarrow{\text{id}_G \times u} G \times G \xrightarrow{\mu} G.$$

Since u is a monomorphism, the pair

$$G \times H \xrightleftharpoons[\lambda]{\text{pr}_1} G$$

is an equivalence pair in $\mathbf{Vaf}/k$ (cf. V, 2.b)). Recall (cf. 1.2.C) that the cokernel G/H of this pair is defined as follows. Let $\mathcal{O}(G)$ and $\mathcal{O}(H)$ be the affine algebras of G and H , $\Delta : \mathcal{O}(G) \rightarrow \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(G)$ the diagonal morphism of $\mathcal{O}(G)$, and I the kernel of the morphism $u^\sharp : \mathcal{O}(G) \rightarrow \mathcal{O}(H)$. (By Proposition 1.3 one knows that $u^\sharp$ induces an

⁽⁸⁹⁾**N.D.E.** The original indicated that “(this) follows from the functorial character of ψ_A ”. We have detailed this in what follows.

⁽⁹⁰⁾**N.D.E.** We have detailed the original in what follows.

isomorphism $\mathcal{O}(G)/I \xrightarrow{\sim} \mathcal{O}(H)$.) Then the affine algebra $\mathcal{O}(G/H)$ of G/H is the kernel of the pair of morphisms

$$\mathcal{O}(G) \xrightarrow[(\text{id} \hat{\otimes} u^\sharp) \Delta]{\tau_1} \mathcal{O}(G) \hat{\otimes}_k \mathcal{O}(H),$$

where $\tau_1(x) = x \hat{\otimes} 1$, i.e.

$$\mathcal{O}(G/H) = \{x \in \mathcal{O}(G) \mid \Delta(x) - x \hat{\otimes} 1 \in \mathcal{O}(G) \hat{\otimes} I\}.$$

If, moreover, H is *topologically flat over k* , pr_1 is topologically flat and from Theorem 1.4 one deduces the following theorem.

Theorem. *Let $u : H \rightarrow G$ be a monomorphism of formal k -groups. Suppose that H is topologically flat over k . Then the projection $p : G \rightarrow G/H$ is surjective and topologically flat, one has an isomorphism*

$$(*) \quad G \times H \xrightarrow{\sim} G \times_{G/H} G,$$

and G/H represents the quotient sheaf for the flat topology. Consequently, G/H is equipped with a canonical structure of an object with group of operators G , such that $p : G \rightarrow G/H$ is a morphism of objects with operators. If, moreover, u identifies H with a normal subgroup of G , then G/H is equipped with a canonical formal k -group structure such that $p : G \rightarrow G/H$ is a morphism of formal k -groups, and H is the kernel of p .

Indeed, the first assertion follows from 1.4; the other two from IV, Corollaries 5.2.2 and 5.2.4.

Corollary. ⁽⁹¹⁾ *Let G be a formal k -group, H a formal subgroup of G , A (resp. $A/J, B$) the affine algebra of G (resp. $H, G/H$), I_A the augmentation ideal of A , and $I_B = B \cap I_A$. Suppose that H is topologically flat over k . Then J equals $\overline{AI_B}$, the closed ideal generated by I_B .*

Indeed, the projection $B \rightarrow B/I_B$ corresponds to the “identity section” $e : \text{Spf}(k) \rightarrow G/H$ of G/H . By (*), H is identified with the fibered product $\text{Spf}(k) \times_{G/H} G$, so its affine algebra A/J is identified with $(B/I_B) \hat{\otimes}_B A \simeq A/\overline{AI_B}$.

2.4.A. — ⁽⁹²⁾ Let G, Q be topologically flat formal k -groups; suppose that there exist homomorphisms $\sigma : Q \rightarrow G$ and $\pi : G \rightarrow Q$ such that $\pi \circ \sigma = \text{id}_Q$. In particular, σ is a monomorphism, so Q is a formal subgroup of G (cf. Remark 1.3). Let $N = \text{Ker}(\pi)$ and σ' be the inclusion $N \hookrightarrow G$. Then G is the *semidirect product* of N by Q (cf. I, 2.3.8), i.e. for every $B \in \text{Ob } \mathbf{AIf}/k$, identifying $N(B)$ and $Q(B)$ with their images in $G(B)$ under σ' and σ , $N(B)$ is a normal subgroup of $G(B)$ and the map

$$(1) \quad \mu : N(B) \times Q(B) \rightarrow G(B), \quad (x, q) \mapsto xq$$

is bijective. Then the morphism of formal k -varieties

$$(2) \quad \theta : G(B) \rightarrow N(B), \quad g \mapsto g \cdot \sigma \pi(g^{-1})$$

is a retraction of σ' , the inverse of μ is the map

$$(3) \quad g \mapsto (\theta(g), \pi(g)),$$

and $\theta \circ \sigma' : N \rightarrow G/Q$ is an isomorphism of formal k -varieties. In particular, N is topologically flat over k , by 1.4(ii). Denote by α (resp. β) the map from $Q \times N$ (resp. $G \times N = N \times Q \times N$)

⁽⁹¹⁾**N.D.E.** We have made explicit this corollary, which is used, for example, in 5.2.1/5.2.3.

⁽⁹²⁾**N.D.E.** We have added paragraphs 2.4.A and 2.4.B, which will be useful in 5.1.3 and 2.9 respectively.

to N defined set-theoretically by $\alpha(q, y) = qyq^{-1}$ (resp. $\beta(x, q, y) = x\alpha(q, y)$). Then one has the commutative diagram below:

$$(4) \quad \begin{array}{ccccc} G \times G & \xrightarrow{\text{id} \times \theta} & G \times N & \xrightarrow{\theta \times \text{id}} & N \times N \\ m_G \downarrow & & \downarrow \beta & \swarrow m_N & \\ G & \xrightarrow{\theta} & N & & \end{array} .$$

This can be expressed as follows in terms of the affine algebras A, A_0, A' of G, Q, N (cf. 5.1.3 below). Let $\rho' : A \rightarrow A'$, $\rho : A \rightarrow A_0$, and $\tau : A_0 \rightarrow A$ be the homomorphisms of k -bialgebras corresponding to σ', σ, π , and let $I = \text{Ker}(\rho)$. Then, by the preceding corollary, A' is identified with $A/\overline{A\tau(J_0)}$, where J_0 denotes the augmentation ideal of A_0 . On the other hand, let B be the affine algebra of G/Q ; it is the kernel of the pair of morphisms

$$A \xrightarrow[\text{(id} \otimes \rho) \Delta_A]{\tau_1} A \hat{\otimes}_k A_0,$$

i.e. $B = \{x \in A \mid \Delta_A(x) - x \hat{\otimes} 1 \in A \hat{\otimes} I\}$. Denote by γ the continuous morphism of k -algebras $\theta^\sharp : A' \rightarrow A$; it is a section of ρ' and an isomorphism of profinite k -algebras from A' to B . Moreover, $\tau = \pi^\sharp$ identifies A_0 with a sub-bialgebra of A , which is precisely the affine algebra of the quotient $N \backslash G$. One then deduces from (1) and (3) that one has an isomorphism of profinite k -algebras

$$(*) \quad A' \hat{\otimes}_k A_0 \xrightarrow{\sim} A, \quad a' \hat{\otimes} a_0 \mapsto \gamma(a')\tau(a_0),$$

whose inverse is the map $a \mapsto (\rho' \otimes \rho)\Delta_A(a)$. Finally, identify A' with its image in A under γ , so that the projection $A \rightarrow A'$ is then $\gamma\rho'$. Denoting by $\Delta_{A'}$ the comultiplication of A' , one then deduces from (4) that $\Delta_A(A') \subset A \hat{\otimes} A'$ and that the diagram below commutes:

$$\begin{array}{ccc} A' & \xrightarrow{\Delta_A} & A \hat{\otimes} A' \\ \parallel & & \downarrow \gamma\rho' \otimes \text{id} \\ A' & \xrightarrow{\Delta_{A'}} & A' \hat{\otimes} A' \end{array}$$

(one therefore also has $\gamma\rho' \circ c_A = c_{A'}$, where c_A (resp. $c_{A'}$) is the antipode of A (resp. A')). Moreover, denoting by m_A multiplication on A , one deduces from (2) that, for every $a \in A$,

$$\gamma\rho'(a) = (m_A \circ (\text{id} \otimes \tau\rho c_A) \circ \Delta_A)(a).$$

2.4.B. — ⁽⁹²⁾For simplicity, suppose that k is artinian. Then the foregoing can be expressed more simply in terms of the covariant bialgebras of G, Q, N . Indeed, since $G = N \times Q$ as formal k -varieties, $H(G) = H(N) \otimes_k H(Q)$ as k -coalgebras. Moreover, since multiplication on G is given by

$$(x, q) \cdot (x', q') = (x\alpha(q, x'), qq'), \quad \text{where } \alpha(q, x') = qxq^{-1},$$

multiplication on $H(G)$ is given as follows: for every $x \in H(N)$ and $q \in H(Q)$,

$$(x \otimes q) \cdot (x' \otimes q') = x\phi(q, x') \otimes qq',$$

where $\phi : H(Q) \otimes_k H(N) \rightarrow H(N)$ is the morphism of k -coalgebras induced by α . Since α is the composite morphism below (where δ_G (resp. m_G) is the diagonal morphism (resp. multiplication) of G , c_Q the inversion morphism of Q , and $v(q \otimes q' \otimes x) = q \otimes x \otimes q'$),

$$Q \times N \xrightarrow{v \circ (\delta_G \times \text{id})} Q \times N \times Q \xrightarrow{\text{id} \times \text{id} \times c_Q} Q \times N \times Q \xrightarrow{m_G} G,$$

one obtains, still denoting by c_Q the antipode of $H(Q)$, that

$$(*) \quad \phi(q \otimes x') = \sum_i q_i x' c_Q(q'_i) \quad \text{if} \quad \Delta_{H(Q)}(q) = \sum_i q_i \otimes q'_i.$$

In particular, if M is an abstract group and $H(Q)$ is the group algebra kM (i.e. $Q = \mathrm{Spf}^* kM$ is the *constant formal k -group* M_k), for every $\gamma \in M$ and $x' \in H(N)$ one has $\phi(\gamma \otimes x') = \gamma x' \gamma^{-1}$, and this defines an action of M on $H(N)$ by automorphisms of Hopf algebras.

2.4.1. Proposition. *Let $f : G \rightarrow K$ be a morphism of formal k -groups. If $H = \mathrm{Ker}(f)$ is topologically flat over k , the homomorphism $f' : G/H \rightarrow K$ induced by f is a monomorphism.*

This is a consequence of the results of Exposé IV;⁽⁹³⁾ nevertheless, we give a direct proof. Let T be a formal variety of finite length over k and t an element of $(G/H)(T)$ such that $f' \circ t$ is the identity element of $K(T)$. We must show that t is the identity element of $(G/H)(T)$. Denote by p the projection $G \rightarrow G/H$ and by X the fibered product $T \times_{G/H} G$. By 2.4, p is surjective and topologically flat, hence so is the morphism $\mathrm{pr}_1 : X \rightarrow T$, so pr_1 is an epimorphism by Proposition 1.3.1; it therefore suffices to show that $t \circ \mathrm{pr}_1$ is the identity element of $(G/H)(X)$. Denote by pr_2 the projection $X \rightarrow G$; one has $t \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2$, whence the equality $1 = f' \circ t \circ \mathrm{pr}_1 = f' \circ p \circ \mathrm{pr}_2 = f \circ \mathrm{pr}_2$; then the exact sequence

$$1 \rightarrow H \rightarrow G \xrightarrow{f} K$$

shows that pr_2 factors through H , so $p \circ \mathrm{pr}_2$ is the zero morphism. Since $p \circ \mathrm{pr}_2 = t \circ \mathrm{pr}_1$, this proves the proposition. From the proposition one deduces the following corollary. Denote by $\mathcal{O}(G), \mathcal{O}(K), \mathcal{O}(G/H)$ the affine algebras of $G, K, G/H$; one has seen (2.4) that p induces an injection from $\mathcal{O}(G/H)$ to $\mathcal{O}(G)$. Moreover, by Proposition 1.3, since $f' : G/H \rightarrow K$ is a monomorphism, the morphism $\mathcal{O}(K) \rightarrow \mathcal{O}(G/H)$ is surjective, whence:

Corollary. *Let $f : G \rightarrow K$ be a morphism of formal k -groups and $H = \mathrm{Ker}(f)$. If H is topologically flat over k , $\mathcal{O}(G/H)$ is the image of $\mathcal{O}(K)$ in $\mathcal{O}(G)$.*

2.4.2. — Retain the preceding notation and suppose that H and G are topologically flat over k . Then G is topologically flat over k and over G/H , so, by 1.3.3, G/H is topologically flat over k . Consequently, by 2.4, the canonical projection q from K to $\mathrm{Coker}(f')$ is topologically flat, and f' is an isomorphism from G/H to $\mathrm{Ker}(q)$. Moreover, it is clear that $\mathrm{Coker}(f) = \mathrm{Coker}(f')$. Thus, under the hypothesis that G and $H = \mathrm{Ker}(f)$ are topologically flat over k , one has obtained an isomorphism between $\mathrm{Ker}(q)$, the *image* of f , and $G/\mathrm{Ker}(f)$, the *coimage* of f . This implies the theorem below.

Theorem. *Let k be a field. Commutative formal k -groups form an abelian category.*

Corollary. *Let k be a field. Affine commutative group k -schemes form an abelian category.⁽⁹⁴⁾*

This follows from the theorem and the equivalence of categories 2.2.2.

2.5. — A formal k -group is said to be *étale* if the underlying formal variety is étale (cf. 1.6); these formal groups have a very simple structure. Indeed, suppose that k is local; let κ be the residue field of k , κ_s a separable closure of κ , and Γ the topological Galois group of κ_s over κ . A Γ -set consists of a set E and a *continuous* action of Γ on E (i.e. the isotropy group of every element $x \in E$ is an open subgroup of Γ). For every formal k -variety X , put

$$X(\kappa_s) = \varinjlim_{\ell} X(\ell),$$

where ℓ runs through the finite extensions of κ contained in κ_s .⁽⁹⁵⁾ Then Γ acts continuously on each $X(\ell)$, hence also on $X(\kappa_s)$. Moreover, let $X_{\kappa} = X \widehat{\otimes}_k \kappa$ (cf. 1.6.C); for every ℓ one has $X(\ell) = X_{\kappa}(\ell)$, whence $X(\kappa_s) = X_{\kappa}(\kappa_s)$. Now suppose that X is étale over k ; then X_{κ} is the formal κ -variety which is the direct sum of the $\mathrm{Spec} \kappa(x)$, for $x \in X$, and if one

⁽⁹³⁾**N.D.E.** Cf. IV, 5.2.6.

⁽⁹⁴⁾**N.D.E.** See also the remarks following VI A, 5.4.3.

⁽⁹⁵⁾**N.D.E.** Note that if $[\kappa_s : \kappa] = \infty$, κ_s is not a profinite κ -algebra, so the preceding notation is an abuse of notation. Moreover, we have detailed the original in what follows.

denotes by X'_κ the κ -scheme which is the direct sum of the $\text{Spec } \kappa(x)$, one sees that $X_\kappa(\kappa_s)$ is precisely $X'_\kappa(\kappa_s) = \text{Hom}_{((\text{Sch})/\kappa)}(\text{Spec } \kappa_s, X'_\kappa)$. Denote by $\mathcal{C} = ((\text{Sch})/\kappa)^{\text{ét}}$ the full subcategory of $((\text{Sch})/\kappa)$ consisting of étale κ -schemes. One knows that the functor $X' \mapsto X'(\kappa_s)$ is an equivalence from $\mathcal{C}$ to the category $\mathcal{C}'$ of Γ -sets (cf. SGA 1, V §§ 7–8 or [DG70], § I.4 6.4); it therefore induces an equivalence between the category of $\mathcal{C}$ -groups and that of $\mathcal{C}'$ -groups, and one sees at once that a $\mathcal{C}'$ -group is the same thing as an abstract group G equipped with a continuous action of Γ by group automorphisms (one then says that G is a Γ -group). In view of the equivalences $\mathbf{Vaf}_{/k}^{\text{ét}} \xrightarrow{\sim} \mathbf{Vaf}_{/\kappa}^{\text{ét}} \xrightarrow{\sim} ((\text{Sch})/\kappa)^{\text{ét}}$ of 1.6.E, one therefore obtains the following:

Proposition. *Let k be a local pseudocompact ring, κ its residue field, κ_s a separable closure of κ , and $\Gamma = \text{Gal}(\kappa_s/\kappa)$.*

- (i) *The functor $X \mapsto X(\kappa_s)$ is an equivalence from the category of étale formal k -varieties to that of Γ -sets.*
- (ii) *It induces an equivalence from the category of étale formal k -groups to that of Γ -groups.*

Remark 2.5.A. ⁽⁹⁶⁾ Let k be a field, k_s a separable closure of k , G an étale formal k -group, and M the abstract group $G(k_s)$. Denote by X a set of representatives of the orbits of $\Gamma = \text{Gal}(k_s/k)$ in M , and for every $x \in X$ let Γ_x be its stabilizer, which is a subgroup of Γ of finite index, and $L_x = k_s^{\Gamma_x}$, which is an extension of k of degree $[\Gamma : \Gamma_x]$ (see, for example, [BAlg], V § 10.10). Then, by the equivalence of categories above, the affine algebra $\mathcal{A}(G)$ of G is the product of the L_x , equipped with the product topology, so the L_x are precisely the simple quotients $\mathcal{A}(G)/\mathfrak{m}$, where $\mathfrak{m}$ is an open maximal ideal of $\mathcal{A}(G)$. Since these correspond by duality to subcoalgebras of $H(G)$, one obtains that $H(G)$ is *pointed* (i.e. $\dim_k(C) = 1$ for every simple subcoalgebra C) if and only if $L_x = k$ for every x ; in this case $\mathcal{A}(G)$ is the topological algebra k^M , hence $H(G)$ is the group algebra kM , and one therefore has $M = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\} = G(k)$.

2.5.1. — Again suppose that the pseudocompact ring k is arbitrary. Since the functor $X \mapsto X_e$ of 1.6.F commutes with finite products, it transforms every formal group G into an étale formal group G_e , and since the morphism $p : X \rightarrow X_e$ of *loc. cit.* is functorial in X , $p : G \rightarrow G_e$ in this case is a morphism of formal groups. ⁽⁹⁷⁾ Consider the kernel $\text{Ker}(p)$; it is the fibered product of the diagram

$$\begin{array}{ccc} & G & \\ & \downarrow p & \\ \text{Spf}(k) & \xrightarrow{\varepsilon} & G_e \end{array}$$

where ε is the identity section of G_e . Since p induces a bijection of the underlying sets, one deduces (cf. 1.2, N.D.E. (41)) that the set underlying $\text{Ker}(p)$ is the image of $\text{Spf}(k)$ under ε , and that for every point s of $\text{Spf}(k)$, the local algebra of $\text{Ker}(p)$ at the point $\varepsilon(s)$ is $\mathcal{O}_{G, \varepsilon(s)} \hat{\otimes} \mathcal{O}_{G_e, \varepsilon(s)} k_s$. Moreover, at each point $\varepsilon(s)$ the residue field of $\mathcal{O}_{G, \varepsilon(s)}$ is $\kappa(s)$, and hence $\mathcal{O}_{G_e, \varepsilon(s)} = k_s$, whence $\mathcal{O}_{\text{Ker}(p), \varepsilon(s)} = \mathcal{O}_{G, \varepsilon(s)}$. For these reasons we shall say that $\text{Ker}(p)$ is the *infinitesimal neighborhood of the origin of G* , and shall write $\text{Ker}(p) = G_0$, thus obtaining an exact sequence

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e.$$

In what follows, we shall say that G is *infinitesimal* if $G = G_0$. ⁽⁹⁸⁾ This is equivalent to saying that, for every $s \in \text{Spf}(k)$, or again for every continuous morphism $k \rightarrow \kappa$, where κ

⁽⁹⁶⁾ N.D.E. We have added this remark.

⁽⁹⁷⁾ N.D.E. We have detailed the original in what follows.

⁽⁹⁸⁾ N.D.E. We have added the following sentence.

is a field equipped with the discrete topology, the identity is the unique element of $G(\kappa(s))$, resp. of $G(\kappa)$. Suppose further that G is *topologically flat over k* .⁽⁹⁹⁾ Then, by 1.6.1, the morphism $p : G \rightarrow G_e$ is topologically flat, and since it is bijective, it is therefore an effective epimorphism by Proposition 1.3.1. Consequently, G_e is identified with the quotient G/G_0 . One has therefore obtained:

Corollary. *Let G be a formal group topologically flat over k . Then G_e is identified with the quotient G/G_0 ; i.e. one has an exact sequence of formal groups*

$$1 \rightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e \rightarrow 1.$$

2.5.2. — Suppose that k is a perfect field. In this case one has seen (cf. 1.6.2) that the morphism $p : X \rightarrow X_e$ has a section $s : X_e \rightarrow X$ depending functorially on X ; this section is therefore a morphism of formal groups when X is a formal group. One therefore obtains the following:

Proposition. *When k is a perfect field, every formal k -group G is canonically isomorphic to the semidirect product of an infinitesimal group G_0 and an étale group G_e acting on G_0 .*⁽¹⁰⁰⁾

If, moreover, G is *commutative*, G is the product of G_0 and G_e . By 2.2.2, this canonical decomposition of commutative formal k -groups corresponds to an analogous decomposition of affine commutative group k -schemes; see paragraph 2.5.3 below.

Remark 2.5.2.A. ⁽¹⁰¹⁾ The exact sequence $1 \rightarrow G_0 \rightarrow G \rightarrow G_e \rightarrow 1$ is not necessarily split when k is not perfect: let us give the following example taken from [DG70], § III.6, 8.6. Let k be a nonperfect field of characteristic $p > 0$. Let $\lambda \in k - k^p$, let L_λ be the abelian Lie p -algebra with basis (x, y) , the symbolic p th power (cf. VII A, 5.2) being given by $x^{(p)} = x$ and $y^{(p)} = \lambda x$, let $U_p(L_\lambda) = k[x, y]/(x^p - x, y^p - \lambda x)$ be the restricted enveloping algebra of L_λ (cf. VII A, 5.3), and let G_λ be the commutative formal k -group with affine algebra $U_p(L_\lambda)$. Then G_λ is a nonsplit extension of the constant étale k -group $(\mathbf{Z}/p\mathbf{Z})_k$ by the infinitesimal k -group $\alpha_{p,k} = \text{Spf } k[x]/(x^p - x)$, i.e. one has a nonsplit exact sequence of commutative formal k -groups:⁽¹⁰²⁾

$$0 \rightarrow \alpha_{p,k} \rightarrow G_\lambda \rightarrow (\mathbf{Z}/p\mathbf{Z})_k \rightarrow 0.$$

By duality, it corresponds to a nonsplit exact sequence of commutative algebraic k -groups

$$0 \rightarrow \mu_{p,k} \rightarrow D'(G_\lambda) \rightarrow \alpha_{p,k} \rightarrow 0,$$

where $\mu_{p,k} = \text{Spec } k[t]/(t^p - 1)$ (and in this way one obtains all the extensions of $\alpha_{p,k}$ by $\mu_{p,k}$; cf. [DG70], § III.6, 8.6).

2.5.3. — ⁽¹⁰³⁾ Let k be a field.

Definition 2.5.3.A. One says that a commutative group k -scheme is *unipotent* if it is isomorphic to $\text{Spec } H(G)$, where G is a commutative *infinitesimal* formal k -group.⁽¹⁰⁴⁾

⁽⁹⁹⁾**N.D.E.** We have detailed what follows.

⁽¹⁰⁰⁾**N.D.E.** The same argument also shows that, for an arbitrary field k , if all the residue fields of G equal k , then G_e is the constant k -group $(M)_k$, where $M = G(k) = \text{Spf}^* H(G)$, and $H(G)$ is the semidirect product of $H(G_0)$ by the group algebra kM ; cf. addition 2.9 below.

⁽¹⁰¹⁾**N.D.E.** We have added this remark.

⁽¹⁰²⁾**N.D.E.** $\alpha_{p,k}$, $(\mathbf{Z}/p\mathbf{Z})_k$, and G_λ are also algebraic k -groups finite and flat over k ; cf. N.D.E. (84) in 2.2.2.

⁽¹⁰³⁾**N.D.E.** We have introduced the numbering 2.5.3 and 2.5.3.A to 2.5.3.C.

⁽¹⁰⁴⁾**N.D.E.** This coincides with the “usual” notion of a unipotent group k -scheme; cf. addition 2.8 at the end of Section 2.

⁽¹⁰⁵⁾Moreover, following Exp. IX, Definition 1.1, one says that a group k -scheme H is of *multiplicative type* if there exists a scheme S , faithfully flat and quasi-compact over $\operatorname{Spec} k$, such that H_S is a diagonalizable S -group, i.e. is isomorphic to $D_S(M)$ for some “abstract” abelian group M . By descent (fpqc), this implies that H is *affine* and commutative. Moreover, since one can replace S by the residue field of one of its points, one sees that H is of multiplicative type if and only if there exists an extension K of k such that H_K is a diagonalizable K -group.

Proposition 2.5.3.B. *Let T be an affine commutative group k -scheme and k_s a separable closure of k .*

- (i) *For T to be of multiplicative type, it is necessary and sufficient that its dual $D(T)$ be a commutative étale formal k -group.*
- (ii) *Consequently, T is of multiplicative type if and only if $T \otimes_k k_s$ is diagonalizable.*

PROOF. Denote by $\mathcal{A}$ the affine algebra of $D(T)$ and by $\mathcal{A}_0$ its local component at the identity; then $D(T)$ is étale over k if and only if $\mathcal{A}_0 = k$. If K is an extension of k , the algebra $\mathcal{A}_0 \hat{\otimes}_k K$ is local (since it is a projective limit of artinian local rings), hence coincides with the local component $(\mathcal{A} \hat{\otimes}_k K)_0$; moreover, since formation of $D(T)$ commutes with base change (cf. 2.2.2), one also has $\mathcal{A} \hat{\otimes}_k K \simeq D(T_K)$. Suppose that T is of multiplicative type; then there exists an extension K of k such that $\mathcal{O}(T) \otimes_k K$ is isomorphic, as a Hopf K -algebra, to the group algebra KM for some abelian group M . Then $\mathcal{A} \hat{\otimes}_k K$ is isomorphic to the algebra K^M , equipped with the product topology, so one has $K = (\mathcal{A} \hat{\otimes}_k K)_0 = \mathcal{A}_0 \hat{\otimes}_k K$, and this implies $\mathcal{A}_0 = k$, hence $D(T)$ is étale. Conversely, suppose that $D(T)$ is étale. Then $D(T) \hat{\otimes}_k k_s = D(T_{k_s})$ is a constant k_s -group M , so its covariant bialgebra is the group algebra $k_s M$ (cf. Remark 2.5.A), hence $\mathcal{O}(T) \otimes_k k_s = k_s M$, which proves that T is of multiplicative type and split by the extension $k \rightarrow k_s$. The proposition follows. $\square$

By 2.2.2, 2.5.1, and 2.5, one obtains:

Corollary 2.5.3.C. *Let k be a field and G an affine commutative group k -scheme.*

- (i) *G contains a subgroup of multiplicative type G_m such that G/G_m is unipotent.*
- (ii) *When k is perfect, there also exists a canonical retraction from G to G_m , so that G is the product of a unipotent group and a group of multiplicative type.*
- (iii) ⁽¹⁰⁶⁾*Let k_s be a separable closure of k and $\Gamma = \operatorname{Gal}(k_s/k)$. The category of group k -schemes of multiplicative type is anti-equivalent to the category of continuous Γ -modules.*

2.6. — We shall now study infinitesimal formal groups, to which the following paragraphs are devoted. Lie algebras play a primordial role in this study. First suppose that the base ring k is artinian, and let G be a formal group over k . One can give three different interpretations of the Lie algebra of G , all of which we shall use:

- (a) Let D be the algebra $k[d]/(d^2)$ of dual numbers over k , and δ the homomorphism from D to k which annihilates d . For every formal group G over k , $\operatorname{Lie}(G)$ is the kernel of $G(\delta)$, so by definition one has an *exact sequence of groups*

$$1 \longrightarrow \operatorname{Lie}(G) \longrightarrow G(D) \xrightarrow{G(\delta)} G(k) \longrightarrow 1.$$

- (b) Let A be the affine algebra of G and $I_A = \operatorname{Ker} \varepsilon_A$ its augmentation ideal. The elements of the group $G(D)$ are the morphisms of profinite k -algebras $f : A \rightarrow D$. The condition $G(\delta)(f) = 1$ is equivalent to $\delta \circ f = \varepsilon_A$. Since $x - \varepsilon_A(x) \cdot 1_A \in I_A$ for

⁽¹⁰⁵⁾**N.D.E.** The original indicated: “let us say that a commutative group k -scheme is of multiplicative type if it is isomorphic to $\operatorname{Spec} H(G)$, where G is a commutative étale formal k -group”. Here we have given the “usual” definition, taken from Exp. IX, 1.1, and shown its equivalence to the preceding condition; see also [DG70], § IV.1, Th. 2.2.

⁽¹⁰⁶⁾**N.D.E.** We have added part (iii), a consequence of Proposition 2.5.

every $x \in A$, this is equivalent to $f(I_A) \subset k \cdot d$, so $\text{Ker}(f)$ contains I_A^2 , hence also $\overline{I_A^2}$, and thus f induces a continuous linear map f' from $I_A/\overline{I_A^2} = \omega_{G/k}$ to k such that for every $x \in A$ one has the equality

$$f(x) = \varepsilon_A(x) \cdot 1_D + f'(\bar{x}) \cdot d,$$

where $\bar{x}$ denotes the image of $x - \varepsilon_A(x) \cdot 1_A$ in $I_A/\overline{I_A^2}$. One then sees that *the map $f \mapsto f'$ defines a bijection from $\text{Lie}(G)$ to the topological dual $\omega_{G/k}^\dagger$ of $\omega_{G/k}$* (cf. 0.2.2). This bijection respects the group structures. Indeed, let f, g be two elements of $\text{Lie}(G)$; their product $f \cdot g$ is the composite map $h \circ \Delta_A$, where $h : A \hat{\otimes} A \rightarrow D$ satisfies $h(b \hat{\otimes} b') = f(b)g(b')$. Now, if $a \in I_A$, by 2.1(ii) one has $\Delta_A(a) - a \hat{\otimes} 1 - 1 \hat{\otimes} a \in I_A \hat{\otimes} I_A$, whence $(f \cdot g)(a) = f(a) + g(a)$ (cf. also II 3.10). In what follows, we identify $\text{Lie}(G)$ with $\omega_{G/k}^\dagger$ by means of the bijection $f \mapsto f'$ described above. *The group $\text{Lie}(G)$ is thus equipped with a k -module structure.*

- (c) Let $A^\dagger, D^\dagger$ be the k -modules dual to A, D , and $\{1_D^\dagger, d^\dagger\}$ the basis dual to the basis $\{1_D, d\}$ of D over k (one has $1_D^\dagger = \delta$). Since D is free of finite rank over k , the canonical map

$$\text{Hom}_c(A, D) \longrightarrow \text{Hom}_k(D^\dagger, A^\dagger), \quad f \longmapsto {}^t f$$

is bijective. On the other hand, f is determined by the values ${}^t f(1_D^\dagger)$ and ${}^t f(d^\dagger) = x$. The condition $G(\delta)(f) = 1$ is equivalent to the equality ${}^t f(1_D^\dagger) = \varepsilon_A$. Moreover, one easily sees that f is compatible with multiplication if and only if one has (cf. 2.3)

$$(*) \quad \delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

Finally, it is clear that a continuous linear map $f : A \rightarrow D$ which is compatible with multiplication and satisfies $\delta \circ f = \varepsilon_A$ sends the identity element of A to that of D .⁽¹⁰⁷⁾ The map $f \mapsto x$ therefore allows us to identify $\text{Lie}(G)$ with the set of “primitive elements” of $\mathbf{H}(G)$ (i.e. the $x \in \mathbf{H}(G)$ satisfying relation $(*)$). If x, y are two such elements, one has

$$\begin{aligned} \delta_G(xy) &= \delta_G(x)\delta_G(y) = \varphi_G((x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y)) \\ &= \varphi_G(xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy), \end{aligned}$$

whence $\delta_G(xy - yx) = \varphi_G((xy - yx) \otimes 1 + 1 \otimes (xy - yx))$. This shows that the k -module $\text{Lie}(G)$ is identified with a Lie subalgebra of $\mathbf{H}(G)$: we shall say that $\text{Lie}(G)$ is *the Lie algebra of G* .⁽¹⁰⁸⁾

2.6.1. — When k is an arbitrary pseudocompact ring and G a formal group over k , we call the functor $\underline{\text{Lie}}(G)$ which associates to every object C of $\mathbf{AIf}/k$ the C -Lie algebra of the formal C -group $G' = G \hat{\otimes}_k C$ the $\mathbf{O}_k$ -Lie algebra of G :⁽¹⁰⁹⁾ put $A' = A \hat{\otimes}_k C$; since I_A is a direct factor of A , $I_{A'}$ equals $I_A \hat{\otimes}_k C = I_A \hat{\otimes}_A A'$, and since $\omega_{G/k} = I_A \hat{\otimes}_A k$ and likewise $\omega_{G'/C} = I_{A'} \hat{\otimes}_{A'} C$, one obtains $\omega_{G'/C} = \omega_{G/k} \hat{\otimes}_A C$, and hence

$$\underline{\text{Lie}}(G)(C) = \text{Hom}_c(\omega_{G/k} \hat{\otimes}_A C, C), \quad \text{i.e.} \quad \underline{\text{Lie}}(G) = \mathbf{V}_k^t(\omega_{G/k})$$

⁽¹⁰⁷⁾**N.D.E.** Indeed, $\delta \circ f = \varepsilon_A$ implies that $f(1) = 1 + \lambda d$, with $\lambda \in k$, and then $f(1) = f(1)^2 = 1 + 2\lambda d$ gives $\lambda = 0$.

⁽¹⁰⁸⁾**N.D.E.** Comparing with VII A 2.5, one sees that if K is a group k -scheme of *finite type* and G is the formal completion of K at the origin (i.e. $\mathcal{A}(G)$ is the completion of the local ring $\mathcal{O}_{K,e}$ for the $\mathfrak{m}$ -adic topology, where $\mathfrak{m}$ is the kernel of the augmentation $\varepsilon : \mathcal{O}_{K,e} \rightarrow k$), then $\mathbf{H}(G)$ is identified with the algebra of distributions $U(K)$, and $\text{Lie}(G)$ with $\text{Lie}(K)$. (The condition that K be of finite type over k is used to ensure that $\mathfrak{m}/\mathfrak{m}^2$ is a k -module of finite length, hence discrete, so that its topological dual coincides with its ordinary dual.) In particular, when G is a *finite* formal k -group (i.e. such that $\mathcal{A}(G)$ is a finite k -module), in which case G can also be considered as the group k -scheme $\text{Spec } \mathcal{A}(G)$, the two definitions of $\text{Lie}(G)$ coincide.

⁽¹⁰⁹⁾**N.D.E.** We have detailed the original in what follows.

(with the notation of 1.2.3.B). Thus, by Proposition 1.2.3.E, $\underline{\text{Lie}}(G)$ is flat over $\mathbf{O}_k$ if and only if $\omega_{G/k}$ is a projective pseudocompact k -module.

2.6.2. — Conversely, every Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a group-valued k -functor. Indeed, denote by $\mathbf{U}(\mathbf{L})$ the functor which associates to every object C of $\mathbf{A}lf/k$ the enveloping algebra $U(\mathbf{L}(C))$ of the C -Lie algebra $\mathbf{L}(C)$. By VII A, 3.2.2, $\mathbf{U}(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$ and therefore determines, by 2.2, a group-valued k -functor $\text{Spf}^* \mathbf{U}(\mathbf{L})$, which we shall henceforth denote by $\mathcal{G}(\mathbf{L})$. Thus $\mathcal{G}(\mathbf{L})(C)$ is the group of elements $z \in U(\mathbf{L}(C))$ of augmentation 1 such that $\Delta_{U(\mathbf{L}(C))}(z) = z \otimes z$. Moreover, when $\mathbf{L}$ is flat over $\mathbf{O}_k$, one has the following proposition.

Proposition. ⁽¹¹⁰⁾ *Let $\mathbf{L}$ be a flat $\mathbf{O}_k$ -Lie algebra.*

- (i) $\mathcal{G}(\mathbf{L})$ is a formal group topologically flat over k , whose $\mathbf{O}_k$ -bialgebra is $\mathbf{U}(\mathbf{L})$.
- (ii) $\mathcal{G}(\mathbf{L})$ is infinitesimal.
- (iii) For every object C of $\mathbf{A}lf/k$, $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))(C)$ is identified with the set

$$\text{Prim } U(\mathbf{L}(C)) = \{x \in U(\mathbf{L}(C)) \mid \varepsilon(x) = 0 \text{ and } \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

of primitive elements of $U(\mathbf{L}(C))$. In particular, one has a natural morphism of $\mathbf{O}_k$ -Lie algebras $\tau_{\mathbf{L}} : \mathbf{L} \rightarrow \underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$.

Indeed, the hypothesis that $\mathbf{L}$ is flat over $\mathbf{O}_k$ means that for every morphism $C \rightarrow D$ of $\mathbf{A}lf/k$, one has $\mathbf{L}(D) = \mathbf{L}(C) \otimes_C D$, and that for each local component C' of C , $\mathbf{L}(C')$ is a free C' -module. The first condition implies $U(\mathbf{L}(D)) = U(\mathbf{L}(C)) \otimes D$ (by the universal property of the tensor product and that of the functor $L \mapsto U(L)$), and the second condition implies, by the Poincaré–Birkhoff–Witt theorem (cf. Bourbaki, *Groupes et algèbres de Lie*, I 2.7), that $U(\mathbf{L}(C'))$ is a free C' -module. Thus the bialgebra $\mathbf{U}(\mathbf{L})$ is flat over $\mathbf{O}_k$. To prove (ii) and (iii), one can suppose that k is artinian. Put $L = \mathbf{L}(k)$, $U = U(L)$, $U_0 = k \cdot 1_U$, and let U^+ be the two-sided ideal of U generated by the image of L . Put, moreover, for every $n \geq 1$,

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

By 1.3.6 it suffices to show that U is the union of the U_n . Now, if one identifies L with its canonical image in U , L is evidently contained in U_1 . If $x_1, \dots, x_n$ are elements of L , one therefore has $\Delta_U(x_1 \cdots x_n) = (x_1 \otimes 1 + 1 \otimes x_1) \cdots (x_n \otimes 1 + 1 \otimes x_n)$, which shows, by induction on n , that the product $x_1 \cdots x_n$ belongs to U_n , hence that $U = \bigcup_n U_n$. This proves (ii). On the other hand, let $D = k[d]/(d^2)$ be the algebra of dual numbers over k . By hypothesis, one has $\mathbf{L}(D) \simeq L \otimes D$, whence $U(\mathbf{L}(D)) \simeq U \otimes D$ by the universal properties of the tensor product and the enveloping algebra. It follows that $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))(k)$ is identified with the set of elements $z = 1 + xd$ of $U \oplus Ud$ (where $x \in U$) such that $\varepsilon(z) = 1$ and $\Delta(z) = z \otimes z$, which is equivalent to $\varepsilon(x) = 0$ and $\Delta(x) = x \otimes 1 + 1 \otimes x$, i.e. $x \in \text{Prim } U$. In particular, the map $\tau_{\mathbf{L}} : x \mapsto 1 + dx$ is a morphism of $\mathbf{O}_k$ -Lie algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$.

2.6.3. — If $\mathbf{L}$ is a flat Lie algebra over $\mathbf{O}_k$, the formal group $\mathcal{G}(\mathbf{L})$ can be characterized by a universal property.⁽¹¹¹⁾ Indeed, every morphism ϕ from $\mathcal{G}(\mathbf{L})$ to a formal group G induces a morphism $\underline{\text{Lie}}(\phi) : \underline{\text{Lie}}(\mathcal{G}(\mathbf{L})) \rightarrow \underline{\text{Lie}}(G)$; composing it with the morphism $\tau_{\mathbf{L}} : \mathbf{L} \rightarrow \underline{\text{Lie}}(\mathcal{G}(\mathbf{L}))$ (cf. 2.6.2), one obtains a morphism $\phi' : \mathbf{L} \rightarrow \underline{\text{Lie}}(G)$, and one has:

Proposition. *If G is a formal k -group and $\mathbf{L}$ a flat $\mathbf{O}_k$ -Lie algebra, the map $\phi \mapsto \phi'$ defined above is a bijection*

$$\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}(\mathbf{L}), G) \xrightarrow{\sim} \text{Hom}_{\text{Lie}}(\mathbf{L}, \underline{\text{Lie}}(G)),$$

where the right-hand term denotes the set of morphisms of $\mathbf{O}_k$ -Lie algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(G)$.

⁽¹¹⁰⁾**N.D.E.** We have brought out parts (i) and (ii), and added part (iii), which will be useful in 2.6.3 and 3.3.2.

⁽¹¹¹⁾**N.D.E.** We have modified what follows, taking advantage of the addition made in 2.6.2.

Indeed, one immediately reduces to the case where k is artinian. Put $L = \mathbf{L}(k)$. By 2.3.1, $\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}(\mathbf{L}), G)$ is in bijection with the set of morphisms of unital algebras $\phi : U(L) \rightarrow H(G)$ such that the following diagram commutes:

$$\begin{array}{ccc} U(L) & \xrightarrow{h} & H(G) \\ \Delta_{U(L)} \downarrow & & \searrow \delta_G \\ U(L) \otimes U(L) & \xrightarrow{h \otimes h} & H(G) \otimes H(G) \end{array} \quad \begin{array}{c} \\ \nearrow \varphi_G \\ H(G \times G) \end{array}$$

Now h is defined by its restriction to L , which is a morphism of Lie algebras from L to the Lie algebra underlying $H(G)$, and commutativity of the diagram means that h maps L into the part of $H(G)$ consisting of the x such that $\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x)$, which is precisely $\text{Lie}(G)$; cf. 2.6 c).

2.7. — We conclude these generalities with a statement going back to S. Lie which will serve us in paragraph 5.1. A *formal monoid over k* is by definition a pair (M, m) consisting of a formal variety M and a morphism $m : M \times M \rightarrow M$ such that $m(C)$ makes $M(C)$ a monoid for every object C of $\mathbf{Alf}/k$.⁽¹¹²⁾ In particular, the “identity section”, which associates to every object C the identity element of $M(C)$, defines a section ε_M of the canonical projection $M \rightarrow \text{Spf}(k)$. We shall say that the formal monoid M is *infinitesimal* if ε_M induces a bijection of the underlying sets.

Proposition. *Every topologically flat infinitesimal formal k -monoid M is a formal k -group.*⁽¹¹³⁾

We must show that $M(C)$ is a group for every object C of $\mathbf{Alf}/k$. One therefore immediately reduces to the case where k is artinian. Let then $U = H(M)$ be the coalgebra of M (1.3.5); the multiplication $m : M \times M \rightarrow M$ induces a morphism of coalgebras $m_U : U \otimes U \rightarrow U$, which makes U an associative algebra over k ; the identity element of this algebra is the image of the identity element of k under the map from k to U induced by the identity section ε_M of M . Similarly, the projection $M \rightarrow \text{Spf}(k)$ induces a homomorphism ε_U from U to k ; we shall denote the kernel of ε_U by U^+ .

We must show that there exists an antipodism, that is, a morphism of coalgebras $c_U : U \rightarrow U$ such that, for every $u \in U$, one has:

$$(*) \quad (m_U \circ (c_U \otimes \text{id}_U) \circ \Delta_U)(x) = \varepsilon_U(u) \cdot 1_U.$$

Let (U_n) be the filtration of U defined in 1.3.6; put $U_n^+ = U^+ \cap U_n$. Since M is infinitesimal, U^+ is the union of the submodules U_n^+ .⁽¹¹⁴⁾ One then puts $c_0(1) = 1$ and $c_1(x) = -x$ if $x \in U_1^+$, i.e. if x is a primitive element. Suppose that $c_{n-1} : U_{n-1} \rightarrow U$ has been constructed so as to satisfy $(*)$ for every $x \in U_{n-1}$, and let $x \in U_n^+$. By the proof of Lemma 1.3.6.A,

⁽¹¹²⁾**N.D.E.** Here and in what follows we have written “monoid” instead of “monoid with identity” (recall that a monoid is by definition a set equipped with an associative composition law and having an identity element).

⁽¹¹³⁾**N.D.E.** By the equivalence of categories of 1.3.5.D, a monoid in the category of topologically flat formal k -varieties “is the same thing” as a monoid in the category of flat cocommutative $\mathbf{O}_k$ -coalgebras, i.e. a cocommutative $\mathbf{O}_k$ -bialgebra (in the usual sense, i.e. not necessarily equipped with an antipode). Moreover, by 1.3.6, the hypothesis that M be infinitesimal is equivalent to saying that the corresponding bialgebra is connected. Thus, if k is an artinian ring, the proposition is equivalent to saying that: every connected cocommutative k -bialgebra, flat over k , is a k -Hopf algebra, i.e. possesses an antipode (and the cocommutativity hypothesis is in fact superfluous, cf. the proof).

⁽¹¹⁴⁾**N.D.E.** The original continued thus: “one then shows easily, by induction on n , that there exists one and only one linear map $c_n : U_n \rightarrow U_n$ such that the composite map $m_U \circ (c_n \otimes \text{id}_U) \circ \Delta_U : U_n^+ \rightarrow U$ is zero”; we have detailed the proof, which rests on that of Lemma 1.3.6.A.

one has $\Delta_U(x) - x \otimes 1 \in U_{n-1} \otimes U^+$ (this is where the hypothesis that U be flat over k enters), so one can write $\Delta_U(x) = x \otimes 1 + \sum_i y_i \otimes z_i$, with $y_i \in U_{n-1}$; one then puts $c_n(x) = -\sum_i c_{n-1}(y_i)z_i$. One thus obtains a k -linear map $c : U \rightarrow U$, which is the left inverse of id_U for the monoid law on $\text{End}_k(U)$ defined by $f \cdot g = m_U \circ (f \otimes g) \circ \Delta_U$ (the identity element being the map $\eta : u \mapsto \varepsilon(u) \cdot 1_U$). It follows that c is uniquely determined, and is also the right inverse of id_U , i.e. one also has $m_U \circ (c_n \otimes \text{id}_U) \circ \Delta_U = \eta$ (without assuming U cocommutative).

2.8. — Unipotent group schemes over a field. — ⁽¹¹⁵⁾Let k be a field. “Recall” that a group k -scheme G is said to be *unipotent* if it satisfies the following two conditions (cf. [DG70], § IV.2, Prop. 2.5):

- (a) G is affine.
- (b) Every simple $\mathcal{O}(G)$ -comodule is trivial, i.e. if $\rho : V \rightarrow V \otimes_k \mathcal{O}(G)$ is an $\mathcal{O}(G)$ -comodule structure on a k -vector space $V \neq 0$, and if there is no nonzero subspace $W \neq V$ such that $\rho(W) \subset W \otimes_k \mathcal{O}(G)$, then V has dimension 1 and $\rho(v) = v \otimes 1$ for every $v \in V$.

According to loc. cit., when G is of finite type over k , this is equivalent to the definition given in Exp. XVII, § 1, namely that G possesses a finite composition series whose successive quotients are isomorphic to k -subgroups of $\mathbf{G}_{a,k}$.

Now, for every affine group k -scheme G , the comultiplication of $\mathcal{O}(G)$ equips the pseudocompact k -module $A = \mathcal{O}(G)^*$ with a structure of profinite k -algebra, not necessarily commutative, the identity element 1_A being the augmentation $\varepsilon : \mathcal{O}(G) \rightarrow k$. On the other hand, let $I = \{f \in A \mid f(1_{\mathcal{O}(G)}) = 0\}$; this is a two-sided ideal of A , and one has $A = k \cdot 1_A \oplus I$, cf. 1.3.6.

Let V be a subspace of A of finite codimension; consider the continuous k -bilinear map $\varphi : A \times A \rightarrow A/V$, $(a, b) \mapsto ab + V$; by Lemma 0.3.1, there exist two subspaces L_1, L_2 of finite codimension in A such that V contains AL_2 and L_1A ; then $L = L_1 \cap L_2$ is a subspace of A of finite codimension, and V contains the two-sided ideal ALA , which has finite codimension. This shows that the two-sided ideals of finite codimension form a basis of neighborhoods of 0. One deduces that an $\mathcal{O}(G)$ -comodule “is the same thing” as a *continuous A -module*, i.e. an A -module V such that the map $A \times V \rightarrow V$ is continuous, V being equipped with the discrete topology. Such a module is clearly the union of submodules V_i of finite dimension over k , each being a module over a quotient k -algebra A_i of A , of finite dimension over k . It follows that if M is a simple continuous module, it has finite dimension over k , and is a faithful simple module over the finite-dimensional k -algebra $A/\text{Ann}(M)$; the latter is therefore a simple finite-dimensional k -algebra, i.e. $\text{Ann}(M)$ is an open maximal ideal of A . Conversely, let P be an open prime ideal⁽¹¹⁶⁾ of A ; then A/P is a finite-dimensional k -algebra in which the ideal (0) is prime, so it is a simple finite-dimensional k -algebra, and hence there exists, up to isomorphism, a unique simple continuous A -module whose annihilator is P . It follows that the map $M \mapsto \text{Ann}(M)$ defines a bijection between the isomorphism classes of simple continuous A -modules and the open prime ideals of A . In particular, one calls the A -module A/I , which has dimension 1 over k , the “trivial module”; it corresponds to the trivial $\mathcal{O}(G)$ -comodule V of dimension 1, i.e. such that $\rho(v) = v \otimes 1_{\mathcal{O}(G)}$ for every $v \in V$. One therefore obtains the following proposition:

Proposition 2.8.1. *Let k be a field, G an affine group k -scheme, and $A = \mathcal{O}(G)^*$.*

⁽¹¹⁵⁾**N.D.E.** We have added this subsection.

⁽¹¹⁶⁾**N.D.E.** Recall that a two-sided ideal P of a ring A is said to be prime if in A/P the product of two nonzero two-sided ideals is nonzero.

- (i) Then G is unipotent if and only if I is the unique open prime ideal of A .⁽¹¹⁷⁾
- (ii) In particular, if G is commutative, so that $\mathcal{O}(G) = H(D(G))$, where $D(G) = \mathrm{Spf}(A)$ denotes the Cartier dual of G , then G is unipotent if and only if $D(G)$ is infinitesimal.

2.9. — Pointed cocommutative Hopf algebras over a field. — ⁽¹¹⁸⁾Let k be a field, C a k -coalgebra, and A the algebra C^* equipped with the structure of profinite k -algebra, not necessarily commutative, described in 2.8; by 0.2.2, one has $C = A^\dagger = \mathrm{Hom}_c(A, k)$. One deduces that the map $D \mapsto D^\perp = \{f \in A \mid f(D) = 0\}$ is a bijection from the set of subcoalgebras of C onto that of closed two-sided ideals (in what follows, we shall simply say “ideals”) of A ; the inverse bijection is given by $I \mapsto I^\perp = \{x \in C = A^\dagger \mid x(I) = 0\}$. Since every maximal closed ideal is an open maximal ideal (cf. 0.2.1), every subcoalgebra therefore contains a simple subcoalgebra, necessarily of finite dimension.

Recall that a subcoalgebra D of C is said to be *irreducible* if it contains only one simple subcoalgebra S_0 , which is equivalent to saying that $\mathfrak{m}_0 = S_0^\perp$ is the unique open maximal ideal containing $D^\perp$, i.e. that $D^\perp + \mathfrak{m} = A$ for every open maximal ideal $\mathfrak{m} \neq \mathfrak{m}_0$. This is in particular the case if $D = S_0$. Then the sum Σ_0 of all the irreducible subcoalgebras C_i containing S_0 is clearly a subcoalgebra, and it is irreducible since, for every $\mathfrak{m} \neq \mathfrak{m}_0$, one has, by 0.2.D:

$$\mathfrak{m} + \bigcap_i C_i^\perp = \bigcap_i (\mathfrak{m} + C_i^\perp) = A.$$

One says that Σ_0 is the *irreducible component* of C corresponding to C_0 .

On the other hand, one says that C is *pointed* if every simple sub- k -coalgebra of C has dimension 1; this is equivalent to saying that for every open maximal ideal $\mathfrak{m}$ of A , one has $A/\mathfrak{m} = k$. Recall also that C is said to be *connected* if it is irreducible and pointed. (Let us note in passing that if C is a bialgebra, it is connected if and only if it is irreducible, since $k \cdot 1_C$ is a simple subcoalgebra.)

Suppose henceforth that C is *cocommutative*. Then A is commutative and is therefore the product of its local components $A_{\mathfrak{m}}$, for $\mathfrak{m} \in \Upsilon(A)$ (cf. 0.1.1); denote by $S_{\mathfrak{m}}$ the simple subcoalgebra $\mathfrak{m}^\perp \simeq (A/\mathfrak{m})^*$ and by $\Sigma_{\mathfrak{m}}$ its irreducible component. One can describe $\Sigma_{\mathfrak{m}}$ as follows. Denote by $J_{\mathfrak{m}}$ the kernel of the projection $A \rightarrow A_{\mathfrak{m}}$; it is contained in $\mathfrak{m}$ and is the smallest closed ideal I of $\mathfrak{m}$ such that $I + \mathfrak{m}' = A$ for every $\mathfrak{m}' \neq \mathfrak{m}$. Indeed, if I has this property, then I contains $A_{\mathfrak{m}'}$ for every $\mathfrak{m}' \neq \mathfrak{m}$, hence contains $J_{\mathfrak{m}}$. Since $A = J_{\mathfrak{m}} \oplus A_{\mathfrak{m}}$, it follows that $\Sigma_{\mathfrak{m}} = J_{\mathfrak{m}}^\perp$ is identified with $A_{\mathfrak{m}}^\dagger$. One can now prove the:

Proposition 2.9.1. *Let k be a field.*

- (i) *Let G be a formal k -group such that all the residue fields of G equal k . Then G_e is the constant k -group M_k , where $M = G(k) = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$, and $H(G)$ is the semidirect product of $H(G_0)$ by kM (cf. 2.4.B).*
- (ii) *Equivalently: let H be a pointed cocommutative k -Hopf algebra. Then H is the semidirect product of the irreducible component H_0 of the identity element 1_H by kM , where $M = \{x \in H \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$.*

Let us prove (i). Since all the residue fields of G equal k , the projection $\pi : G \rightarrow G_e$ admits the section $s : G_e \rightarrow G$ defined by $\mathcal{O}_{G,g} \rightarrow \kappa(g) = k$, for every $g \in G$; moreover, for every $g, h \in G$, $\mathcal{O}_{G,g} \hat{\otimes}_k \mathcal{O}_{G,h}$ is local with residue field k , and one therefore obtains that $s \times s$ is a section of the projection $\pi \times \pi : G \times G \rightarrow (G \times G)_e = G_e \times G_e$. Since π is a morphism of groups, it follows that $\pi \circ m_G \circ (s \times s) = m_{G_e} = \pi \circ s \circ m_{G_e}$, and since π is an epimorphism this implies that s is a morphism of groups. One therefore obtains that $G = G_0 \rtimes G_e$, and

⁽¹¹⁷⁾**N.D.E.** This is also equivalent to saying that $k \cdot 1_{\mathcal{O}(G)}$ is the unique simple k -subcoalgebra of $\mathcal{O}(G)$; see for example [Ab80], 3.1.4; let us mention in passing a misprint in loc. cit., p. 130, line 4: $M \simeq C^*/\mathrm{ann} M$ is to be replaced by $C^*/\mathrm{ann} M \simeq \mathrm{End}_k(M)$.

⁽¹¹⁸⁾**N.D.E.** We have added this paragraph, to translate Proposition 2.5.2, as well as the variant mentioned in N.D.E. (100), into the language of cocommutative Hopf algebras.

hence $H(G)$ is the semidirect product of $H(G_0)$ by $H(G_e)$. Moreover, since all the residue fields of G equal k , $H(G_e)$ is the group algebra kM , where $M = G_e(k)$ (cf. 2.5.A). Finally, since G_0 is infinitesimal, the morphism $G(k) \rightarrow G_e(k)$ is injective; it is therefore bijective (since it admits a section), so $M = G(k)$. Point (i) follows.

To prove (ii), it only remains to see that H_0 equals $H(G_0)$. Now the identity element 1_H of H is none other than the augmentation $\varepsilon_A : A \rightarrow k$, which is the identity section of $G(k)$; thus the local component of $\mathcal{A}(G)$ corresponding to H_0 is none other than $\mathcal{A}(G_0)$ and hence, by what we have seen above, one has $H_0 = \mathcal{A}(G_0)^\dagger = H(G_0)$. This proves the proposition.

Remarks 2.9.2. (a) The above proposition, implicitly contained in 2.5.2, was obtained independently by B. Kostant (cf. [Sw69], Preface). Combined with Cartier's Theorem 3.3 below (cf. [Ca62], § 12, Th. 3), also obtained by Kostant (cf. [Sw69], loc. cit.), this result is often called the "Cartier–Gabriel–Kostant theorem".

(b) In the form (ii), 2.9.1 was extended by R. G. Heyneman and M. E. Sweedler to the case where one assumes that H is pointed and a direct sum of its irreducible components (but is not necessarily cocommutative), cf. [HS69], Th. 3.5.8.

3. Phenomena particular to characteristic 0

Throughout Section 3, we assume that the pseudocompact ring k contains the field of rational numbers $\mathbf{Q}$.

Lemma 3.1. *Let C be a commutative unital $\mathbf{Q}$ -algebra and L a Lie algebra over C whose underlying C -module is free. Then the canonical map $L \rightarrow U(L)$ is an isomorphism from L onto the set of primitive elements of $U(L)$.*

Indeed, identify L with its canonical image in $U(L)$; let I be a totally ordered set and $(x_i)_{i \in I}$ a basis of L indexed by I ; denote by $\mathbf{N}^{(I)}$ the set of families $n = (n_i)_{i \in I}$ of natural numbers such that n_i is zero except perhaps for a finite number of indices $i_1 < i_2 < \dots < i_s$ (these indices depend on n); finally, put $x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \dots x_{i_s}^{n_{i_s}}$ and $n! = (n_{i_1}!)(n_{i_2}!) \dots (n_{i_s}!)$.

One then knows that the x^n form a basis of $U(L)$ (the Poincaré–Birkhoff–Witt theorem) and one sees easily that one has

$$(*) \quad \Delta_{U(L)} \left(\frac{x^n}{n!} \right) = \sum \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!},$$

the sum being taken over all the elements m of $\mathbf{N}^{(I)}$ such that $0 \leq m \leq n$ (i.e. such that $0 \leq m_i \leq n_i$ for every i). It clearly follows that one has $\Delta_{U(L)}(u) = u \otimes 1 + 1 \otimes u$ if and only if u is a linear combination of the x_i .

3.2. — Suppose now that C is *artinian*, with radical $\mathfrak{r}$. For every (associative, unital) C -algebra U , the ideal $\mathfrak{r}U$ therefore consists of nilpotent elements; if x belongs to $\mathfrak{r}U$, we shall put

$$\exp_U x = 1 + x + \frac{x^2}{2!} + \dots \quad (119)$$

One thus obtains a bijection from $\mathfrak{r}U$ onto $1 + \mathfrak{r}U$; the inverse bijection maps an element $1 + y$ of $1 + \mathfrak{r}U$ to

$$\log_U(1 + y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \dots$$

Moreover, it is clear that the map $\exp_U$ is functorial in U .⁽¹²⁰⁾

⁽¹¹⁹⁾**N.D.E.** If $x, x' \in \mathfrak{r}U$ commute, one therefore has $\exp_U(x + x') = (\exp_U x)(\exp_U x')$.

⁽¹²⁰⁾**N.D.E.** i.e. for every morphism $\phi : U \rightarrow V$ of C -algebras, one has $\phi(\exp_U(x)) = \exp_V \phi(x)$.

The ring C still being artinian, suppose that U is equipped with a bialgebra structure over C (cf. 2.2). For every primitive element x of $\mathfrak{r}U$ (cf. VII A 3.2.3), one then has

$$\begin{aligned}\Delta_U(\exp_U x) &= \exp_{U \otimes U}(\Delta_U(x)) \\ &= \exp_{U \otimes U}(x \otimes 1 + 1 \otimes x) \\ &= \exp_{U \otimes U}(x \otimes 1) \cdot \exp_{U \otimes U}(1 \otimes x) \\ &= ((\exp_U x) \otimes 1) \cdot (1 \otimes (\exp_U x)) \\ &= (\exp_U x) \otimes (\exp_U x).\end{aligned}$$

One therefore sees that the bijection $\exp_U$ transforms a primitive element of $\mathfrak{r}U$ into an element z of $1 + \mathfrak{r}U$ such that $\Delta_U(z) = z \otimes z$. Conversely, if z satisfies these conditions then, putting $x = \log_U(z)$, the preceding calculation shows that $\exp_{U \otimes U}(x \otimes 1 + 1 \otimes x)$ equals $z \otimes z = \Delta(\exp_U x) = \exp_{U \otimes U}(\Delta_U(x))$, whence $x \otimes 1 + 1 \otimes x = \Delta_U(x)$.⁽¹²¹⁾ Note further that if $z \in 1 + \mathfrak{r}U$ satisfies $\Delta_U(z) = z \otimes z$, then $\varepsilon_U(z)^2 = \varepsilon_U(z)$ and since $\varepsilon_U(z)$ is invertible (since z is, $\mathfrak{r}$ being nilpotent), it follows that $\varepsilon_U(z) = 1$.

Consider in particular a Lie algebra L flat over C , take for U the enveloping algebra $U(L)$ of L over C and identify L with its canonical image in U . By Lemma 3.1, L is therefore the set of primitive elements of U (indeed L is a product of free modules over the local components of C). Then consider the formal C -group $\mathcal{G}(L) = \mathrm{Spf}^* U(L)$, which has $U = U(L)$ for its covariant bialgebra (cf. 2.6.2). Let $\mathfrak{m}$ be a maximal ideal of C and $\overline{C} = C/\mathfrak{m}$. Since $\mathcal{G}(L)$ is infinitesimal (loc. cit.), the identity element of $\overline{U} = U \otimes_C \overline{C}$ is the only element $\bar{z}$ of $\overline{U}$ such that⁽¹²²⁾ $\varepsilon_{\overline{U}}(\bar{z}) = 1$ and $\Delta_{\overline{U}}(\bar{z}) = \bar{z} \otimes \bar{z}$. It follows that the elements z of U such that $\varepsilon_U(z) = 1$ and $\Delta_U(z) = z \otimes z$ necessarily belong to $1 + \mathfrak{r}U$.

Finally, since $L \cap \mathfrak{r}U$ is identified with $\mathfrak{r}L = L \otimes_C \mathfrak{r}$, one sees at last that: $\exp_U$ defines a bijection from $L \otimes_C \mathfrak{r}$ onto the group $\mathcal{G}(L)(C)$. We summarize:

Proposition. *Let k be a pseudocompact ring containing $\mathbf{Q}$ and $\mathbf{L}$ a flat $\mathbf{O}_k$ -Lie algebra.*

- (i) *For every object C of $\mathbf{Alf}/k$, denote its radical by $\mathfrak{r}(C)$; then the map*

$$\exp_{U(\mathbf{L}(C))} : \mathbf{L}(C) \otimes_C \mathfrak{r}(C) \longrightarrow \mathcal{G}(\mathbf{L})(C)$$

is bijective and functorial in C and $\mathbf{L}$.

- (ii) *The natural morphism $\mathbf{L} \longrightarrow \underline{\mathrm{Lie}}(\mathcal{G}(\mathbf{L}))$ (cf. 2.6.2) is an isomorphism of $\mathbf{O}_k$ -Lie algebras.*⁽¹²³⁾

3.2.1. — The bijection $\exp_{U(\mathbf{L}(C))}$ allows one to define, by transport of structure, a group law on the set $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$ (which one identifies with a subset of $U(\mathbf{L}(C))$ as in 3.2). If x and y are two elements of $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$, this law is such that

$$\begin{aligned}x \cdot y &= \log((\exp x)(\exp y)) = \log \left(1 + \sum_{p+q>0} \frac{x^p y^q}{p! q!} \right) \\ &= \sum_{m \geq 1} \sum_{p_i + q_i > 0} \frac{(-1)^{m-1}}{m} \frac{x^{p_1} y^{q_1}}{p_1! q_1!} \cdots \frac{x^{p_m} y^{q_m}}{p_m! q_m!} = \sum_{\ell \geq 1} P_\ell(x, y),\end{aligned}$$

⁽¹²¹⁾**N.D.E.** We have detailed the preceding passage and added the following sentence.

⁽¹²²⁾**N.D.E.** We have added the condition $\varepsilon_{\overline{U}}(\bar{z}) = 1$, omitted in the original.

⁽¹²³⁾**N.D.E.** We have added point (ii), an immediate consequence of 3.1.

where $P_\ell(x, y)$ denotes the sum of the monomials of total degree ℓ in x and y . One has for example:

$$\begin{aligned} P_1(x, y) &= x + y, \\ P_2(x, y) &= \underbrace{\frac{x^2}{2} + xy + \frac{y^2}{2}}_{m=1} - \underbrace{\frac{1}{2}(x^2 + xy + yx + y^2)}_{m=2} \\ &= \frac{1}{2}(xy - yx) = \frac{1}{2}[x, y], \end{aligned}$$

and $P_3(x, y)$ is the sum of the following three terms:

$$\begin{aligned} &\underbrace{\frac{x^3}{6} + \frac{x^2y}{2} + \frac{xy^2}{2} + \frac{y^3}{6}}_{m=1} \\ &- \frac{1}{2} \underbrace{\left(x^3 + \frac{3}{2}x^2y + \frac{1}{2}yx^2 + xyx + yxy + \frac{1}{2}y^2x + \frac{3}{2}xy^2 + y^3 \right)}_{m=2} \\ &+ \frac{1}{3} \underbrace{(x^3 + x^2y + yx^2 + xyx + yxy + y^2x + xy^2 + y^3)}_{m=3}, \end{aligned}$$

whence

$$\begin{aligned} P_3(x, y) &= \frac{1}{12}(x^2y + yx^2 - 2xyx - 2yxy + y^2x + xy^2) \\ &= \frac{1}{12}([y, x], x) + [y, [y, x]]. \end{aligned}$$

One can show more generally that one has the *Campbell–Hausdorff formula*:⁽¹²⁴⁾

$$\begin{aligned} P_\ell(x, y) &= \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m \cdot \ell} \sum_{\substack{p_1, \dots, p_m \\ q_1, \dots, q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right) \frac{(\operatorname{ad} x)^{p_m}}{p_m!} (y) \\ &\quad + \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m \cdot \ell} \sum_{\substack{p_1, \dots, p_{m-1} \\ q_1, \dots, q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!} \frac{(\operatorname{ad} y)^{q_i}}{q_i!} \right) (x), \end{aligned}$$

where the $p_j, q_i \in \mathbf{N}$ satisfy $p_i + q_i \geq 1$ for $i = 1, \dots, m-1$ and $p_m + \sum_{i=1}^{m-1} (p_i + q_i) = \ell - 1$ (i.e. in the above sums, every nonzero “Lie monomial” has total degree ℓ); for a proof, see N. Jacobson, *Lie Algebras* (Interscience, 1962), § V.5, or [BLie], II § 6.4, Th. 2.

3.3. — If G is a formal k -group with affine algebra A , recall that one denotes by I_A the augmentation ideal of A and by $\omega_{G/k}$ the pseudocompact k -module $I_A/\overline{I_A^2} \simeq I_A \hat{\otimes}_A k$.

Theorem. (Cartier). *Let k be a pseudocompact ring⁽¹²⁵⁾ containing $\mathbf{Q}$ and G a formal k -group. The following assertions are equivalent:*

- (i) *There exists a flat $\mathbf{O}_k$ -Lie algebra $\mathbf{L}$ such that G is isomorphic to $\mathcal{G}(\mathbf{L})$ (and in this case $\mathbf{L} = \underline{\operatorname{Lie}}(G)$ by 3.2).*
- (ii) *There exists a projective pseudocompact k -module ω such that the formal variety underlying G is isomorphic to the formal variety $V_k^{f,0}(\omega)$ (cf. 1.2.5) with affine algebra $k[[\omega]]$ (and in this case $\omega \simeq \omega_{G/k}$).*
- (iii) *G is infinitesimal and $\omega_{G/k}$ is a projective pseudocompact k -module.*
- (iv) *G is infinitesimal and topologically flat over k .*

⁽¹²⁴⁾ **N.D.E.** We have corrected the formula given, which was erroneous, and added the reference [BLie].

⁽¹²⁵⁾ **N.D.E.** We have removed the hypothesis that k be local (the proof reduces to this case).

PROOF. (i) $\Rightarrow$ (ii): Let $\omega = \Gamma^*(\mathbf{L})$ be the pseudocompact k -module dual to $\mathbf{L}$ (cf. 1.2.3.D). For every object C of $\mathbf{AIf}/k$, we must exhibit an isomorphism from $V_k^{f,0}(\omega)(C)$ onto $\mathcal{G}(\mathbf{L})(C)$ which is functorial in C . By 1.2.5, $V_k^{f,0}(\omega)(C)$ is identified with the set $\text{Hom}_c(\omega, \mathfrak{r}(C))$ of continuous k -linear maps from ω to the radical of C . This set is naturally isomorphic to the set $\text{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$ of continuous C -linear maps from $\omega \widehat{\otimes}_k C$ to $\mathfrak{r}(C)$; finally, since $\omega \widehat{\otimes}_k C$ is a projective pseudocompact C -module, the canonical map

$$(\omega \widehat{\otimes}_k C)^\dagger \otimes_C \mathfrak{r}(C) \longrightarrow \text{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

is bijective (cf. 0.3.6.A). Since, by 1.2.3.E, $\mathbf{L}(C)$ is identified with $V_k^f(\Gamma^*(\mathbf{L}))(C) = (\omega \widehat{\otimes}_k C)^\dagger$, one obtains that $V_k^{f,0}(\omega)(C)$ is canonically isomorphic to $\mathbf{L}(C) \otimes_C \mathfrak{r}(C)$, which is canonically isomorphic to $\mathcal{G}(\mathbf{L})(C)$ by Proposition 3.2. This proves the implication (i) $\Rightarrow$ (ii).

(ii) $\Rightarrow$ (iii): Let ω be a projective object of $\mathbf{PC}(k)$ and h an isomorphism from $k[[\omega]]$ onto the affine algebra A of G . Composing h with the augmentation $\varepsilon_A : A \rightarrow k$, one obtains a homomorphism $\varepsilon_A \circ h : k[[\omega]] \rightarrow k$, which is determined by its restriction λ to ω ; the latter sends ω into the radical $\mathfrak{r}$ of k . Thus the map $x \mapsto x - \lambda(x)$, from ω to the radical of $k[[\omega]]$, extends, by the universal property of $k[[\omega]]$ (cf. 1.2.5), to an endomorphism ℓ_λ of $k[[\omega]]$. The equalities $\ell_\lambda \circ \ell_{-\lambda} = \ell_{-\lambda} \circ \ell_\lambda = \text{id}$ show that ℓ_λ is an automorphism of $k[[\omega]]$. Consequently, $h \circ \ell_\lambda$ is, like h , an isomorphism from $k[[\omega]]$ onto A and moreover $\varepsilon_A \circ h \circ \ell_\lambda$ maps ω to 0. Replacing h by $h \circ \ell_\lambda$ if necessary, one can therefore suppose that $\varepsilon_A \circ h$ vanishes on the closed ideal I of $k[[\omega]]$ generated by ω . In this case, h induces an isomorphism from $I/\overline{I^2}$ onto $I_A/\overline{I_A^2}$; since $I/\overline{I^2} \simeq \omega$, it follows that $\omega_{G/k} = I_A/\overline{I_A^2}$ is isomorphic to ω , hence projective. It is clear, on the other hand, that $V_k^{f,0}(\omega)$ is infinitesimal, as is G .

(iii) $\Rightarrow$ (i): Suppose that G is infinitesimal and that $\omega_{G/k}$ is projective. Let $\mathbf{L}$ be the $\mathbf{O}_k$ -Lie algebra of G ; its underlying $\mathbf{O}_k$ -module is $V_k^f(\omega_{G/k})$, by 2.6.1. Consequently, $\mathbf{L}$ is flat over $\mathbf{O}_k$, and $\Gamma^*(\mathbf{L}) \simeq \omega_{G/k}$, by Proposition 1.2.3.E. Thus, by the proof of (i) $\Rightarrow$ (ii), the affine algebra of the formal k -group $\mathcal{G}(\mathbf{L})$ is identified with $k[[\omega_{G/k}]]$. On the other hand, by 2.6.3, the identity morphism of $\mathbf{L}$ is canonically associated with a morphism of formal groups $\mathcal{G}(\mathbf{L}) \rightarrow G$, hence with a continuous morphism of k -algebras

$$\phi : A \rightarrow k[[\omega_{G/k}]].$$

Let I be the closed ideal of $k[[\omega_{G/k}]]$ generated by $\omega_{G/k}$; filter $k[[\omega_{G/k}]]$ (resp. A) by the closures of the ideals I^n (resp. I_A^n). We have to show that ϕ , which by definition induces the identity on $\omega_{G/k} = I_A/\overline{I_A^2} = I/\overline{I^2}$, is an isomorphism.

Since $\omega_{G/k}$ is a projective object of $\mathbf{PC}(k)$, there exists a section τ of the canonical projection $I_A \rightarrow \omega_{G/k}$. By the universal property of $k[[\omega_{G/k}]]$ (cf. 1.2.5), τ defines a continuous morphism of algebras

$$\psi : k[[\omega_{G/k}]] \rightarrow A$$

and $\phi \circ \psi$ induces the identity map on $\omega_{G/k}$, hence also on the associated graded of $k[[\omega_{G/k}]]$. It follows that $\phi \circ \psi$ is an isomorphism, by [CA], § V.7, Lemma 1.⁽¹²⁶⁾

Moreover, ψ induces an isomorphism from $I/\overline{I^2}$ onto $I_A/\overline{I_A^2}$, hence a *surjection* of the associated graded of $k[[\omega_{G/k}]]$ and A . On the other hand, since G is radicial, I_A is contained in the radical of A , so that the intersection of the $\overline{I_A^n}$ is zero. Thus, by loc. cit., ψ is surjective. Then, since $\phi \circ \psi$ is an isomorphism and ψ a surjection, ψ and ϕ are isomorphisms. This proves that (iii) $\Rightarrow$ (i).

Note finally that it is clear that (i) or (ii) imply (iv), so that it remains to prove the implication (iv) $\Rightarrow$ (ii). For this, one can suppose that k is local, with residue field k_0 . Put then $G_0 = G \widehat{\otimes}_k k_0$, $\omega = \omega_{G/k}$, $\omega_0 = \omega \widehat{\otimes}_k k_0$, etc.⁽¹²⁷⁾ Since k_0 is a field, the pseudocompact

⁽¹²⁶⁾**N.D.E.** Cf. 5.1.5 below; see also [BAC], III, § 2.8, Th. 1 and corollaries. On the other hand, we have detailed the original in what follows.

⁽¹²⁷⁾**N.D.E.** We have detailed the original in what follows.

k_0 -module ω_0 possesses a pseudobasis $(y_\lambda)_{\lambda \in \Lambda}$; denoting by ω' the topologically free k -module which is the product of copies k_λ of k , for $\lambda \in \Lambda$, and lifting each y_λ to an element x_λ of ω , one obtains a continuous k -linear map $f : \omega' \rightarrow \omega$ such that $f_0 = f \otimes_k k_0$ is invertible.⁽¹²⁸⁾ Since ω' is a projective pseudocompact k -module, f lifts to a continuous k -linear map $g : \omega' \rightarrow I_A$ such that $\pi \circ g = f$, where π is the projection $I_A \rightarrow \omega$. By the universal property of $k[[\omega']]$ (cf. 1.2.5), g induces a morphism of topological algebras $\phi : k[[\omega']] \rightarrow A$.

Now $\omega' \otimes_k k_0$ is identified with $\omega \otimes_k k_0 = \omega_{G_0/k_0}$ and hence $k[[\omega']] \otimes_k k_0$ is identified with $k_0[[\omega_{G_0/k_0}]]$. Since the hypotheses (iii) hold for k_0 and G_0 , the proof of (iii) $\Rightarrow$ (i) shows that $\phi_0 = \phi \otimes_k k_0$ is invertible. Since, on the other hand, $k[[\omega']]$ and A are projective pseudocompact k -modules, ϕ is invertible by 0.3.4. (In particular, denoting by I the augmentation ideal of $k[[\omega']]$, ϕ induces an isomorphism from $\omega' = I/\overline{I^2}$ onto $I_A/\overline{I_A^2} = \omega$.) $\square$

Corollary 3.3.1. *Let S be a locally noetherian scheme over $\mathbf{Q}$ and G a flat group S -scheme locally of finite type.⁽¹²⁹⁾ If $\omega_{G/S}$ is a locally free $\mathcal{O}_S$ -module, G is smooth over S at every point of the identity section.*

Indeed, let x be a point of the identity section, s its image in S , and $\mathcal{O}_x$ and $\mathcal{O}_s$ the local algebras of x and s .⁽¹³⁰⁾ Since, by hypothesis, $\mathcal{O}_s$ and $\mathcal{O}_x$ are noetherian local rings, the $\mathfrak{m}$ -adic topology on each of these rings coincides with the “pseudocompact” topology defined by the ideals of finite codimension. Denote by $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ the completions for this topology. By EGA IV₄, 17.5.3, G is smooth over S at x if and only if $\widehat{\mathcal{O}}_x$ is formally smooth over $\widehat{\mathcal{O}}_s$, both these algebras being equipped with the $\mathfrak{m}$ -adic topology; it therefore suffices to show this last property. Now $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ are the local rings of x and s in the formal varieties $\widehat{G}/\widehat{S}$ and $\widehat{S}$ defined in 1.2.6. Put $k = \widehat{\mathcal{O}}_s$ and $H = \mathrm{Spf}(\widehat{\mathcal{O}}_x)$; then H is an infinitesimal formal k -variety and, since the formation of $\widehat{G}/\widehat{S}$ commutes with products (loc. cit.), H is an infinitesimal formal k -group. Denote by I the augmentation ideal of $\mathcal{O}_x$. By the hypotheses, $\omega_{G/S,x} = I/I^2$ is a locally free $\mathcal{O}_s$ -module of finite rank n . Since $\mathcal{O}_s$ is noetherian, it follows that $\omega_{H/k}$, which is the completion of $\omega_{G/S,x}$, is identified with $\omega_{G/S} \otimes_{\mathcal{O}_s} \widehat{\mathcal{O}}_s$, hence is a topologically free k -module of rank n . Thus, by the implication (iv) $\Rightarrow$ (ii) of 3.3, $\widehat{\mathcal{O}}_x$ is isomorphic to $k[[\omega_{H/k}]]$, i.e. to a formal power series algebra $k[[t_1, \dots, t_n]]$. Finally, the latter is formally smooth over k , by EGA 0_{IV}, 19.3.3. This proves the corollary.

We thus recover a result obtained elsewhere for group schemes locally of finite presentation over an arbitrary scheme S , cf. VI B, 1.6.

Corollary 3.3.2. *Let k be a field of characteristic 0. The functor $L \mapsto \mathcal{G}(L)$ is an equivalence from the category of k -Lie algebras onto that of infinitesimal formal k -groups.⁽¹³¹⁾*

⁽¹²⁸⁾ **N.D.E.** One does not know a priori that ω is a projective pseudocompact k -module, but this will follow from what comes next: compare with the proof of (iv) $\Rightarrow$ (iii) in VII A, 7.4.

⁽¹²⁹⁾ **N.D.E.** We have added the flatness hypothesis, omitted in the original.

⁽¹³⁰⁾ **N.D.E.** We have detailed the original in what follows.

⁽¹³¹⁾ **N.D.E.** Since, by 2.7, 2.2.1 and 1.3.6, an infinitesimal formal k -group G is “the same thing” as a connected cocommutative k -bialgebra H (cf. 2.9), this statement is equivalent to the theorem below, obtained independently by Kostant (cf. 2.9.2). This theorem had previously been obtained by J. W. Milnor and J. C. Moore ([MM65]), under the additional hypothesis that H be generated as an algebra by its primitive elements (although published in 1965, this text had circulated before 1960, cf. the review [Ja65]), so that it is often called the “Cartier–Kostant–Milnor–Moore theorem”.

Theorem (Cartier–Kostant–Milnor–Moore). — *Let k be a field of characteristic 0. The functors $L \mapsto U(L)$ and $H \mapsto \mathrm{Prim}(H)$ define an equivalence between the category of k -Lie algebras and that of connected cocommutative k -bialgebras.*

On the other hand, if k is an artinian ring containing $\mathbf{Q}$, then 3.2 (ii) and 3.3 (combined with 2.7, 2.2.1 and 1.3.6) likewise show that the functor $L \mapsto \mathcal{G}(L)$ (resp. $L \mapsto U(L)$) is an equivalence from the category of flat k -Lie algebras onto that of topologically flat infinitesimal formal k -groups (resp. onto that of flat connected cocommutative k -bialgebras).

Indeed, when G ranges over the infinitesimal formal k -groups, the functor $G \mapsto \mathcal{G}(\text{Lie } G)$ is isomorphic, by Theorem 3.3, to the identity functor of the category of infinitesimal k -groups. Similarly, by 3.2 (ii), the functor $L \mapsto \text{Lie}(\mathcal{G}(L)) = \text{Prim } U(L)$ is isomorphic to the identity functor of the category of k -Lie algebras.

3.3.3. — Still suppose that k is a field of characteristic 0. Let $\bar{k}$ be an algebraic closure of k and Γ the topological Galois group of $\bar{k}$ over k .

For every formal k -group G , we denote by $\underline{\text{Aut}}_k(G)$ the k -group functor which associates to every finite-dimensional commutative k -algebra C the group of automorphisms of the formal C -group $G \hat{\otimes}_k C$.⁽¹³²⁾ Since G is topologically flat over k (since k is a field), i.e. its affine algebra $A = A(G)$ is topologically flat over k , or, equivalently, its covariant bialgebra $H = H(G)$ is flat over k , this k -functor is a formal k -group. Indeed, since $\underline{\text{Aut}}_k(G)$ is identified with the fibered product of the following diagram (cf. Exp. I, 1.7.3):

$$\begin{array}{ccc} & \underline{\text{End}}_k(G) \times \underline{\text{End}}_k(G) & \\ & \downarrow & \\ \text{Spf}(k) & \longrightarrow & \underline{\text{End}}_k(G) \times \underline{\text{End}}_k(G), \end{array}$$

where the vertical (resp. horizontal) arrow is given by $(\phi, \psi) \mapsto (\phi \circ \psi, \psi \circ \phi)$ (resp. is the identity section), it suffices to see that the k -functor $\underline{\text{End}}_k(G)$ is represented by an element of $\mathbf{Vaf}/k$, i.e. (cf. 1.1 and 0.4.2) that it is left exact, i.e. commutes with fibered products of k -algebras. Now giving an element of $\underline{\text{End}}_k(G)(C)$ is equivalent to giving, say, a morphism of k -algebras $\phi : H \rightarrow H \otimes_k C$ which also respects the coalgebra structure, i.e. such that the well-known diagrams commute; since H is flat over k , $H \otimes_k -$ commutes with fibered products of k -algebras, and one deduces that the k -functor $\underline{\text{End}}_k(G)$ is left exact, so that we can treat it as a formal k -group.

If L is a k -Lie algebra and G the formal group $\mathcal{G}(L)$, Theorem 3.3 shows that $\underline{\text{Aut}}_k(G)$ is isomorphic to the k -group functor $\underline{\text{Aut}}_k(L)$ which associates to a finite-dimensional k -algebra C the group of automorphisms of the C -Lie algebra $C \otimes_k L$.

If G is an arbitrary formal k -group, we have seen in 2.5.2 that G decomposes canonically into a semidirect product of an étale formal group G_e and an infinitesimal formal group G_0 . This semidirect product is determined by the data of $L = \text{Lie}(G) = \text{Lie}(G_0)$, of the Γ -group $M = G_e(\bar{k}) = G(\bar{k})$, and of a morphism of formal k -groups

$$\Phi : G_e \rightarrow \underline{\text{Aut}}_k(G_0) \simeq \underline{\text{Aut}}_k(L).$$

Such a morphism sends G_e into the “étale part” $\underline{\text{Aut}}_k(L)_e$ of $\underline{\text{Aut}}_k(L)$, cf. 2.5.1. It is therefore determined by the data of a morphism of Γ -groups:

$$\phi = \Phi(\bar{k}) : M \rightarrow (\underline{\text{Aut}}_k L)(\bar{k}) = \text{Aut}_{\bar{k}}(L \otimes_k \bar{k}).$$

If one makes Γ act on $L \otimes_k \bar{k}$ by means of the formula $\gamma(\ell \otimes t) = \ell \otimes \gamma(t)$, then ϕ makes M act on $L \otimes_k \bar{k}$ by automorphisms of $\bar{k}$ -Lie algebra in such a way that one has $\phi(\gamma(m)) = \gamma \circ \phi(m) \circ \gamma^{-1}$, i.e. :

$$\gamma(m)(\ell \otimes t) = \gamma(m(\ell \otimes \gamma^{-1}(t)))$$

for every $m \in M$, $\gamma \in \Gamma$ and $\ell \otimes t \in L \otimes_k \bar{k}$. We express this last condition by saying that M acts on $L \otimes_k \bar{k}$ compatibly with Γ .

One can summarize the situation with a “learned” statement: call a Γ -Lie algebra over k the data of a triple (L, M, ϕ) consisting of a k -Lie algebra L , a Γ -group M , and an action ϕ of M on $L \otimes_k \bar{k}$ which is “compatible with the action of Γ on M and on $L \otimes_k \bar{k}$ ”.

If (L, M, ϕ) and (L', M', ϕ') are two such Γ -Lie algebras, a morphism from the former to the latter is a pair (f, θ) consisting of a morphism $f : L \rightarrow L'$ of k -Lie algebras and a

⁽¹³²⁾**N.D.E.** The original indicated: “This k -functor is manifestly left exact (H is topologically flat over k !)”. We have detailed this in what follows.

morphism $\theta : M \longrightarrow M'$ of Γ -groups such that

$$(f \otimes 1)(m \cdot x) = \theta(m) \cdot (f \otimes 1)(x)$$

for every $x \in L \otimes_k \bar{k}$ and $m \in M$. One then obtains:

Theorem. *Let k be a field of characteristic 0. Then the functor which associates to G the triple $(\text{Lie}(G), G(\bar{k}), \Phi(\bar{k}))$ is an equivalence from the category of formal k -groups onto that of Γ -Lie algebras.⁽¹³³⁾*

4. Phenomena particular to characteristic $p > 0$

Throughout Section 4, we denote by p a prime number, by k a pseudocompact ring of characteristic p , and by π the continuous endomorphism $x \mapsto x^p$ of k .

4.1. — Let X be a formal k -variety with affine algebra A ; one denotes by $X^{(p/k)}$ or $X^{(p)}$ the formal k -variety $X \hat{\otimes}_{\pi} k$ obtained by the base change $\pi : k \longrightarrow k$ (cf. 1.2.D); its affine algebra is the completed tensor product $A \hat{\otimes}_{\pi} k$. Then the continuous morphism $A \hat{\otimes}_{\pi} k \longrightarrow A$ which maps $a \hat{\otimes}_{\pi} \lambda$ to $a^p \lambda$ induces a morphism of formal k -varieties

$$\text{Fr}(X/k) : X \longrightarrow X^{(p/k)}.$$

In what follows, we shall say that $\text{Fr}(X/k)$ is the *Frobenius morphism of X relative to k* and we shall often write Fr instead of $\text{Fr}(X/k)$.

4.1.1. — ⁽¹³⁴⁾Now consider a scheme S over the prime field $\mathbf{F}_p$ and an S -scheme $\eta : X \longrightarrow S$; let $\text{fr}(S)$ be the “absolute” Frobenius morphism of S (it induces the identity on the underlying topological space and maps every section f of the structure sheaf to f^p ; cf. VII A 4.1) and let $X^{(p)}$ be the S -scheme $X \times_{\text{fr}(S)} S$ (VII A, loc. cit.), i.e. the structure morphism $X^{(p)} \longrightarrow S$ is $\text{fr}(S) \circ \eta$.

Let $\hat{S}$ be the formal scheme whose underlying topological space is the set of closed points of S , equipped with the discrete topology, the local ring $\mathcal{O}_{\hat{S},s}$ at such a closed point s being the separated completion $\hat{\mathcal{O}}_{S,s}$ of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength (cf. 1.2.6); its affine algebra $k = \mathcal{A}(\hat{S})$ is therefore the product of the $\hat{\mathcal{O}}_{S,s}$, for s ranging over the closed points of S . Recall (loc. cit.) that one denotes by $\hat{X}/\hat{S}$ the formal k -variety consisting of the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S , the local ring $\mathcal{O}_{\hat{X}/\hat{S},x}$ being the separated completion $\hat{\mathcal{O}}_{X,x}$ of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$ -module (and hence also as an $\mathcal{O}_{S,s}$ -module). One can therefore form, by base change, the formal k -variety $(\hat{X}/\hat{S})^{(p)} = (\hat{X}/\hat{S}) \hat{\otimes}_{\pi} k$.

One can also consider the formal k -variety $\widehat{X^{(p)}}/\hat{S}$: the underlying set consists of the $x \in X^{(p)} = X$ such that, denoting by s the image of x in S , the morphism

$$\kappa(s) \xrightarrow{\text{fr}} \kappa(s) \xrightarrow{\eta^{\sharp}} \kappa(x)$$

makes $\kappa(x)$ a finite-degree extension of $\kappa(s)$; in this case, the same holds for $\eta^{\sharp} : \kappa(s) \longrightarrow \kappa(x)$, i.e. x is a point of $\hat{X}/\hat{S}$, and one then has

$$\mathcal{O}_{\widehat{X^{(p)}}/\hat{S},x} = \hat{\mathcal{O}}_{X,x} \hat{\otimes}_{\pi} \hat{\mathcal{O}}_{S,s} = \hat{\mathcal{O}}_{X^{(p)},x}$$

(the second equality because $X \mapsto \hat{X}/\hat{S}$ commutes with products, cf. 1.2.6). One therefore sees that: $\widehat{X^{(p)}}/\hat{S}$ is identified with a formal subvariety of $(\hat{X}/\hat{S})^{(p)}$. Moreover, one has

⁽¹³³⁾**N.D.E.** In particular, when $\bar{k} = k$, one thus recovers the “Cartier–Gabriel–Kostant theorem” mentioned in 2.9.2.

⁽¹³⁴⁾**N.D.E.** We have corrected the original, which gave the inclusion $(\hat{X}/\hat{S})^{(p)} \subset \widehat{X^{(p)}}/\hat{S}$ instead of the reverse inclusion. Let us mention, on the other hand, that this paragraph is not used in what follows.

equality if and only if for every closed point s of S , $\kappa(s)$ has finite degree over $\kappa(s)^p$, which is the case, for example, if S is a scheme locally of finite type over a field κ such that $[\kappa : \kappa^p] < +\infty$.

4.1.2. — Let G be a formal k -group. Since the functor $X \mapsto X^{(p)} = X \widehat{\otimes}_{\pi} k$ commutes with finite products, it transforms a formal k -group into a formal k -group. Moreover, since $\text{Fr}(X/k)$ is functorial in X , the morphism

$$\text{Fr} : G \longrightarrow G^{(p)}$$

is a homomorphism of formal k -groups. If one puts $G^{(p^{n+1})} = (G^{(p^n)})^{(p)}$, the same holds for the composite morphism

$$\text{Fr}^n = \text{Fr}^n(G/k) : G \xrightarrow{\text{Fr}} G^{(p)} \xrightarrow{\text{Fr}} G^{(p^2)} \xrightarrow{\text{Fr}} \dots \xrightarrow{\text{Fr}} G^{(p^n)}.$$

Denote by A the affine algebra of G and by I_A its augmentation ideal.

Definitions. (a) The kernel of Fr^n will be denoted by ${}_{\text{Fr}^n}G$. It follows directly from the definitions that ${}_{\text{Fr}^n}G$ is infinitesimal and has affine algebra the quotient $A/I_A^{\{p^n\}}$, where $I_A^{\{p^n\}}$ denotes the closed ideal of A generated by the p^n th powers of the elements of I_A .

(b) One says that G has *height* $\leq n$ if $I_A^{\{p^n\}} = 0$, that is, if one has ${}_{\text{Fr}^n}G = G$.

4.2. — It is clear that the Lie algebra $\underline{\text{Lie}}(G)$ of a formal k -group G is a Lie p -subalgebra of the algebra $H(G)$ (cf. 2.3). Indeed, one immediately reduces to the case where k is artinian; in this case, $\underline{\text{Lie}}(G)$ is the subset of $H(G)$ consisting of the elements x such that $\varphi_G(x \otimes 1 + 1 \otimes x) = \Delta_G(x)$ with the notation of 2.3 and 2.6 (c). One then has

$$\begin{aligned} \varphi_G(x^p \otimes 1 + 1 \otimes x^p) &= \varphi_G((x \otimes 1 + 1 \otimes x)^p) \\ &= \varphi_G(x \otimes 1 + 1 \otimes x)^p = \Delta_G(x)^p = \Delta_G(x^p). \end{aligned}$$

4.2.1. — Conversely, every Lie p -algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k -group functor. Indeed, denote by $U_p(\mathbf{L})$ the functor which associates to every object C of $\mathbf{AIf}/k$ the restricted enveloping algebra $U_p(\mathbf{L}(C))$ of the Lie p -algebra $\mathbf{L}(C)$ over C (VII A 5.3). By VII A 5.4, $U_p(\mathbf{L})$ is an $\mathbf{O}_k$ -bialgebra and therefore determines, by 2.2, a k -group functor $\text{Spf}^* U_p(\mathbf{L})$ which we shall henceforth denote by $\mathcal{G}_p(\mathbf{L})$.

Thus, $\mathcal{G}_p(\mathbf{L})(C)$ is the group of elements $z \in U_p(\mathbf{L}(C))$ of augmentation 1 and such that $\Delta_{U_p(\mathbf{L}(C))}(z) = z \otimes z$.

4.2.2. — Suppose that $\mathbf{L}$ is a Lie p -algebra flat over $\mathbf{O}_k$. Then, taking account of VII A 5.3.3, one shows as in point (i) of Proposition 2.6.2 that $U_p(\mathbf{L})$ is flat over $\mathbf{O}_k$; thus $\mathcal{G}_p(\mathbf{L})$ is a topologically flat formal k -group which has $U_p(\mathbf{L})$ for its covariant $\mathbf{O}_k$ -bialgebra.

⁽¹³⁵⁾By the proof of 2.6.2 (iii), for every k -algebra C of finite length, $\underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L}))(C)$ is the set of primitive elements of $U_p(\mathbf{L})(C) = U_p(\mathbf{L}(C))$ (see also VII A 3.2.3); moreover, by VII A 5.5.3, the canonical morphism $\mathbf{L} \longrightarrow U_p(\mathbf{L})$ induces an isomorphism of Lie p -algebras

$$\tau_{p,\mathbf{L}} : \mathbf{L} \xrightarrow{\sim} \underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L}))$$

(compare with 3.1 in characteristic 0).

The formal group $\mathcal{G}_p(\mathbf{L})$ can be characterized by a universal property. Indeed, every morphism h from $\mathcal{G}_p(\mathbf{L})$ to a formal group G induces a morphism $\underline{\text{Lie}}(h) : \underline{\text{Lie}}(\mathcal{G}_p(\mathbf{L})) \longrightarrow \underline{\text{Lie}}(G)$. Composing the latter with the isomorphism $\tau_{p,\mathbf{L}}$, one obtains a morphism $h' : \mathbf{L} \longrightarrow \underline{\text{Lie}}(G)$.

Proposition. *If k is a pseudocompact ring of characteristic $p > 0$, G a formal k -group and $\mathbf{L}$ a Lie p -algebra flat over $\mathbf{O}_k$, the map $h \mapsto h'$ defined above is a bijection*

$$\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}_p(\mathbf{L}), G) \xrightarrow{\sim} \text{Hom}_{p\text{-Lie}}(\mathbf{L}, \underline{\text{Lie}}(G)),$$

where the right-hand term denotes the set of morphisms of Lie p -algebras from $\mathbf{L}$ to $\underline{\text{Lie}}(G)$.

⁽¹³⁵⁾**N.D.E.** We have detailed what follows.

(136) Indeed, one immediately reduces to the case where k is artinian. Put $L = \mathbf{L}(k)$. By 2.3.1, $\text{Hom}_{\mathbf{Grf}/k}(\mathcal{G}_p(\mathbf{L}), G)$ is in bijection with the set of morphisms of unital algebras $h : U_p(L) \rightarrow H(G)$ such that the following diagram commutes:

$$\begin{array}{ccc} U_p(L) & \xrightarrow{h} & H(G) \\ \Delta_{U_p(L)} \downarrow & & \searrow \delta_G \\ U_p(L) \otimes U_p(L) & \xrightarrow{h \otimes h} & H(G) \otimes H(G) \end{array} \quad \begin{array}{c} \\ \\ \nearrow \varphi_G \end{array} \quad \begin{array}{c} \\ \\ H(G \times G) \end{array}.$$

Now h is defined by its restriction to L , which is a morphism of Lie p -algebras from L to the Lie p -algebra underlying $H(G)$, and commutativity of the diagram means that h maps L into the subset of $H(G)$ consisting of the x such that $\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x)$, which is none other than $\underline{\text{Lie}}(G)$, cf. 2.6 (c).

4.3. — We now propose to study the formal k -group $\mathcal{G}_p(\mathbf{L})$ in greater detail when $\mathbf{L}$ is a Lie p -algebra flat over $\mathbf{O}_k$.

For this, we first consider a ring C of characteristic p and a Lie p -algebra L over C whose underlying module is free with basis $(x_i)_{i \in I}$. Denote, moreover, by P the set of families $n = (n_i)_{i \in I}$ consisting of natural numbers such that $0 \leq n_i < p$ and such that the n_i are zero except perhaps for a finite number of them. If we equip I with a total order and call $i_1, i_2, \dots, i_r$ ($i_1 < i_2 < \dots < i_r$) the indices i such that $n_i \neq 0$, we can therefore put $|n| = n_{i_1} + \dots + n_{i_r}$ and

$$x^n = x_{i_1}^{n_{i_1}} \cdot x_{i_2}^{n_{i_2}} \cdots x_{i_r}^{n_{i_r}}, \quad n! = (n_{i_1}!) \cdots (n_{i_r}!).$$

One knows that the monomials $x^n/n!$ form a basis of $U_p(L)$ (VII A 5.3.3) and it is clear that one has

$$(*) \quad \Delta \left(\frac{x^n}{n!} \right) = \sum_{m \leq n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!},$$

the sum being taken over all $m \in P$ such that one has $m \leq n$ (i.e. $m_i \leq n_i$ for every $i \in I$). Formula (*) allows an easy determination of the natural filtration of $U_p(L)$ (cf. 1.3.6). Put $U = U_p(L)$, let U^+ be the two-sided ideal generated by L , and let $U_0 = C \cdot 1_U$. As in 1.3.6, one defines a filtration of U (by sub- C -coalgebras) by putting, for $n \geq 1$:

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

Formula (*) then implies that U_n is the free C -module with basis the monomials x^m such that $|m| \leq n$.

4.3.1. — With the notation of 4.3, let us determine the elements ξ of $U = U_p(L)$ such that $\varepsilon_U(\xi) = 1$ and $\Delta_U(\xi) = \xi \otimes \xi$. Every element ξ of U is written $\xi = \sum_{n \in P} \xi_n (x^n/n!)$, with $\xi_n \in C$. The condition $\varepsilon_U(\xi) = 1$ implies the equality $\xi_0 = 1$, where 0 denotes the element of P all of whose components are zero. The condition $\Delta_{U_p(L)}(\xi) = \xi \otimes \xi$ implies:

$$\sum_{m \leq n} \xi_n \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!} = \sum_{q,r} \xi_q \xi_r \frac{x^q}{q!} \otimes \frac{x^r}{r!},$$

that is,

$$\xi_{q+r} = \xi_q \xi_r \quad \text{if } q_i + r_i < p, \quad \xi_q \xi_r = 0 \quad \text{otherwise.}$$

If one denotes by ξ_i the value of ξ_n when $n_i = 1$ and $n_j = 0$ for $j \neq i$, these conditions mean that one has

$$\xi_n = \prod_i \xi_i^{n_i} \quad \text{if } n \in P, \quad \xi_i^p = 0 \quad \forall i \in I.$$

(136) **N.D.E.** The proof is identical to that of 2.6.3.

One derives from this:

Proposition. *Let k be a local pseudocompact ring⁽¹³⁷⁾ of characteristic $p > 0$, $\mathbf{L}$ a Lie p -algebra flat over $\mathbf{O}_k$, C an object of $\mathbf{Alf}/k$ and $\sqrt[p]{0_C}$ the ideal of C consisting of the elements x such that $x^p = 0$. There exists a bijection “functorial in C ”:*

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathcal{G}_p(\mathbf{L})(C).$$

By Remark 1.2.3.F, one can indeed choose a basis $({}^C x_i)_{i \in I}$ of $\mathbf{L}(C)$ over C in such a way that, for every morphism $\varphi : C \rightarrow D$ of $\mathbf{Alf}/k$, $\mathbf{L}(\varphi)$ maps ${}^C x_i$ to ${}^D x_i$. By the preceding, the map

$$\sum_i {}^C x_i \otimes \xi_i \mapsto \sum_{n \in P} \left(\prod_i \xi_i^{n_i} \right) \frac{{}^C x^n}{n!}$$

is a functorial bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C)$.

4.3.2. — “There is no Campbell–Hausdorff formula in characteristic $p > 0$.” Let us explain: the functorial isomorphism of 4.3.1 depends on the choice of the bases $({}^C x_i)_{i \in I}$. Unlike what happens in 3.2 (the characteristic 0 case), there is, in general, no bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C)$ which is “functorial in both C and $\mathbf{L}$ ”. The argument which follows shows, for example, that such an isomorphism does not exist when k is a field containing the $(p^2 - 1)$ th roots of unity.

Indeed, choose $\mathbf{L}$ in such a way that, for every $C \in \mathbf{Alf}/k$, $\mathbf{L}(C)$ is the abelian Lie p -algebra over C generated by an element x subject to the relation $x^{(p^2)} = 0$. One therefore has

$$\mathbf{L}(C) = Cx \oplus Cx^{(p)}, \quad U_p(\mathbf{L}(C)) \simeq k[x]/(x^{p^2}).$$

We shall show that the only functorial morphism

$$\chi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \rightarrow U_p(\mathbf{L}(C))$$

which is compatible with the endomorphisms of $\mathbf{L}$ and maps $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ into $\mathcal{G}_p(\mathbf{L})(C)$ is the constant morphism with value 1.

Indeed, let ψ_C be the bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathcal{G}_p(\mathbf{L})(C) = \text{Prim } U_p(\mathbf{L}(C))$ given by 4.3.1, i.e. :

$$x \otimes a + x^{(p)} \otimes b \mapsto \sum_{0 \leq i, j < p} a^i b^j x^{i+pj}.$$

Composing χ_C with ψ_C^{-1} , one obtains a morphism functorial (in C):

$$\begin{aligned} \theta_C : \quad \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} &\longrightarrow \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \\ x \otimes a + x^{(p)} \otimes b &\longmapsto x \otimes P(a, b) + x^{(p)} \otimes Q(a, b). \end{aligned}$$

Functoriality in C implies that $P(a, b)$ and $Q(a, b)$ are the values at (a, b) of two polynomials $P, Q \in k[x, y]$ which one can suppose to have degree $< p$ in X as well as in Y .⁽¹³⁸⁾ Let α be an element of k and ℓ_α the endomorphism of Lie p -algebra of $\mathbf{L}$ which maps x to αx (and hence $x^{(p)}$ to $\alpha^p x^{(p)}$). Then $U_p(\ell_\alpha)$ is the algebra endomorphism which sends x to αx (and hence each x^i to $\alpha^i x^i$, for $i < p^2$), and one sees easily that the square below

$$\begin{array}{ccc} \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \\ \ell_\alpha(C) \otimes \text{id} \downarrow & & \downarrow U_p(\ell_\alpha)(C) \\ \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(\mathbf{L}(C)) \end{array}$$

⁽¹³⁷⁾**N.D.E.** We have added the hypothesis that k be local, so that every topologically flat pseudocompact k -module is topologically free, cf. the proof.

⁽¹³⁸⁾**N.D.E.** We have detailed the original in what follows.

commutes. The hypothesis $\chi_C \circ (\ell_\alpha(C) \otimes \text{id}) = U_p(\ell_\alpha)(C) \circ \chi_C$ then implies the equalities:

$$P(\alpha a, \alpha^p b) = \alpha P(a, b) \quad \text{and} \quad Q(\alpha a, \alpha^p b) = \alpha^p Q(a, b).$$

Writing $P = \sum_{i,j < p} \lambda_{ij} X^i Y^j$ and $Q = \sum_{i,j < p} \mu_{ij} X^i Y^j$, and taking for C the algebra $k[X, Y]/(X^p, Y^p)$, one deduces that if $\lambda_{ij} \neq 0$ (resp. $\mu_{ij} \neq 0$) then $\alpha^{i-1+pj} = 1$ (resp. $\alpha^{i+p(j-1)} = 1$). Thus, taking for α a primitive root of unity of order $p^2 - 1$, one deduces that $P = \lambda X$ and $Q = \mu Y$, with $\lambda, \mu \in k$. Consequently, one has:

$$\chi_C(x \otimes a + x^{(p)} \otimes b) = \sum_{0 \leq i, j < p} (\lambda a)^i (\mu b)^j x^{i+pj}.$$

(138) Finally, consider the endomorphism f of $\mathbf{L}$ which sends x to $x + x^{(p)}$; taking $C = k[a]/(a^3)$ and comparing the coefficients of x^p and x^{p+1} in $(\chi_C \circ f(C))(x \otimes a)$ and in $(U_p(f)(C) \circ \chi_C)(x \otimes a)$, one obtains that $\lambda = \mu$ and $\lambda^2 a^2 = 0$, whence $\lambda = \mu = 0$.

Theorem 4.4. *Let k be a pseudocompact ring of characteristic $p > 0$ and G a formal k -group. The following assertions are equivalent:*

- (i) *There exists a Lie p -algebra $\mathbf{L}$ flat over $\mathbf{O}_k$ such that G is isomorphic to $\mathcal{G}_p(\mathbf{L})$ (and in this case $\mathbf{L} = \underline{\text{Lie}}(G)$ by 4.2.2).*
- (ii) *There exists a projective pseudocompact k -module ω such that the affine algebra of G is isomorphic to the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p , for $x \in \omega$ (and in this case $\omega \simeq \omega_{G/k}$).*
- (iii) *G has height ≤ 1 and $\omega_{G/k}$ is a projective pseudocompact k -module.*
- (iv) *G has height ≤ 1 and is topologically flat over k .*

PROOF. (i) $\Rightarrow$ (ii): Let $\omega = \Gamma^*(\mathbf{L})$ (cf. 1.2.3.D) and let A be the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p , for $x \in \omega$. Every continuous morphism $h : A \rightarrow C$, where C is an object of $\mathbf{AIf}/k$, is determined by its restriction h' to ω ; this restriction h' sends ω into $\sqrt[p]{0_C}$. One thus obtains a canonical bijection from $\text{Hom}_{\mathbf{AIf}/k}(A, C)$ onto the set $\text{Hom}_c(\omega, \sqrt[p]{0_C})$ of continuous k -linear maps from ω to $\sqrt[p]{0_C}$. This last set is canonically isomorphic to $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ (see the proof of 3.3). The implication (i) $\Rightarrow$ (ii) therefore follows from the functorial bijection $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathcal{G}_p(\mathbf{L})(C)$ established in 4.3.1.

For the other implications, consult the proofs of Theorems 3.3 and VII A 7.4.1, which are analogous. $\square$

Remark 4.4.A. (139) Let G be an infinitesimal formal k -group, with affine algebra A , such that $\omega_{G/k} = I_A/\overline{I_A^2}$ is a projective pseudocompact k -module. Then there exists a section $\tau : \omega_{G/k} \rightarrow I_A$ of the projection $I_A \rightarrow \omega_{G/k}$, and τ induces a continuous morphism of algebras $\psi : k[[\omega_{G/k}]] \rightarrow A$ which is *surjective*, cf. the proof of the implication (iii) $\Rightarrow$ (i) in 3.3.

Corollary 4.4.1. *If k is a field of characteristic $p > 0$, the functor $L \mapsto \mathcal{G}_p(L)$ is an equivalence from the category of Lie p -algebras over k onto that of formal k -groups of height ≤ 1 .*

Indeed, when G ranges over the formal groups of height ≤ 1 , the functor $G \mapsto \mathcal{G}_p(\text{Lie } G)$ is isomorphic to the identity functor by Theorem 4.4; similarly, we have seen in 4.2.2 that the functor $L \mapsto \text{Lie } \mathcal{G}_p(L)$ is isomorphic to the identity functor (see also VII A, 5.5.3).⁽¹⁴⁰⁾

4.4.2. — Still take for k a field of characteristic p . Let G be an infinitesimal formal k -group, with affine algebra A . Since G is infinitesimal, every open ideal of A contains $I_A^{\{p^n\}}$ for n large enough, so G is the projective limit of the affine algebras $A/I_A^{\{p^n\}}$ of the groups $\mathbb{F}_n G$ (cf.

(139) **N.D.E.** We have added this remark, used in 4.4.2.

(140) **N.D.E.** If k is an artinian ring of characteristic $p > 0$, the same proof gives an equivalence between the category of flat Lie p -algebras over k and that of topologically flat formal k -groups of height ≤ 1 .

4.1.2). *Every infinitesimal formal k -group is therefore an inductive limit of formal k -groups of finite height.*

Suppose that G has height $\leq n$ and put $H = G/\mathrm{Fr}G$.⁽¹⁴¹⁾ By 2.4 and 2.4.1, $\mathrm{Fr} : G \rightarrow G^{(p)}$ factors into an epimorphism $\pi : G \rightarrow H$ followed by a monomorphism $i : H \rightarrow G^{(p)}$. One then has the following commutative diagram:

$$\begin{array}{ccccc} G & \xrightarrow{\pi} & H & \xrightarrow{\mathrm{Fr}^{n-1}} & H^{(p^{n-1})} \\ & \searrow \mathrm{Fr} & \downarrow i & & \downarrow i^{(p^{n-1})} \\ & & G^{(p)} & \xrightarrow{\mathrm{Fr}^{n-1}} & G^{(p^n)} \end{array}$$

and since the functor $X \mapsto X^{(p)}$ commutes with fibered products, $i^{(p^{n-1})}$ is again a monomorphism. Since $\mathrm{Fr}^n(G/k)$ is zero and π is an epimorphism, $\mathrm{Fr}^{n-1}(G^{(p)}/k) \circ i = i^{(p^{n-1})} \circ \mathrm{Fr}^{n-1}(H/k)$ is zero and hence, since $i^{(p^{n-1})}$ is a monomorphism, $\mathrm{Fr}^{n-1}(H/k)$ is zero, so H has height $\leq n-1$. One therefore sees that: *every formal k -group of finite height possesses a composition series whose quotients have height ≤ 1 , hence can be described by Lie p -algebras over k .*

Finally, the affine algebra A of G is a quotient of $k[[\omega_{G/k}]]$, cf. 4.4.A, so if $\omega_{G/k}$ has finite dimension over k , all the algebras $A/I_A^{(p^n)}$ are finite-dimensional k -vector spaces. One therefore sees that: *every infinitesimal formal group G over a field of characteristic $p > 0$, such that $\omega_{G/k}$ has finite dimension over k , is an inductive limit of finite formal groups (i.e. of finite length, cf. 1.2.6).*

5. Homogeneous spaces of infinitesimal formal groups over a field

5.0. — ⁽¹⁴²⁾Suppose, for simplicity, that k is a field. Let G be a formal k -group with affine algebra $A = \mathcal{A}(G)$. Let A^+ be the augmentation ideal of A ; for every $a \in A$, denote by $\bar{a} = a - \varepsilon_A(a)1_A$ its projection onto A^+ . Let F be a formal subgroup of G , defined by the closed ideal J of A , and let π (resp. $\bar{\pi}$) be the projection $A \rightarrow A/J$ (resp. the composite of the projections $A \rightarrow A^+ \rightarrow A^+/J$). Note that, for every $a \in A$, the projection of $\Delta_A(a)$ onto $A \hat{\otimes} A^+$ is $\Delta_A(a) - a \hat{\otimes} 1$.

By Theorem 2.4, one can form the quotient formal k -variety $X = G/F$; its affine algebra B is

$$\begin{aligned} B &= \mathrm{Ker} \left(A \xrightarrow[\mathrm{id}_A \hat{\otimes} \pi \Delta_A]{\tau_1} A \hat{\otimes}_k (A/J) \right) \\ &= \{a \in A \mid (\mathrm{id}_A \hat{\otimes} \pi) \Delta_A(a) = a \hat{\otimes} \pi(1)\} \\ &= \mathrm{Ker}((\mathrm{id}_A \hat{\otimes} \bar{\pi}) \Delta_A). \end{aligned}$$

It is also the subalgebra of A consisting of the elements ϕ such that, for every $C \in \mathrm{Ob} \mathbf{AIf}/k$ and $g \in G(C)$, $h \in F(C)$, one has $\phi(gh) = \phi(g)$. For every $g, g' \in G(C)$ and $h \in F(C)$, one has $\phi(gg'h) = \phi(gg')$, so $\Delta(\phi)$ belongs to the kernel of $\mathrm{id}_A \hat{\otimes}_k (\mathrm{id}_A \hat{\otimes} \bar{\pi}) \Delta_A$, which equals $A \hat{\otimes}_k B$ since A is topologically flat over k . One therefore has $\Delta_A(B) \subset A \hat{\otimes}_k B$, i.e. the closed subalgebra B is also a *left coideal*.

On the other hand, B determines F since, by Corollary 2.4.1, one has $J = \overline{AB^+}$, i.e. J is the closed ideal generated by $B^+ = B \cap A^+$. One thus obtains an *injective* map $\mathcal{Q}$ from the set $\mathcal{F}$ of formal subgroups of G into the set $\mathcal{B}$ of closed subalgebras B of A which are left coideals. There then arises the question of determining the image of this map, and

⁽¹⁴¹⁾**N.D.E.** We have detailed the original in what follows.

⁽¹⁴²⁾**N.D.E.** We have added subsection 5.0, to express the following Proposition 5.1 in the language of cocommutative Hopf algebras, and to cite the results obtained since in this direction.

Proposition 5.1 below shows that $\mathcal{Q}$ is bijective when G is *infinitesimal*. In fact, this is true for *every* formal k -group G .

Indeed, recall (cf. 2.2.1) that the functor $G \rightarrow H(G)$ is an *equivalence* between the category of formal k -groups and that of cocommutative k -Hopf algebras; if F is a formal subgroup of G , defined by the closed ideal J of A , then the Hopf subalgebra $H(F)$ of $H = H(G)$ is the orthogonal of J for the duality between $A = H^*$ and $H = A^\dagger$ (cf. 0.2.2). On the other hand, if B is a closed subalgebra of A which is also a left coideal, then its orthogonal $I = B^\perp$ is a coideal of H (i.e. $\Delta_H(I) \subset I \otimes H + H \otimes I$ and $\varepsilon_H(I) = 0$) and a left ideal. Denote by $\mathcal{H}$ (resp. $\mathcal{I}$) the set of Hopf subalgebras (resp. left ideals which are coideals) of H . For every $I \in \mathcal{I}$, denote by π_I (resp. $\bar{\pi}_I$) the projection $H \rightarrow H/I$ (resp. the composite of the projections $H \rightarrow H^+ \rightarrow H^+/I$), where H^+ is the augmentation ideal of H .

Let K be a Hopf subalgebra of H and $K^+ = K \cap H^+$. If F is the formal subgroup corresponding to K , then $J = K^\perp$ and A^+/J is identified with the dual of K^+ , and since $B = \mathcal{Q}(F)$ is the kernel of the map

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\text{id} \hat{\otimes} \bar{\pi}} A \hat{\otimes}_k (A^+/J),$$

one obtains that $\mathcal{Q}$ corresponds by duality to the map Φ which associates to K the image of

$$H \xleftarrow{m_H} H \otimes_k H \xleftarrow{\text{id} \hat{\otimes} \text{can.}} H \otimes_k K^+,$$

i.e. the left ideal HK^+ , which is also a coideal. One sees similarly that the map which associates to $B \in \mathcal{B}$ the ideal $\overline{AB^+}$ corresponds by duality to the map Ψ which associates to every $I \in \mathcal{I}$ the kernel of $(\text{id} \otimes_k \bar{\pi}_I) \circ \Delta_H$, i.e. one has

$$\Psi(I) = \{x \in H \mid (\text{id} \otimes_k \pi_I)\Delta_H(x) = x \otimes \pi_I(1)\}.$$

One then has the following theorem:

Theorem 5.0.1. *Let k be a field and H a cocommutative Hopf algebra. Then the maps Φ and Ψ above are inverse bijections between the set of Hopf subalgebras of H and that of left ideals which are coideals.*

This theorem was first proved by K. Newman, cf. [Ne75], Th. 4.1 (where the word “cocommutative” was omitted). His proof uses “the Cartier–Gabriel–Kostant theorem” (cf. 2.9) to reduce to the case where H is connected, then the existence in this case of a “Sweedler basis” (cf. [Sw67], Th. 3), a result dual to the Dieudonné–Cartier Theorem 5.2.2 below. Another, shorter, proof was given by H. J. Schneider [Sch90], Th. 4.15. A generalization was subsequently obtained by A. Masuoka when one assumes only that the coradical H_0 of H (i.e. the sum of the simple subcoalgebras) is commutative, [Ma91], Th. 1.3 (3).

Let us mention finally that for a commutative k -Hopf algebra, thus corresponding to an affine group k -scheme G , one cannot expect an analogue of 5.0.1 without additional hypotheses, since for a k -subgroup F of G the quotient G/F is not necessarily affine. But M. Takeuchi established in [Tak72], Th. 4.3 (resp. [Tak79], Th. 3), an analogous bijection between the set of k -subgroups F of G which are invariant (resp. such that G/F is affine), and that of subalgebras B of $\mathcal{O}(G)$ such that $\Delta(B) \subset A \otimes B$ and which are stable under the antipode (resp. and such that $B \rightarrow A$ is faithfully flat).

5.1. — Let k be a pseudocompact ring.⁽¹⁴³⁾ Let G be a topologically flat infinitesimal formal k -group, A its affine algebra, B a closed subalgebra of A , $X = \text{Spf}(B)$ and $r : G \rightarrow X$ the

⁽¹⁴³⁾**N.D.E.** In the original, k is assumed in 5.1 to be a field. In fact, this hypothesis can be replaced by flatness hypotheses; we have modified nos. 5.1 to 5.1.5 accordingly.

epimorphism induced by the inclusion of B into A . One proposes to see under what condition r makes X the right quotient of G by a subgroup H (cf. 2.4).⁽¹⁴⁴⁾

Proposition. *Let G be a topologically flat infinitesimal formal k -group, A its affine algebra, I_A the augmentation ideal of A , B a closed subalgebra of A , and $J_B = \overline{AI_B}$, where $I_B = B \cap I_A$. Suppose that A is topologically flat over k , as is $\overline{J_B^n}/\overline{J_B^{n+1}}$ for every $n \geq 0$. Then the following two assertions are equivalent:*

- (i) *For every $x \in B$, $\Delta_A(x) - x \hat{\otimes} 1$ belongs to $A \hat{\otimes}_k J_B$.*
- (ii) *The sequence below (where $\tau_1(a) = a \hat{\otimes} 1$ and π is the projection $A \rightarrow A/J_B$) is exact:*

$$(*) \quad B \longrightarrow A \xrightarrow[\substack{\tau_1 \\ (\pi \hat{\otimes} \text{id}_A) \Delta_A}]{} A \hat{\otimes}_k (A/J_B),$$

i.e. B is the set of all $x \in A$ such that $\Delta_A(x) - x \hat{\otimes} 1$ belongs to $A \hat{\otimes}_k J_B$.

In this case, $H = \text{Spf}(A/J_B)$ is a formal subgroup of G , and the sequence below (where λ is the restriction to $G \times H$ of the multiplication of G) is exact:

$$(**) \quad G \times H \xrightarrow[\lambda]{\text{pr}_1} G \longrightarrow \text{Spf}(B),$$

i.e. $\text{Spf}(B)$ is isomorphic to G/H .

Put $\bar{A} = A/J_B$ and $H = \text{Spf}(\bar{A})$; then H is a formal subvariety of G . Since $J_B \subset I_A$, the augmentation ε_A induces a continuous morphism of k -algebras $\bar{\varepsilon} : \bar{A} \rightarrow k$.

If (i) holds, then $\Delta_A(I_B) \subset I_B \hat{\otimes} 1 + A \hat{\otimes}_k J_B$, and hence Δ_A induces, on passing to the quotient, a diagonal morphism $\bar{\Delta}$. Then $\bar{\Delta}$ and $\bar{\varepsilon}$ equip H with a structure of formal submonoid of G . Since G is infinitesimal, the same holds for H ; thus, by Proposition 2.7, H is a formal subgroup of G . It then follows from the definition of G/H (cf. 2.4) that (ii) implies the last assertion of the proposition.

On the other hand, it is clear that (ii) implies (i). The proof of the converse occupies paragraphs 5.1.1 to 5.1.5.

5.1.1. — First consider the following category $\mathcal{C}$: an object of $\mathcal{C}$ is a pair (A, J) consisting of a profinite k -algebra A and a closed ideal J of A ; a morphism $\psi : (A, J) \rightarrow (A', J')$ of $\mathcal{C}$ is a continuous homomorphism of k -algebras $A \rightarrow A'$ which maps J into J' . If one associates to (A, J) the pair $(\text{Spf}(A/J), \text{Spf}(A))$, one clearly obtains an antiequivalence from $\mathcal{C}$ onto the category of pairs (Z, Y) consisting of a formal k -variety Y and a formal subvariety Z , a morphism $\phi : (Z, Y) \rightarrow (Z', Y')$ being a morphism of formal k -varieties $Y \rightarrow Y'$ which maps Z into Z' .

A *cogroup* structure on an object (A, J) of $\mathcal{C}$ consists of the data of a formal group structure on $\text{Spf}(A)$ such that the following conditions hold (notation of 2.1):

- (1) $\Delta_A(J) \subset J \hat{\otimes}_k A + A \hat{\otimes}_k J$;
- (2) $\varepsilon_A(J) = 0$;
- (3) $c_A(J) \subset J$.

These conditions also mean that $H = \text{Spf}(A/J)$ is a formal subgroup of $G = \text{Spf}(A)$.⁽¹⁴⁵⁾

Suppose, moreover, that A is *local*, i.e. that $\text{Spf}(A)$ is an infinitesimal formal group. Then, if $J \neq A$, conditions (2) and (3) are consequences of (1). Indeed, if J is a closed ideal distinct from A , it is contained in the augmentation ideal I_A , so (2) holds, and $M = \text{Spf}(A/J)$ is

⁽¹⁴⁴⁾**N.D.E.** We have replaced “left” by “right” and modified the statement of Proposition 5.1, to bring out more clearly, on the one hand, the equivalent conditions (i), (ii), and, on the other hand, the conclusion $\text{Spf}(B) \simeq G/H$.

⁽¹⁴⁵⁾**N.D.E.** Note that if (A', J') is a second cogroup of $\mathcal{C}$, corresponding to a pair $H' \subset G'$ of formal k -groups, then giving a morphism of cogroups $(A', J') \rightarrow (A, J)$ is equivalent to giving a morphism of formal k -groups $G \rightarrow G'$ which maps H into H' .

a formal submonoid of G . Since G is infinitesimal, it follows from 2.7 that M is a formal subgroup of G , i.e. condition (3) holds.

5.1.2. — Denote by $\mathbf{Alpg}/k$ the category of “graded profinite” k -algebras: an object of this category consists of the data of a sequence $A_0, A_1, \dots, A_n, \dots$ of pseudocompact k -modules and a profinite algebra structure on the product $\prod_{n \geq 0} A_n$ such that one has $A_n \cdot A_m \subset A_{m+n}$ (A_n is identified with a subset of $\prod_{i \geq 0} A_i$ by means of the canonical injection); a morphism $\psi : (A_n) \rightarrow (B_n)$ is a sequence of continuous linear maps $\psi_n : A_n \rightarrow B_n$ such that one has $\psi_{m+n}(a \cdot a') = \psi_m(a) \cdot \psi_n(a')$ if $a \in A_m$ and $a' \in A_n$.

Definitions. It is clear that two graded profinite k -algebras (A_n) and (B_n) have a *coproduct*⁽¹⁴⁶⁾ in $\mathbf{Alpg}/k$, whose n th component is the product $\prod_{i=0}^n A_i \hat{\otimes}_k B_{n-i}$ of the pseudocompact k -modules $A_i \hat{\otimes}_k B_{n-i}$. This coproduct will be denoted by $(A_n) \hat{\otimes}_k (B_n)$.

Then a cogroup structure on an object (A_n) of $\mathbf{Alpg}/k$ is the data of continuous k -linear maps $\Delta_n : A_n \rightarrow \prod_{i=0}^n A_i \hat{\otimes}_k A_{n-i}$ and $\varepsilon : A_0 \rightarrow k$ which induce on $\prod_{n \geq 0} A_n$ (putting $\varepsilon(A_i) = 0$ for $i \geq 1$) a cogroup structure in $\mathbf{Alp}/k$.

Finally, for every object (A, J) of $\mathcal{C}$, one denotes by $\mathrm{Gr}_J(A)$ the graded profinite algebra associated with the filtration of A by the closures $\overline{J^n}$ of the powers of J ; one therefore has $\mathrm{Gr}_J(A)_n = \overline{J^n} / \overline{J^{n+1}}$ and the multiplication of $\mathrm{Gr}_J(A)$ is induced by that of A .

Lemma.⁽¹⁴⁷⁾ *Let U, V be two pseudocompact k -modules, with U topologically flat, and let $U = U_0 \supset U_1 \supset \dots$ and $V = V_0 \supset V_1 \supset \dots$ be two decreasing sequences of closed sub- k -modules. Filter the completed tensor product $W = U \hat{\otimes}_k V$ by means of the closed submodules*

$$W_n = U_n \hat{\otimes}_k V_0 + U_{n-1} \hat{\otimes}_k V_1 + \dots + U_0 \hat{\otimes}_k V_n.$$

Suppose that each U_i/U_{i+1} is topologically flat over k (so that U/U_n , hence also U_n , is as well, for every n). Then, for every n , one has an isomorphism

$$W_n/W_{n+1} \simeq \bigoplus_{i+j=n} (U_i/U_{i+1}) \hat{\otimes}_k (V_j/V_{j+1}).$$

PROOF. Put $W_{i,j} = U_i \hat{\otimes}_k V_j$ and $\overline{W}_{i,j} = (U_i/U_{i+1}) \hat{\otimes}_k (V_j/V_{j+1})$, for every $i, j \geq 0$. Let us show by induction on n that the natural map

$$\pi_n : W_n \rightarrow \bigoplus_{i+j=n} \overline{W}_{i,j}$$

is surjective and that the inclusion $W_{n+1} \subset \mathrm{Ker}(\pi_n)$ is an equality. For $n = 0$, the projection

$$\pi_0 : U_0 \hat{\otimes}_k V_0 \rightarrow (U_0/U_1) \hat{\otimes}_k (V_0/V_1)$$

is surjective and, since $U_0, U_0/U_1$ and hence U_1 are topologically flat over k , one sees that $\mathrm{Ker}(\pi_0) = U_0 \hat{\otimes}_k V_1 + U_1 \hat{\otimes}_k V_0$ and that, moreover, $U_0 \hat{\otimes}_k V_1 \cap U_1 \hat{\otimes}_k V_0 = U_1 \hat{\otimes}_k V_1$.

Suppose therefore that $n > 0$ and the result is established for $n - 1$. Put $M_0 = U_0 \hat{\otimes}_k V_n$ and $S_0 = \sum_{i=1}^n U_i \hat{\otimes}_k V_{n-i}$. One has $S_0 \subset U_1 \hat{\otimes}_k V_0$ and hence, by the preceding passage applied to $V_0 \supset V_n$ instead of $V_0 \supset V_1$, one has

$$M_0 \cap S_0 \subset U_0 \hat{\otimes}_k V_n \cap U_1 \hat{\otimes}_k V = U_1 \hat{\otimes}_k V_n,$$

whence one deduces that $M_0 \cap S_0 = U_1 \hat{\otimes}_k V_n$. Since $W_n = M_0 + S_0$, one obtains a commutative diagram with exact rows, where one has put $U'_i = U_{i+1}$ and $\overline{W}'_{i,n-1-i} = \overline{W}_{i+1,n-1-i}$ for

⁽¹⁴⁶⁾**N.D.E.** We have replaced “direct sum” by “coproduct”.

⁽¹⁴⁷⁾**N.D.E.** In the original, the lemma is stated when k is a field, the proof being left to the reader in this case.

$i = 0, \dots, n-1$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & S_0 & \longrightarrow & W_n & \longrightarrow & (U_0/U_1) \hat{\otimes}_k V_n \longrightarrow 0 \\ & & \downarrow \pi'_{n-1} & & \downarrow \pi_n & & \downarrow p \\ 0 & \longrightarrow & \bigoplus_{i=0}^{n-1} \overline{W}'_{i,n-1-i} & \longrightarrow & \bigoplus_{i=0}^n \overline{W}_{i,n-1} & \longrightarrow & \overline{W}_{0,n} \longrightarrow 0. \end{array}$$

Then p is surjective, with kernel $(U_0/U_1) \hat{\otimes}_k V_{n+1}$. Moreover, by the induction hypothesis applied to the sequence (U'_i) , π'_{n-1} is surjective, with kernel equal to $W'_n = \sum_{i=1}^n W_{i,n+1-i}$. It follows that π_n is surjective, and that the inclusion $W_{n+1} \subset \text{Ker}(\pi_n)$ is an equality. This proves the lemma. $\square$

Return to an object (A, J) of $\mathcal{C}$ and note that, by 0.2.G, the hypothesis that each $\overline{J^n}/\overline{J^{n+1}}$ be topologically flat over k is equivalent to saying that $\text{Gr}_J(A)$ is topologically flat over k .

Corollary. *Let $\mathcal{P}$ be the full subcategory of $\mathcal{C}$ consisting of the objects (A, J) such that A and $\text{Gr}_J(A)$ are topologically flat over k . Then the functor $\mathcal{P} \rightarrow \mathbf{Alpg}/k$, $(A, J) \mapsto \text{Gr}_J(A)$ commutes with finite coproducts, hence transforms a cogroup of $\mathcal{P}$ into a cogroup of $\mathbf{Alpg}/k$.*

In particular, if k is a field then, for every cogroup (A, J) of $\mathcal{C}$, $\text{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$.⁽¹⁴⁸⁾

5.1.3. — Identify every profinite k -algebra Γ with the graded profinite k -algebra $(\Gamma_n)_{n \geq 0}$ such that $\Gamma_0 = \Gamma$ and $\Gamma_n = 0$ if $n > 0$. In particular, if $(A_n)_{n \geq 0}$ is a graded profinite k -algebra, we shall consider A_0 indifferently as a profinite k -algebra or as a graded profinite k -algebra. We shall then denote by $\rho : (A_n) \rightarrow A_0$ the morphism of $\mathbf{Alpg}/k$ such that $\rho_0 = \text{id}_{A_0}$ and $\rho_n = 0$ if $n > 0$. Similarly, $\tau : A_0 \rightarrow (A_n)$ will denote the section of ρ such that $\tau_0 = \text{id}_{A_0}$ and $\tau_n = 0$ if $n > 0$.

Every cogroup structure on $(A_n)_{n \in \mathbf{N}}$ induces a cogroup structure on A_0 such that ρ and τ are homomorphisms of cogroups. In this case, denote by I_0 the augmentation ideal of A_0 and put $A'_n = A_n / \overline{I_0 A_n}$ for every $n \geq 0$ (so that $A'_0 = k$).

Then $(A'_n)_{n \in \mathbf{N}}$ is a cogroup in $\mathbf{Alpg}/k$ (note that, since $A_0 = I_0 \oplus k \cdot 1$, $I_0 \hat{\otimes}_k A_n \simeq \overline{I_0 A_n}$ is a direct summand of A_n , for every n). Since τ is a section of ρ , by 2.4.A the cogroup $(A_n)_{n \in \mathbf{N}}$ is the “semidirect coproduct” of A_0 and the cogroup $(A'_n)_{n \in \mathbf{N}}$. More precisely, $(A'_n)_{n \in \mathbf{N}}$ is isomorphic, as an object of $\mathbf{Alpg}/k$, to the kernel of the pair:

$$(A_n) \xrightarrow[\text{(id} \hat{\otimes} \rho) \Delta]{\tau_1} (A_n) \hat{\otimes}_k A_0$$

(where $\Delta : (A_n) \rightarrow (A_n) \hat{\otimes} (A_n)$ is the comultiplication of (A_n) and $\tau_1(x) = x \hat{\otimes} 1$), and, identifying A'_n with its image in A_n , the map

$$(A'_n \hat{\otimes}_k A_0) \rightarrow (A_n), \quad a'_n \hat{\otimes} a_0 \mapsto a'_n a_0$$

is an isomorphism of $\mathbf{Alpg}/k$. (N.B. This is not an isomorphism of cogroups, but $\Delta(A') \subset A \hat{\otimes} A'$ and $(\gamma \rho' \hat{\otimes} \text{id}) \circ \Delta|_{A'} = \Delta'$, where Δ' is the comultiplication of A' and $\gamma \rho'$ the projection $A \rightarrow A'$, cf. 2.4.A.)

5.1.4. — Let (A, J) be an object of $\mathcal{C}$ and $(A_n) = \text{Gr}_J(A)$ the associated object of $\mathbf{Alpg}/k$, i.e. $A_n = \overline{J^n}/\overline{J^{n+1}}$ for every $n \geq 0$. It is clear that the algebra $\mathcal{A} = \prod_{n \geq 0} A_n$ is generated by A_0 and A_1 , that is, for $n \geq 1$, the map

$$(1) \quad A_1 \hat{\otimes}_{A_0} A_1 \hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0} A_1 \rightarrow A_n$$

⁽¹⁴⁸⁾**N.D.E.** To a pair $H \subset G$ of formal k -groups, one therefore associates the “formal completion $\hat{G}_H$ of G along H ”, which is a formal k -group; moreover, we shall see in 5.1.3–5.1.4 that the inclusion $\sigma : H \hookrightarrow \hat{G}_H$ possesses a retraction $\pi : \hat{G}_H \rightarrow H$ and that the formal k -group $N = \text{Ker}(\pi)$ is identified, as a formal variety, with the completion of the homogeneous space G/H along the identity section. This will be useful in 5.2.2.

defined by multiplication is surjective.

Suppose, moreover, that (A, J) is a cogroup of $\mathcal{C}$ and that A and the quotients $\overline{J^n}/\overline{J^{n+1}}$ are flat over k . Then, by Corollary 5.1.2, $\text{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$. Thus, by 5.1.3, if one puts

$$(2) \quad A'_n = \left\{ x \in \overline{J^n}/\overline{J^{n+1}} \mid \Delta(x) - x \hat{\otimes} 1 \in \bigoplus_{i=1}^n (\overline{J^{n-i}}/\overline{J^{n-i+1}}) \hat{\otimes} (\overline{J^i}/\overline{J^{i+1}}) \right\},$$

then the map $(A'_n \hat{\otimes} A_0) \longrightarrow (A_n)$, $a'_n \hat{\otimes} a'_n \mapsto a'_n a_0$ is an isomorphism of $\mathbf{Alpg}/k$.⁽¹⁴⁹⁾

Denoting by I_0 the augmentation ideal of A_0 , one deduces from (1) and from the commutative diagram below, where $A_1^{\hat{\otimes} n}$ denotes $A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1$ (n factors):

$$\begin{array}{ccccc} A_1 \hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0} A_1 & \xleftarrow{\sim} & A_1^{\hat{\otimes} n} \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (A_1^{\hat{\otimes} n} \hat{\otimes}_k I_0) \oplus A_1^{\hat{\otimes} n} \\ \downarrow m & & \downarrow m' \hat{\otimes} \text{id} & & \downarrow m' \hat{\otimes} \text{id} \oplus m' \\ A_n & \xleftarrow{\sim} & A'_n \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (A'_n \hat{\otimes}_k I_0) \oplus A'_n \end{array}$$

that the map

$$(3) \quad m' : A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1 \longrightarrow A'_n$$

induced by multiplication is surjective; in other words, the profinite k -algebra $\mathcal{A}' = \prod_{n \geq 0} A'_n$ is generated by its degree 1 terms.

⁽¹⁵⁰⁾Return now to hypothesis (i) of Proposition 5.1: let G be a topologically flat infinitesimal formal k -group, A its affine algebra, I_A the augmentation ideal of A , B a closed subalgebra of A , and $J = \overline{AI_B}$, where $I_B = B \cap I_A$. Denote by H the formal subgroup $\text{Spf}(A/J)$ and by π the projection $A \longrightarrow A/J$. Suppose that $A/\overline{J^n}$ is topologically flat over k , for every $n \geq 1$, and that B is contained in the kernel $\tilde{B}$ of the pair:

$$A \xrightarrow[\text{(id}_A \hat{\otimes}_k \pi) \Delta_A]{\tau_1} A \hat{\otimes}_k (A/J).$$

Let $(A_n) = \text{Gr}_J(A)$, let $I_0 = I_A/J$ be the augmentation ideal of $A_0 = A/J$, and define $(A'_n)_{n \in \mathbf{N}}$ as in (2) above. Denote by $(B_n)_{n \in \mathbf{N}}$ (resp. $(\tilde{B}_n)_{n \in \mathbf{N}}$) the object of $\mathbf{Alpg}/k$ associated with the filtration of B (resp. $\tilde{B}$) induced by that of A , i.e. defined by the ideals $B \cap \overline{J^n}$ (resp. $\tilde{B} \cap \overline{J^n}$). Then it is clear that $B_n \subset \tilde{B}_n \subset A'_n$ for every n , and that

$$\mathcal{B} = \prod_{n \geq 0} B_n \subset \tilde{\mathcal{B}} = \prod_{n \geq 0} \tilde{B}_n$$

are subalgebras of $\mathcal{A}' = \prod_{n \geq 0} A'_n$.

On the other hand, J (resp. $\overline{J^2}$) is the image in A of $I_B \hat{\otimes}_k A$ (resp. $\overline{I_B^2} \hat{\otimes}_k A$). Consequently, the map

$$(I_B/\overline{I_B^2}) \hat{\otimes}_k A \longrightarrow J/\overline{J^2} = A_1 \simeq A'_1 \hat{\otimes}_k A_0$$

⁽¹⁴⁹⁾**N.D.E.** Let G, H and $\hat{G}_H$ be the formal k -groups corresponding to $A, A_0 = A/J$ and $\text{Gr}_J(A)$; then $\tau : A_0 \hookrightarrow \text{Gr}_J(A)$ corresponds to a retraction $\pi : \hat{G}_H \longrightarrow H$ of the inclusion $H \hookrightarrow \hat{G}_H$, and the preceding passage means that $\hat{G}_H$ is the semidirect product of $N = \text{Ker}(\pi)$ by H .

⁽¹⁵⁰⁾**N.D.E.** We have detailed the original in what follows, and placed at the end the “supplement” $\overline{I_B^n} = B \cap \overline{J_B^n}$ (which is not necessary to establish Proposition 5.1).

is surjective, and factors through $(I_B/\overline{I_B^2}) \hat{\otimes}_k A_0$. Since $A_0 = k \cdot 1 \oplus I_0$, one deduces from the commutative exact diagram:

$$\begin{array}{ccc} (I_B/\overline{I_B^2}) \hat{\otimes}_k A_0 & \xleftarrow{\sim} & (I_B/\overline{I_B^2}) \hat{\otimes}_k I_0 \oplus (I_B/\overline{I_B^2}) \\ \downarrow & & \downarrow \\ J/\overline{J^2} & \xleftarrow{\sim} & A'_1 \hat{\otimes}_k I_0 \oplus A'_1 \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

that A'_1 is the image of $I_B/\overline{I_B^2}$, so that one has $B_1 = A'_1$. Since $\mathcal{A}'$ is generated by A'_1 , it follows that, for every $n \geq 1$, one has

$$A'_n \subset B_n \subset \tilde{B}_n \subset A'_n,$$

whence $B_n = \tilde{B}_n = A'_n$.⁽¹⁵¹⁾

Finally, since the formal group G is *infinitesimal*, one has $\mathfrak{r}(A) = I_A$ and hence the ideals $\overline{I_A^n}$ tend to 0 (cf. 0.1.2); a fortiori, the ideals $\overline{J^n}$ tend to 0, and hence the induced filtrations on $\tilde{B}$ and B are separated. Moreover, since B is a *closed* subalgebra of A , it is complete for the topology defined by the ideals $B \cap \overline{J^n}$. Consequently, it follows from [CA], § V.7, Lemma 1 (see also 5.1.5 below) that $B = \tilde{B}$. This completes the proof of Proposition 5.1.

One has, moreover, the following supplement. For every n , $\overline{J^n} = \overline{AI_B^n}$ is the image in A of $A \hat{\otimes}_B \overline{I_B^n}$ and also of $A \hat{\otimes}_B (B \cap \overline{J^n})$. Now, by hypothesis, the affine algebra A/J of the formal subgroup H is topologically flat over k . Thus, by Theorem 2.4, the morphism $G = \mathrm{Spf}(A) \rightarrow G/H = \mathrm{Spf}(B)$ is *surjective and topologically flat*; one therefore has

$$A \hat{\otimes}_B \overline{I_B^n} = \overline{J^n} = A \hat{\otimes}_B (B \cap \overline{J^n}),$$

and this implies that $\overline{I_B^n} = B \cap \overline{J^n}$ for every n . This also follows from the fact that the maps

$$\overline{I_B^n}/\overline{I_B^{n+1}} \rightarrow A'_i = (B \cap \overline{J^n})/(B \cap \overline{J^{n+1}})$$

are surjective, and from 5.1.5 (ii) below, applied to $B'_n = \overline{I_B^n}$ and $B_n = B \cap \overline{J^n}$.

Lemma 5.1.5. ⁽¹⁵²⁾ (i) Let M and N be two abelian groups filtered by decreasing sequences of subgroups $(M_n)_{n \in \mathbf{Z}}$ and $(N_n)_{n \in \mathbf{Z}}$. Suppose that the union of the M_n (resp. N_n) equals M (resp. N), that the intersection of the M_n (resp. N_n) is zero, and that M is complete for the topology defined by the M_n . Let $f : M \rightarrow N$ be a morphism of filtered groups.

- a) If f induces a surjection of the associated graded, then f is a surjection and N is complete for the topology defined by the N_n .
- b) If f induces an injection of the associated graded, then f is an injection.

(ii) Let B be an abelian group, and let $B = B'_0 \supset B'_1 \supset \cdots$ and $B = B_0 \supset B_1 \supset \cdots$ be two separated filtrations of B by subgroups such that $B'_n \subset B_n$ for every n . Suppose that B is complete for the topology defined by the filtration (B'_n) .

If the map $B'_i/B'_{i+1} \rightarrow B_i/B_{i+1}$ is surjective for every i , then $B'_n = B_n$ for every n .

Indeed, (i) is Lemma 1 of [CA], § V.7 (see also [BAC], III, § 2.8), and (ii) follows by taking $M = B'_n \supset B'_{n+1} \supset \cdots$ and $N = B_n \supset B_{n+1} \supset \cdots$.

5.2. — Throughout the remainder of Section 5, k denotes a perfect field of characteristic $p > 0$.

⁽¹⁵¹⁾ **N.D.E.** With the notation of N.D.E. (149), this implies that the formal completion of G/H along the identity section (whose affine algebra is $\prod_n B_n$) is isomorphic, as a formal variety, to the formal k -group N .

⁽¹⁵²⁾ **N.D.E.** The original stated only point (ii); for the reader's convenience, we have stated Lemma 1 of [CA], § V.7, in (i).

We put $\overline{\mathbf{N}} = \mathbf{N} \cup \{\infty\}$. If B is a profinite k -algebra and $r \in \overline{\mathbf{N}}$, we denote by $((x^{p^r}))_{x \in \mathfrak{r}(B)}$ the closed ideal of B generated by the elements x^{p^r} , where x ranges over the radical $\mathfrak{r}(B)$ of B . If $r = \infty$, we use the same notation, agreeing that $((x^{p^\infty}))_{x \in \mathfrak{r}(B)}$ is the zero ideal. In both cases, B_r denotes the quotient $B/((x^{p^r}))_{x \in \mathfrak{r}(B)}$.

We say that B has *height* $\leq r$ if $((x^{p^r}))_{x \in \mathfrak{r}(B)}$ is the zero ideal; if this holds and r is finite, we say that B has *finite height*.

Consider in particular the case where B is of the form $k[[\omega]]$, ω being a pseudocompact k -vector space (cf. 1.2.5).⁽¹⁵³⁾ We then say that B is a *formal power series algebra* and that B_r is a *truncated formal power series algebra* ($r \in \overline{\mathbf{N}}$; we therefore agree to say that $B = B_\infty$ is also “truncated”). If $B = k[[\omega]]$, we also write $((x^{p^r}))_{x \in \omega}$ instead of $((x^{p^r}))_{x \in \mathfrak{r}(B)}$.

Notation. Let ω be a pseudocompact k -vector space filtered by an increasing sequence of closed vector subspaces

$$0 = \omega_0 \subset \omega_1 \subset \omega_2 \subset \omega_3 \subset \cdots.$$

(a) The closed ideal of $k[[\omega]]$ generated by the elements x^{p^r} , where r ranges over $\mathbf{N}$ and x ranges over ω_r , will be denoted by $((x^{p^r}))_{r \in \omega_r}$.

(b) On the other hand, we shall denote by ${}_r\omega$ the filtered pseudocompact vector space such that

$${}_r\omega_i = \omega_i \quad \text{if } i < r, \quad {}_r\omega_i = \omega \quad \text{if } i \geq r.$$

Theorem. (Dieudonné–Cartier). *Let $H \rightarrow G$ be a monomorphism of infinitesimal formal groups over a perfect field k of characteristic $p > 0$. Let B be the affine algebra of the homogeneous space G/H and suppose that one of the following three conditions holds:*⁽¹⁵⁴⁾

- (i) *B has finite height (this holds in particular if G has finite height).*
- (ii) *B is a complete noetherian local ring.*
- (iii) *B is a reduced ring.*

Then B is isomorphic to the completed tensor product of a finite family of truncated formal power series algebras.

The proof of this theorem occupies paragraphs 5.2.1 to 5.2.5.

5.2.1. — Let A be the affine algebra of G , I_A its augmentation ideal, and $I = B \cap I_A$. By 2.4, one has $H = \text{Spf}(A/\overline{AI})$ and $B = \{x \in A \mid \Delta(x) - 1 \hat{\otimes} x \in A \hat{\otimes} \overline{AI}\}$. Put $\omega = I/\overline{I^2}$. One denotes by I_r the closed subideal of I consisting of the x such that $x^{p^r} = 0$, and by ω_r the canonical image of I_r in ω . We shall prove:

Proposition. *If there exists a continuous section $\sigma : \omega \rightarrow I$ of the projection $I \rightarrow I/\overline{I^2}$ such that $\sigma(\omega_r) \subset I_r$ for every r , then B is isomorphic to $k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$.*

Such a section indeed extends to a continuous morphism $k[[\omega]] \rightarrow B$, which factors through $B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$. We prove in 5.2.2 to 5.2.5 that the morphism

$$\phi : B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r} \rightarrow B$$

thus obtained is an isomorphism.

⁽¹⁵³⁾**N.D.E.** In the original, the author uses “linearly compact vector space”, which is equivalent to “pseudocompact vector space” (cf. [BAC], § III.2, Exercises 15 a), 19 a) and 20 d)). We have preferred to retain the terminology “pseudocompact”, used up to this point.

⁽¹⁵⁴⁾**N.D.E.** On the one hand, we have replaced $H \setminus G$ by G/H , and likewise in the proof; on the other hand, we have added condition (iii).

5.2.1.A. — ⁽¹⁵⁵⁾For each $r \in \mathbf{N}$, put $\mathcal{I}_r = \overline{I^2} + I_r$, so that $\mathcal{I}_r/\overline{I^2} \simeq \omega_r$; one then has a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{I}_{r-1} & \longrightarrow & \mathcal{I}_r & \longrightarrow & \mathcal{I}_r/\mathcal{I}_{r-1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \wr \\ 0 & \longrightarrow & \omega_{r-1} & \longrightarrow & \omega_r & \longrightarrow & \omega_r/\omega_{r-1} \longrightarrow 0 \end{array}$$

and since k is a field, the rows are split: one can extend a pseudobasis $\mathcal{B}_{r-1}$ of $\mathcal{I}_{r-1}$ to a pseudobasis $\mathcal{B}_{r-1} \cup \mathcal{B}'_r$ of $\mathcal{I}_r$, and then the closed subspace S_r with pseudobasis $\mathcal{B}'_r$ is a complement to $\mathcal{I}_{r-1}$ in $\mathcal{I}_r$, and the projection $\pi : I \rightarrow \omega$ induces an isomorphism from S_r onto a complement ω'_r to ω_{r-1} in ω_r . Denote by $\mathcal{I}_\infty$ the closed ideal $\bigcup_r \mathcal{I}_r$; it likewise admits a complement S_∞ in I , and π induces an isomorphism from $\mathcal{I}_\infty/\overline{I^2}$ (resp. from S_∞) onto the closure ω_∞ of the union of the ω_r (resp. onto a complement ω'_∞ to ω_∞ in ω). Denote by η the isomorphism $S_\infty \xrightarrow{\sim} \omega'_\infty$. One then obtains continuous linear maps:

$$\begin{array}{c} \overline{I^2} \times S_\infty \times \widehat{\bigoplus_r S_r}^\phi \longrightarrow I \\ \eta \times \theta \downarrow \\ \omega = \omega'_\infty \times \omega_\infty \end{array}$$

where $\widehat{\bigoplus_r S_r}$ is the direct sum of the S_r in $\mathbf{PC}(k)$, i.e. $(\prod_r S_r^\dagger)^*$ (cf. N.D.E. (16) of 0.2.2), and where $\theta : \widehat{\bigoplus_r S_r} \rightarrow \omega_\infty$ is induced by the maps $S_r \xrightarrow{\sim} \omega'_r \hookrightarrow \omega_\infty$. One therefore sees that a sufficient condition (but not a necessary one, see below) for obtaining a section $\sigma : \omega \rightarrow I$ as desired is that θ be an isomorphism. By duality (cf. 0.2.2), this is equivalent to saying that the linear map $\omega_\infty^\dagger \rightarrow \prod_r S_r^\dagger$ is bijective.

5.2.1.B. — Put, as before, $\omega_\infty = \overline{\bigcup_{n \in \mathbf{N}} \omega_n}$. A second case where a section $\sigma : \omega \rightarrow I$ as desired exists is the case where ω_∞ possesses a pseudobasis $\mathcal{B}_\infty$ which is a union of pseudobases of the ω_n/ω_{n-1} , for $n \in \mathbf{N}^*$ (one can then extend it by a pseudobasis $\mathcal{B}'_\infty$ of ω/ω_∞ to obtain a pseudobasis of ω compatible with the filtration). Putting $V = \omega_\infty^\dagger$ and denoting by V_n the orthogonal in V of ω_n , this is equivalent to saying that, in the category of “ordinary” k -vector spaces, the separated decreasing filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \dots$$

is split, i.e. that V is the direct sum, for $n \in \mathbf{N}$, of subspaces F_n such that $F_n \simeq V_n/V_{n+1}$. This is not necessarily the case: for example, if V is the space $\mathcal{S} = k^{\mathbf{N}}$ of sequences of elements of k and $\mathcal{S}_n$ the subspace of sequences (u_i) such that $u_i = 0$ for $i < n$, so that $\dim \mathcal{S}_n/\mathcal{S}_{n+1} = 1$, then $\mathcal{S}$ is not isomorphic to the direct sum of the $\mathcal{S}_n/\mathcal{S}_{n+1}$ since $\mathcal{S}$ does not have countable dimension (on the other hand, $\mathcal{S}$ here is the *product* of the $\mathcal{S}_n/\mathcal{S}_{n+1}$, cf. 5.2.1.A). This is, however, the case if V has countable dimension.⁽¹⁵⁶⁾ Indeed, let $(e_n)_{n \in \mathbf{N}}$ be a basis of V ; we shall construct by induction on n an increasing function $g : \mathbf{N} \rightarrow \mathbf{N}$ and subspaces F_i , for $i = 0, \dots, g(n)$, such that $F_i \simeq V_i/V_{i+1}$ and such that $F_{\leq g(n)} = \bigoplus_{i=0}^{g(n)} F_i$ is a complement to $V_{g(n)+1}$ containing $e_0, \dots, e_n$; one will then have $V = \bigoplus_{i \geq 0} F_i$. Let $n+1 \in \mathbf{N}$; one can suppose that the assertion is established for n (the assertion being empty for $n = -1$). If $e_{n+1} \in F_{\leq g(n)}$, one puts $g(n+1) = g(n)$; otherwise one writes $e_{n+1} = f + x$ with $f \in F_{\leq g(n)}$ and $x \in V_{g(n)+1}$ nonzero. Let then j be the smallest integer such that $x \in V_j - V_{j+1}$; for $i = g(n) + 1, \dots, j$, choose a complement F_i to V_{i+1} in V_i , so that $e_{n+1} \in F_j$, and put $g(n+1) = j$.

⁽¹⁵⁵⁾**N.D.E.** We have added paragraphs 5.2.1.A and 5.2.1.B.

⁽¹⁵⁶⁾**N.D.E.** This paragraph is the fruit of discussions with J.-M. Fontaine and E. Bouscaren; in particular, Bouscaren indicated the following proof to us.

5.2.1.C. — ⁽¹⁵⁷⁾In particular, the two preceding conditions (5.2.1.A and B) hold when the filtration of ω is *stationary*, i.e. when there exists an integer n_0 such that $\omega_n = \omega_{n_0}$ for $n_0 \leq n < +\infty$. In this case, one obtains an isomorphism from $k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$ onto the completed tensor product:

$$\frac{k[[\omega_1]]}{((x^p))_{x \in \omega_1}} \widehat{\otimes} \frac{k[[\omega'_2]]}{((x^{p^2}))_{x \in \omega'_2}} \widehat{\otimes} \cdots \widehat{\otimes} \frac{k[[\omega'_{n_0}]]}{((x^{p^{n_0}}))_{x \in \omega'_{n_0}}} \widehat{\otimes} k[[\omega'_\infty]],$$

where ω'_n (resp. ω'_∞) is a complement to ω_{n-1} in ω_n (resp. to $\omega_\infty = \omega_{n_0}$ in ω). The filtration of ω is clearly stationary in case (i), i.e. if $\omega_r = \omega$ for r large enough, and in case (ii), i.e. if ω has finite dimension, and also in case (iii), i.e. if $I_r = 0$ for every r (and in this case B will be isomorphic to the formal power series algebra $k[[\omega]]$). The above remarks therefore imply our theorem, modulo points 5.2.2–5.2.5 below.

5.2.2. — *First suppose that B has height ≤ 1* , i.e. that $x^p = 0$ if $x \in I$. By 5.1.4, the graded $\text{Gr}_I(B)$ associated with B for the filtration $I \supset \overline{I^2} \supset \overline{I^3} \supset \cdots$ is equipped with a cogroup structure in the category **Alg**/ k , i.e. the profinite algebra $C = \prod_{n \geq 0} \text{Gr}_I(B)_n$ is the affine algebra of a formal k -group N . It is clear that one has $\omega_{N/k} = I/\overline{I^2}$ and that N is infinitesimal of height ≤ 1 . By 4.4, the identity map of $\omega_{N/k}$ therefore induces an isomorphism from $B' = k[[\omega_{N/k}]]/((x^p))_{x \in \omega_{N/k}}$ onto C . This implies in particular that the map ϕ of 5.2.1 induces an isomorphism of the associated gradeds of B' and B when one filters B' and B by the powers of the augmentation ideal. Thus ϕ is an isomorphism, by [CA], § V.7, Lemma 1 (see also 5.1.5).

5.2.3. — *Suppose now that B has finite height $\leq r$* . Let π be the isomorphism $x \mapsto x^p$ from k onto k . The linear map from $B \widehat{\otimes}_\pi k$ to B which sends $b \widehat{\otimes}_\pi x$ to $b^p x = (bx^{1/p})^p$ has a closed image which is none other than the closed subalgebra $B^p = \{b^p \mid b \in B\}$ of B . Put $J = B^p \cap I = B^p \cap I_A$.

⁽¹⁵⁸⁾Denote by G_1 the kernel of the Frobenius morphism $\text{Fr} : G \rightarrow G^{(p)}$ and by HG_1 the formal subgroup of G which is the inverse image of the formal subgroup $H^{(p)}$ of $G^{(p)}$. Then HG_1 is defined by the closed ideal generated by the p th powers of elements of $\overline{AI}$, which equals $\overline{AJ}$. On the other hand, since the formation of G/H commutes with base change (since G/H represents the quotient sheaf for the flat topology, cf. 2.4), $(G/H)^{(p)} = G^{(p)}/H^{(p)}$ and one therefore has commutative diagrams:

$$\begin{array}{ccc} G & \xrightarrow{\text{Fr}} & G^{(p)} \\ \downarrow & & \downarrow \\ G/H & \xrightarrow{\text{Fr}} & G^{(p)}/H^{(p)} \end{array} \quad \begin{array}{ccc} A & \xleftarrow{a \widehat{\otimes} 1 \mapsto a^p} & A \widehat{\otimes}_\pi k \\ \uparrow & & \uparrow \\ B & \xleftarrow{b \widehat{\otimes} 1 \mapsto b^p} & B \widehat{\otimes}_\pi k \end{array}$$

from which one deduces that B^p is the affine algebra of the quotient G/HG_1 .⁽¹⁵⁹⁾ Temporarily denote by C the affine algebra of the quotient HG_1/H . Since the formation of G/H commutes with base change, one has a cartesian square:

$$\begin{array}{ccc} HG_1 & \longrightarrow & G \\ \downarrow & & \downarrow \\ HG_1/H & \longrightarrow & G/H \end{array}$$

whence an isomorphism $A \widehat{\otimes}_B C \simeq A/\overline{AJ} = A \widehat{\otimes}_B (B/\overline{BJ})$, and since A is topologically free over B (by 2.4, since A and B are local), it follows that the natural morphism $B/\overline{BJ} \rightarrow C$

⁽¹⁵⁷⁾**N.D.E.** We return here to the original, which we have shortened in view of the preceding additions.

⁽¹⁵⁸⁾**N.D.E.** We have added what follows.

⁽¹⁵⁹⁾**N.D.E.** As indicated in the original, this also follows from Proposition 5.1, but we have preferred to give the argument above, which does not use the implication (i) $\Rightarrow$ (ii) of loc. cit.

is an isomorphism, so $B/\overline{B\overline{J}}$ is the affine algebra of $HG_1/H^{(159)}$ and of course $B/\overline{B\overline{J}} = B_1$ has height ≤ 1 since $J = ((x^p))_{x \in \tau(B)}$.

Let $B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$, $\phi : B' \rightarrow B$ the morphism introduced in 5.2.1, B'^p the subalgebra $\{x^p \mid x \in B'\}$, and J' the augmentation ideal of B'^p . Then one has a commutative diagram:

$$\begin{array}{ccc} B' & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ B'_1 = B'/\overline{B'J'} & \xrightarrow[\phi_1]{} & B_1 = B/\overline{B\overline{J}} \end{array}$$

and, by 5.2.2, ϕ_1 is an isomorphism.

On the other hand, by 2.4, A is topologically flat over $B = \mathcal{A}(G/H)$ and over $B^p = \mathcal{A}(G/HG_1)$, hence, by 1.3.3, B is topologically flat over B^p . Moreover, by 5.2.4 below, the morphism $B'^p \rightarrow B^p$ induced by ϕ is an isomorphism. One can then apply 0.3.4 to the pseudocompact ring $B'^p = B^p$ and the pseudocompact B^p -modules $M = B'$, $N = B$: by the preceding, $\phi_1 = \phi \hat{\otimes}_{B^p} k$ is an isomorphism, and it follows that ϕ is an isomorphism. This proves 5.2 when B has finite height, modulo point 5.2.4 below.

5.2.4. — For every pseudocompact vector space V over k , we denote by πV the space $V \hat{\otimes}_{\pi} k$ obtained from V by the extension $x \mapsto x^p$ of the scalar field.⁽¹⁶⁰⁾ One then has a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi \overline{I^2} & \xrightarrow{\alpha} & \pi I & \xrightarrow{\beta} & \pi \omega \longrightarrow 0 \\ & & \downarrow u & & \downarrow v & & \downarrow w \\ 0 & \longrightarrow & \overline{J^2} & \xrightarrow{\gamma} & J & \xrightarrow[\delta]{} & \overline{\omega} \longrightarrow 0, \end{array}$$

where one has put $\overline{\omega} = J/\overline{J^2}$ and where the maps u, v, w are induced by the linear map $x \hat{\otimes} a \mapsto x^p a$ from πB to B^p . Since u and v are surjections, w is a surjection and has kernel the image $\pi \omega_1$ of $\pi I_1 = \text{Ker}(v)$.

Then, putting $J_n = \{x \in J \mid x^{p^n} = 0\}$ and $\overline{\omega}_n = \delta(I_n)$, one has $J_n = v(\pi I_{n+1})$ and $\overline{\omega}_n = w(\pi \omega_{n+1})$, for every $n \geq 0$. The section $\pi \sigma : \pi \omega \rightarrow \pi I$, induced by the section σ of 5.2.1, therefore defines, on passing to the quotient, a section $\tau : \overline{\omega} \rightarrow J$ compatible with the filtrations of J and $\overline{\omega}$. Since B^p has height $\leq r - 1$, this section induces, by the induction hypothesis, an isomorphism

$$\psi : B'' = k[[\overline{\omega}]]/((x^{p^n}))_{x \in \overline{\omega}_n} \xrightarrow{\sim} B^p.$$

Now B'' is identified with B'^p and ψ with the morphism $B'^p \rightarrow B^p$ induced by ϕ , and hence *our theorem is proved when B has finite height*.

Remark 5.2.4.A. ⁽¹⁶¹⁾ Suppose that B has height $\leq r + 1$ (with $r \in \mathbf{N}^*$); then $I_{r+1} = I$ and one has an isomorphism

$$(1) \quad B \simeq \frac{k[[S_1]]}{((x_1^p))_{x_1 \in S_1}} \hat{\otimes}_k \cdots \hat{\otimes}_k \frac{k[[S_r]]}{((x_r^{p^r}))_{x_r \in S_r}} \hat{\otimes}_k \frac{k[[S_{r+1}]]}{((x_{r+1}^{p^{r+1}}))_{x_{r+1} \in S_{r+1}}}$$

where each S_n is a complement to $\overline{I^2} + I_{n-1}$ in $\overline{I^2} + I_n$. Then $\omega = I/\overline{I^2}$ is identified with $\prod_{i=1}^{r+1} S_i$, and one sees easily that, for $n = 1, \dots, r + 1$, the image ω_n of I_n in ω is identified with $\prod_{i=1}^n S_i$.

⁽¹⁶⁰⁾**N.D.E.** Since k is perfect, one can identify πV with the abelian group V on which k acts by $\lambda \cdot v = \lambda^{1/p} v$.

⁽¹⁶¹⁾**N.D.E.** We have added this remark, used in 5.2.5.

This has the following consequence. Let $B_r = B/J_r$, where $J_r = ((x^{p^r}))_{x \in \mathfrak{t}(B)}$, and let $\mathfrak{m} = I/J_r$ be the augmentation ideal of B_r ; since $J_r \subset \overline{I^2}$, $\omega(r) = \mathfrak{m}/\overline{\mathfrak{m}^2}$ is identified with ω . For $n = 1, \dots, r$, denote by $\omega(r)_n$ the image in $\omega(r)$ of $\mathfrak{m}_n$; it is also the image in ω of $\{x \in I \mid x^{p^n} \in J_r\}$, so $\omega(r)_n$ contains ω_n . On the other hand, it follows from isomorphism (1) that one has $J_r = ((x_{r+1}^{p^r}))_{x_{r+1} \in S_{r+1}}$, whence

$$(2) \quad B_r \simeq \frac{k[[S_1]]}{((x_1^p))_{x_1 \in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{((x_r^{p^r}))_{x_r \in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{((x_{r+1}^{p^r}))_{x_{r+1} \in S_{r+1}}}$$

and hence, by the preceding, $\omega(r)_n$ is identified with $\prod_{i=1}^n S_i$ for $n = 1, \dots, r-1$. One therefore obtains that the inclusion $\omega_n \subset \omega(r)_n$ is an equality, for $n = 1, \dots, r-1$.

5.2.5. — *It remains to consider the case where B has infinite height, and where the projection $I \rightarrow \omega$ possesses a section σ compatible with the filtrations of ω and I . Consider the morphism*

$$\phi : B' = k[[\omega]]/((x^{p^n}))_{x \in \omega_n} \rightarrow B$$

induced by σ ; it suffices to show that for every $r \in \mathbf{N}$, the map $\phi_r : B'_r \rightarrow B_r$ induced by ϕ is invertible.⁽¹⁶²⁾

For every $r \in \mathbf{N}^*$, denote by G_r the kernel of the iterated Frobenius morphism $G \rightarrow G^{(p^r)}$ and by HG_r the formal subgroup of G which is the inverse image of the formal subgroup $H^{(p^r)}$ of $G^{(p^r)}$, so that HG_r is defined by the closed ideal generated by the p^r th powers of elements of $\overline{AI}$, which equals $\overline{AJ_r}$, where $J_r = \{x^{p^r} \mid x \in I\}$. One then obtains, exactly as in 5.2.3, that $B_r = B/\overline{BJ_r}$ is the affine algebra of HG_r/H (and has, of course, height $\leq r$).

Denote by $\mathfrak{m}(r) = I/\overline{BJ_r}$ the augmentation ideal of B_r ; the canonical projection from B onto B_r clearly induces an isomorphism from $\omega = I/\overline{I^2}$ onto $\omega(r) = \mathfrak{m}(r)/\overline{\mathfrak{m}(r)^2}$, which allows us to identify these two spaces. Let $\omega(r)_n$ be the image in $\omega(r)$ of the closed ideal $\mathfrak{m}(r)_n = \{y \in \mathfrak{m}(r) \mid y^{p^n} = 0\}$; it is also the image in ω of the closed ideal $I(r)_n = \{x \in I \mid x^{p^n} \in \overline{BJ_r}\}$.⁽¹⁶³⁾ It is clear that $\omega_n(r) = \omega$ if $n \geq r$; let us show that $\omega_n(r) = \omega_n$ if $n < r$. For every r, n , the sequence below is exact:

$$0 \rightarrow I(r)_n \cap \overline{I^2} \rightarrow I(r)_n \rightarrow \omega(r)_n \rightarrow 0.$$

Moreover, for fixed n , one has $\bigcap_r I(r)_n = I_n$, since $\bigcap_r \overline{BJ_r} = 0$. Since in $\mathbf{PC}(k)$ filtered projective limits are exact (cf. 0.2), it follows that, for every n , one has

$$(*) \quad \omega_n = \bigcap_r \omega(r)_n.$$

On the other hand, by Remark 5.2.4.A, one has $\omega(r)_n = \omega(r+1)_n$ if $n \leq r$. Combined with (*), this implies that $\omega(r)_n = \omega_n$ if $n < r$.

Consequently, the vector space ω filtered by the subspaces $(\omega(r)_n)_{n \geq 0}$ is none other than ${}_r\omega$ (Notation 5.2). A fortiori, the map $\sigma(r)$ which is the composite of $\sigma : \omega \rightarrow I$ and the projection $I \rightarrow \mathfrak{m}(r)$ is compatible with the filtrations $(\omega(r)_n)$ and $(\mathfrak{m}(r)_n)$ of ω and $\mathfrak{m}(r)$. Since $k[[{}_r\omega]]/((x^{p^n}))_{x \in {}_r\omega_n}$ is none other than B'_r and $\phi_r : B'_r \rightarrow B_r$ is the morphism induced by $\sigma(r)$, the result already established for algebras of finite height shows that ϕ_r is an isomorphism.

Definition 5.2.6. ⁽¹⁶⁴⁾ Let $(A_\lambda)_{\lambda \in \Lambda}$ be a family of profinite k -algebras, each equipped with an augmentation $\varepsilon_\lambda : A_\lambda \rightarrow k$ (this is the case, in particular, if each A_λ is local with residue field k). One then defines the infinite completed tensor product $\mathcal{A} = \widehat{\bigotimes}_{\lambda \in \Lambda} A_\lambda$ as the projective limit in $\mathbf{Alp}/k$ of the $A_F = \widehat{\bigotimes}_{\lambda \in F} A_\lambda$, for F ranging over the finite subsets of Λ , the transition morphisms $A_{F'} \rightarrow A_F$, for $F' = F \cup \{\lambda\}$, being $\text{id} \otimes \varepsilon_\lambda$. In particular,

⁽¹⁶²⁾**N.D.E.** Indeed, B' (resp. B) is the projective limit of the B'_r (resp. B_r). On the other hand, we have modified the original in what follows, taking account of the addition made in 5.2.3.

⁽¹⁶³⁾**N.D.E.** We have detailed the original in what follows.

⁽¹⁶⁴⁾**N.D.E.** We have added this number, to define the infinite tensor products used in 5.3 (a).

if $\Lambda = \mathbf{N}^*$ and one denotes by X_n the formal k -variety $\mathrm{Spf}(A_n)$, then $\mathrm{Spf}(\mathcal{A})$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of “finite” sequences of elements of $\prod_{n \geq 1} X_n(C)$, i.e. sequences

$$(x_1, x_2, \dots) \in \prod_{n \geq 1} X_n(C)$$

such that $x_n = \varepsilon_n$ for n large enough, where ε_n denotes, by abuse of notation, the composite of $\varepsilon_n : A_n \rightarrow k$ and the structure morphism $k \rightarrow C$. (If, moreover, each A_n is a quotient of an algebra $k[[\omega'_n]]$, one can denote by 0 the unique morphism $A_n \rightarrow C$ which vanishes on ω'_n , and one therefore obtains the set of sequences such that “ $x_n = 0$ ” for n large enough.)

Remarks 5.3. (a) Call *stationary* every profinite k -algebra which is the completed tensor product of a family of truncated formal power series algebras.⁽¹⁶⁵⁾

If G is an infinitesimal formal k -group and B the affine algebra of a homogeneous space of G , it follows from Theorem 5.2 that the algebra $B/((x^{p^r}))_{x \in \mathfrak{r}(B)}$ is stationary for every integer $r \in \mathbf{N}$. This implies in particular that B is a projective limit of stationary algebras.⁽¹⁶⁶⁾

(b) I do not know whether, with the notation of 5.2.1, one can choose A and B in such a way that there is no section $\sigma : \omega \rightarrow I$ compatible with the filtrations.⁽¹⁶⁷⁾ One will note, however, that one can have for ω any pseudocompact vector space filtered by an increasing sequence of closed subspaces. Indeed, if $\omega_1 \subset \omega_2 \subset \dots \subset \omega$ is such a filtered space, one can define in the algebra $B = A = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$ a diagonal morphism $\Delta_A : A \rightarrow A \hat{\otimes}_k A$ satisfying conditions (i), (ii), (iii) of 2.1; it suffices to put $\Delta_A(y) = y \hat{\otimes} 1 + 1 \hat{\otimes} y$ when y is the image in A of an element of ω .

Corollary 5.4. *Let G be an algebraic group over a perfect field k of characteristic $p > 0$, H an algebraic subgroup of G , e the image of the identity element of G in G/H and A the local algebra of G/H at e . Then $\hat{A}$ is isomorphic to an algebra of the form*

$$k[[X_1, \dots, X_r, \dots, X_s]]/(X_1^{p^{n_1}}, \dots, X_r^{p^{n_r}}).$$

Indeed, consider the infinitesimal formal groups $\hat{G} = \mathrm{Spf}(\hat{\mathcal{O}}_{G,e})$ and $\hat{H} = \mathrm{Spf}(\hat{\mathcal{O}}_{H,e})$; by 1.3.4, the completion $\hat{A}$ of $A = \mathcal{O}_{G/H,e}$ is isomorphic to $\mathcal{A}(\hat{G}/\hat{H})$, and the corollary therefore follows from Theorem 5.2 (ii).⁽¹⁶⁸⁾

5.5. — Complements. — ⁽¹⁶⁹⁾Recall the following definitions. On the one hand, one says that a noetherian local ring A is a *complete intersection* if the completion $\hat{A}$ is a quotient of

⁽¹⁶⁵⁾**N.D.E.** The author was doubtless thinking of a tensor product $\mathcal{A} = \widehat{\bigotimes}_{n \in \mathbf{N}^*} k[[\omega'_n]]/((x^{p^n}))_{x \in \omega'_n}$, where the ω'_n are arbitrary pseudocompact k -vector spaces. In this case, one sees without difficulty that $\omega = \omega_{\mathcal{A}/k}$ is identified with the product of the ω'_n , and the filtration $(\omega_n)_{n \in \mathbf{N}^*}$ is given by $\omega_n = \prod_{i=1}^n \omega'_i$, i.e. one is in case 5.2.1.B. For this reason, it would be preferable to call these algebras *stable* (rather than “stationary”), cf. [Di73], II § 2.9, p. 75. If, for example, $\dim \omega'_n = 1$ for every $n \in \mathbf{N}^*$, then $\mathcal{A}$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of sequences $(x_n)_{n \in \mathbf{N}^*}$ of elements of C such that $x_n^{p^n} = 0$, and $x_n = 0$ for n large enough. Note finally that this case (i.e. the case where $\omega = \prod_{n \in \mathbf{N}^*} \omega'_n$) corresponds to the case studied, in the dual situation of connected cocommutative Hopf algebras, by M. E. Sweedler, cf. [Sw67], Th. 3.

⁽¹⁶⁶⁾**N.D.E.** But such a projective limit is not necessarily a stable profinite k -algebra (in the sense of the preceding N.D.E.). For example, let $\mathcal{S}$ be the “ordinary” k -vector space of sequences $(u_1, u_2, \dots)$ of elements of k and let $\omega = \mathcal{S}^*$; then ω is the direct sum in $\mathbf{PC}(k)$ of copies k_n of k , for $n \in \mathbf{N}^*$, i.e. one is in case 5.2.1.A. If one denotes by x_n the element of ω defined by $x_n(u) = u_n$, for every sequence $u = (u_i)_{i \in \mathbf{N}^*}$, then the k -algebra $A = k[[\omega]]/((x_n^{p^n}))_{n \in \mathbf{N}^*}$ is such that $\omega_{A/k} = \omega$ and $\omega_n = \prod_{i=1}^n kx_i$, but is not stable: $\mathrm{Spf}(A)$ represents the functor which associates to every $C \in \mathbf{Alf}/k$ the set of “infinite” sequences $(x_n)_{n \in \mathbf{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ for every $n \in \mathbf{N}^*$.

⁽¹⁶⁷⁾**N.D.E.** The editors do not know either, outside the cases considered in 5.2.1.A and B.

⁽¹⁶⁸⁾**N.D.E.** See also [DG70], § III.3, Th. 6.1.

⁽¹⁶⁹⁾**N.D.E.** We have added this subsection, to give some consequences of 5.4, mentioned in Exposés III and VI A.

a complete regular noetherian local ring B by an ideal I generated by a regular sequence of elements of B (cf. EGA IV₄, 19.3.1). On the other hand, let $\tau : Y \hookrightarrow X$ be a closed immersion of schemes. If $y \in Y$, one says that τ is a *regular immersion at y* if the kernel of $\mathcal{O}_{X,y} \rightarrow \mathcal{O}_{Y,y}$ is generated by a regular sequence; if, moreover, X is locally noetherian and τ is a regular immersion at every point, one says that τ is a *regular immersion* (cf. loc. cit., Prop. 16.9.10 and Def. 16.9.2).

Corollary 5.5.1. *If G is an algebraic group over a field k , the local ring $\mathcal{O}_{G,e}$ is a complete intersection.*

Indeed, by EGA IV₄, 19.3.4, one can suppose that k is algebraically closed. If $\text{char}(k) = 0$, one already knows that G is smooth (cf. 3.3.1 or VI B, 1.6.1) and hence $\mathcal{O}_{G,e}$ is a formal power series k -algebra, by EGA IV₄, 17.5.3 (d''). If $\text{char}(k) = p > 0$, it follows from 5.4, applied to $H = \{e\}$, that $\mathcal{O}_{G,e}$ is a complete intersection.

Remarks 5.5.2. Let k be a field, G a smooth algebraic k -group, and H a closed subgroup of G .

a) We have seen in Exp. III, 4.15, that the immersion $H \hookrightarrow G$ is regular; this can also be deduced from 5.4, as follows. As in loc. cit., one can suppose that k is algebraically closed, and it suffices to show that the kernel I of $\mathcal{O}_{G,e} \rightarrow \mathcal{O}_{H,e}$ is generated by a regular sequence. Put $A = \mathcal{O}_{G,e}$, $\hat{G} = \text{Spf}(\hat{A})$ and $\hat{H} = \text{Spf}(\hat{\mathcal{O}}_{H,e})$. Since A is noetherian, one has an exact sequence

$$0 \rightarrow I \otimes_A \hat{A} \rightarrow \hat{A} \xrightarrow{\pi} \mathcal{A}(H) \rightarrow 0$$

and $\hat{A}$ is faithfully flat over A . Thus, by EGA IV₄, 16.9.10 (ii) and 19.1.5 (ii), it suffices to show that the kernel $\hat{I} = I \otimes_A \hat{A}$ of π is generated by a regular sequence of elements of $\hat{A}$.

Now, since G is smooth, $\hat{A}$ is reduced; by 5.4, the subalgebra $B = \mathcal{A}(\hat{G}/\hat{H})$ is therefore isomorphic to a formal power series algebra $k[[x_1, \dots, x_n]]$, and hence the identity section of $\hat{G}/\hat{H}$ is defined in B by the regular sequence $(x_1, \dots, x_n)$. Since $\hat{A}$ is noetherian, the ideal J of $\hat{A}$ generated by $x_1, \dots, x_n$ is closed and hence equal to $\hat{I}$, by Corollary 1.4. Moreover, since $\hat{A}$ is topologically flat, hence flat over B (cf. 0.3.8), $(x_1, \dots, x_n)$ is a regular sequence in $\hat{A}$, by EGA IV₄, 19.1.5 (ii).

b) One can also deduce from 5.2 (ii) the following more precise result. Suppose that k is perfect. By 5.2 (ii) applied to the algebra $C = \mathcal{A}(H)$, there exist a basis $(y_1, \dots, y_{r+s})$ of ω_H and integers $1 \leq n_1 \leq \dots \leq n_r$ such that $\mathcal{A}(H)$ is isomorphic to the quotient of $k[[y_1, \dots, y_{r+s}]]$ by the ideal generated by the $y_i^{p^{n_i}}$ for $i = 1, \dots, r$. Lift the y_i to elements x_i of ω_G and extend $(x_1, \dots, x_{r+s})$ to a basis $(x_1, \dots, x_n)$ of ω_G . Since $\mathcal{A}(G)$ is reduced, the morphism $k[[x_1, \dots, x_n]] \rightarrow \mathcal{A}(G)$ is an isomorphism, by 5.2 (iii). One therefore obtains: *there exists a “coordinate system” $(x_1, \dots, x_n)$ of G (i.e. an isomorphism $\mathcal{A}(G) \simeq k[[x_1, \dots, x_n]]$) such that H is defined by the equations $x_i^{p^{n_i}} = 0$ for $i = 1, \dots, r$ and $x_i = 0$ for $i > r + s$ (“Dieudonné’s theorem”, compare with [Di55], § 19, Th. 6, and [Di73], II § 3.2, Prop. 3, and what precedes it).*

Bibliography

- [CA] P. Gabriel, *Des catégories abéliennes*, Bull. Soc. Math. France **90** (1962), 323–448.⁽¹⁷⁰⁾
- [Ab80] E. Abe, *Hopf algebras*, Cambridge Univ. Press, 1980.
- [Br00] C. Breuil, *Groupes p -divisibles, groupes finis et modules filtrés*, Ann. of Math. **152** (2000), no. 2, 489–549.
- [BAI] N. Bourbaki, *Algèbre*, Chap. I–III and IV–VII, Hermann, 1970, and Masson, 1981.
- [BAC] N. Bourbaki, *Algèbre commutative*, Chap. I–IV, Masson, 1985.
- [BEns] N. Bourbaki, *Théorie des ensembles*, Chap. I–IV, Hermann, 1970.
- [BLie] N. Bourbaki, *Groupes et algèbres de Lie*, Chap. II–III, Hermann, 1970.
- [Ca62] P. Cartier, *Groupes algébriques et groupes formels*, pp. 87–111 in: Colloque sur la théorie des groupes algébriques (Bruxelles, 1962), Gauthier-Villars, 1962.
- [DG70] M. Demazure, P. Gabriel, *Groupes algébriques*, Masson & North-Holland, 1970.
- [De72] M. Demazure, *Lectures on p -divisible groups*, Lect. Notes Math. **302**, Springer-Verlag, 1972.
- [Di55] J. Dieudonné, *Groupes de Lie et hyperalgèbres de Lie sur un corps de caractéristique $p > 0$* , Comm. Math. Helv. **28** (1955), 87–118.
- [Di73] J. Dieudonné, *Introduction to the theory of formal groups*, Marcel Dekker, 1973.
- [Fo77] J.-M. Fontaine, *Groupes p -divisibles sur les corps locaux*, Astérisque **47–48**, Soc. Math. France, 1977.
- [Gr57] A. Grothendieck, *Sur quelques points d’algèbre homologique*, Tôhoku Math. J. **9** (1957), 119–221.
- [Gr74] A. Grothendieck, *Groupes de Barsotti-Tate et cristaux de Dieudonné*, Presses Univ. Montréal, 1974.
- [HS69] R. G. Heyneman, M. E. Sweedler, *Affine Hopf Algebras I*, J. Algebra **13** (1969), 194–241.
- [Il85] L. Illusie, *Déformations de groupes de Barsotti-Tate, d’après A. Grothendieck*, pp. 151–198 in: Séminaire sur les pinceaux arithmétiques: la conjecture de Mordell, Astérisque **127**, Soc. Math. France, 1985.
- [Ja65] I. M. James, review of [MM65] in Math. Reviews, MR0174052.
- [La75] M. Lazard, *Commutative formal groups*, Lect. Notes Math. **443**, Springer-Verlag, 1975.
- [LT65] J. Lubin, J. Tate, *Formal complex multiplication in local fields*, Ann. of Math. **81** (1965), 380–387.
- [LT66] J. Lubin, J. Tate, *Formal moduli for one-parameter formal Lie groups*, Bull. Soc. Math. France **94** (1966), 49–60.
- [Ma91] A. Masuoka, *On Hopf algebras with cocommutative coradicals*, J. Algebra **144** (1991), 451–466.
- [Me72] W. Messing, *The crystals associated with Barsotti-Tate Groups: with applications to abelian schemes*, Lect. Notes Math. **264**, Springer-Verlag, 1972.
- [MM65] J. W. Milnor, J. C. Moore, *On the structure of Hopf algebras*, Ann. of Math. **81** (1965), 211–264.
- [Mi65] B. Mitchell, *Theory of categories*, Academic Press, 1965.
- [Ne75] K. Newman, *A correspondence between bi-ideals and sub-Hopf algebras in cocommutative Hopf algebras*, J. Algebra **36** (1975), 1–15.
- [Po73] N. Popescu, *Abelian categories with applications to rings and modules*, Academic Press, 1973.
- [Sch90] H. J. Schneider, *Principal homogeneous spaces for arbitrary Hopf algebras*, Israel J. Math. **72** (1990), nos. 1–2, 167–195.
- [Sw67] M. Sweedler, *Hopf algebras with one grouplike element*, Trans. Amer. Math. Soc. **127** (1967), no. 3, 515–526.
- [Sw69] M. Sweedler, *Hopf algebras*, Benjamin, 1969.
- [Tak72] M. Takeuchi, *A correspondence between Hopf ideals and sub-Hopf algebras*, Manuscripta Math. **7** (1972), 251–270.
- [Tak79] M. Takeuchi, *Relative Hopf modules—Equivalence and freeness criteria*, J. Algebra **60** (1979), 452–471.
- [Ta67] J. Tate, *p -divisible groups*, pp. 158–183 in: Proc. Conf. Local Fields (Driebergen, 1966) (ed. T. A. Springer), Springer-Verlag, 1967.
- [We95] C. A. Weibel, *An introduction to homological algebra*, Cambridge Univ. Press, 1995.

⁽¹⁷⁰⁾N.D.E. To this reference, which appeared in the original, we have added the references which follow.